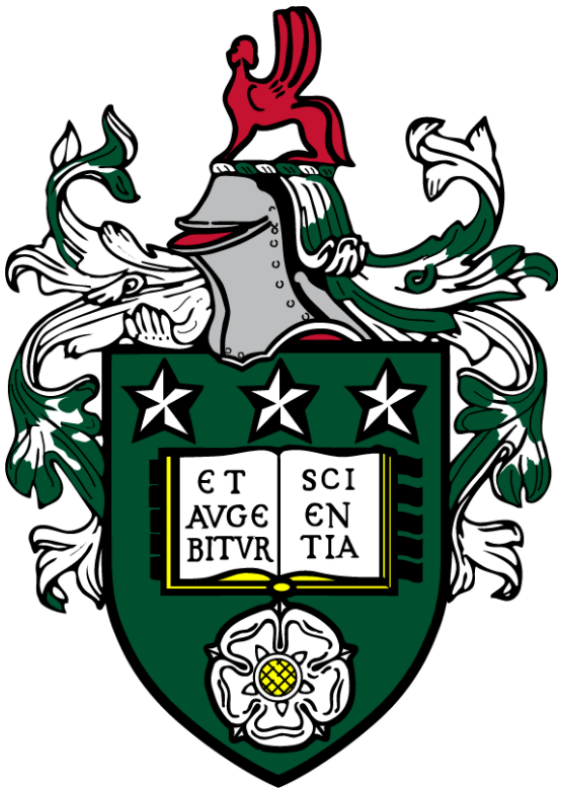
NEURONS VS NOISE: USING FEEDFORWARD NEURAL NETWORKS FOR STOCK MARKET INDEX PRICING AND PREDICTION



KAVI PATEL

BSC ACCOUNTING AND FINANCE | THE UNIVERSITY OF LEEDS

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2 ABSTRACT

2.1 Abstract

This paper builds, trains, and tests a Feed Forward Neural Network for stock market prediction. Using macroeconomic variables as indicators for the FTSE100 Index price, the model outputs predictions compared to a holdout sample to evaluate performance and reflect on the seminal theory around market efficiency and investor behaviour. The paper aims to provide an alternative modelling approach to traditional models such as the Capital Asset Price Model and ARIMA forecasting. The study collects and processes macroeconomic variable data between 1984 and 2024 from reputable sources, contributing to a complete, clean, and reliable data frame. A core model named "BASE" was used as a starting point, an unlagged neural network with one input layer, four hidden layers, and one output layer. The study compares BASE to a duplicated but lagged model in intervals of one and twelve months, and an ARIMA model. This study finds that the markets are predictable by developing and testing various models with varying accuracy to discover that L1M, a duplicate of BASE with MEVs lagged by one month, was the most accurate. This dissertation, therefore, questions market predictability, accepting both hypotheses regarding price prediction and price change direction at a given time interval. This alternative approach has implications in financial academia and financial policymaking and has the potential to be applied as a component of a successful trading strategy.

2.2 Definitions

ARIMA	"Auto Regressive Integrated Moving Average" Model refers to this paper's second control variable
Artificial Intelligence	A branch of computer science dedicated to algorithms that can perform tasks that typically require human intelligence
BASE	The original, unlagged neural network used as the core model in this paper
Correlation	A statistical measure of the extent to which two variables move together
Efficient Market Hypothesis	A 3-tiered theoretical model, suggesting that markets immediately adjust prices to new information as it is released
Feed Forward Neural Network	A type of neural network where information moves in one direction only
FTSE100 Index	Benchmark index of the 100 largest companies listed on the London Stock Exchange by market capitalisation
Fully Connected Dense Network	A neural network where every neuron in every layer is connected to every neuron in the previous and subsequent layers
London Stock Exchange	Stock exchange for Equity in the United Kingdom
LxM	"Lagged (by) x months", the lagged models used as control variables in this paper, where "x" in LxM refers to the number of months lagged.
Holdout Sample	A subset of data is excluded from training to test the model once trained on historical data
Market Anomaly	Evidence of the market pricing equity differently from the value suggested by fundamental analysis

Market Noise	Movement in stock prices is not driven by fundamental information
Market Player	An individual or entity that buys and sells equity in the market
Mean Square Error	Statistical measure of the extent to which two variables are equal across a time series
MEV	"Macroeconomic variable" refers to each of the 26 independent variables that span January 1984 to July 2024 across a range of core indicators such as GDP, Unemployment, and Inflation
Neural Network	A computational model inspired by the structure of the human brain, in which layers of interconnected "neurons" process data and learn patterns through training
Random Walk Theory	Theoretical model suggesting that information is random, and therefore, in conjunction with the Efficient Market Hypothesis, prices should also follow and be random

3 INTRODUCTION

3.1 Purpose

Since the earliest forms of civilisation, trade has played a pivotal role in developing communities across the globe. In modern economies, trade is centred around exchanges, evolving into what now operate as complex financial markets, where buyers and sellers convene and exchange their goods at prices derived from supply and demand. Central to the academic debate on markets is the notion of price unpredictability, formalised by the Efficient Market Hypothesis and Random Walk Theory, in conjunction suggesting that asset prices follow a pricing mechanism that is impossible to forecast.

Yet, with the rapid advancement of Artificial Intelligence over recent years, these long-standing theories have been called into question. The ability of complex algorithms, which supersede human capacity in detecting complex, nonlinear relationships in data, presents a compelling opportunity to disrupt academic assumptions and challenge the idea of fully efficient markets.

This dissertation investigates whether a Feedforward Neural Network, trained on 26 macroeconomic variables, can accurately and consistently predict FTSE100 Index prices as a proxy for the equity market in the United Kingdom. By constructing a model and comparing it to traditional methods such as lagged and ARIMA models, this study aims to explore whether AI can provide meaningful insight into market pricing dynamics.

This paper contributes to the growing literature on the intersection between Machine Learning and Finance. It also explores theoretical implications should the model succeed and considers how neural networks might influence future investment strategies and financial modelling.

3.2 Findings

BASE achieved 95.28% MSE accuracy and 88.25% correlation accuracy. However, it was found that L1M outperformed all other models, with 96.97% MSE accuracy and 90.58% correlation accuracy. The ARIMA model failed to predict prices, reinforcing nonlinear and non-stationary relationships between MEVs and the FTSE100 Index price, which remained hidden even after variable analysis conducted in 5.7. As a result, both hypotheses are strongly supported, suggesting that neural networks can predict stock market prices and the direction of price changes.

3.3 Motivation

Models like the Efficient Market Hypothesis and Random Walk Theory rely on unrealistic assumptions such as rational player behaviour and perfect public information, struggling to reconcile theoretical prices with observed anomalies, behavioural patterns, and general volatility in modern markets. AI offers a powerful alternative; a fresh, non-linear, data-driven approach to complex financial modelling. These models are uniquely suited to capturing intricate relationships between variables in massive datasets, far beyond the capability of classical econometric models and human judgment. If this approach demonstrated predictive accuracy, we could challenge the notion that prices fully reflect all available information, leading to significant implications for investment strategy, policymaking, and the future role of AI in financial decision-making.

3.4 Gap in the Literature

AI modelling aligns with a gap between theory and practice. Many academic papers diagnose inefficiencies and abnormal patterns in the market but fail to offer a tangible model or pricing tool as an alternative, such as Fama (1970), Schwert (2003), and Malkiel (2003), either reinforcing or critiquing the EMH, with no practical advice. Therefore, this paper seeks to bridge the gap by building, testing, and deploying the model using real-world data. Secondly, macroeconomic variables are underrepresented as market indicators in academia. Studies like Nti et al. (2020) and Lam (2004) acknowledge the role of MEVs, but very few go on to use them alone without firm-specific or technical data. While technical indicators dominate Machine Learning Finance research, macroeconomic variable-driven prediction of index-level pricing is relatively novel. Therefore, this study adds to an emerging area of study with high relevance to national and institutional investors.

In addition, while a wide range of work explores different models and technologies, no single, robust, and replicable model can systematically and reliably convert data into accurate prices. This study aims to identify whether a neural network could satisfy this requirement. Furthermore, there is limited research on using neural networks in an EMH context. Studies like Dunis & Jalilov (2001) and Murekachiro (2020) show that neural networks outperform traditional models. However, they rarely challenge the EMH directly or test whether AI can identify pricing errors in real index-level data like the FTSE100 Index. Furthermore, behavioural insights from Lo & MacKinlay (1988), Barberis et al. (1998), and the Adaptive Markets Hypothesis (Neely et al., 2007) suggest that inefficiencies fade over time. However, although academics have suggested that the markets are dynamic, models remain static and outdated.

3.5 Contribution

This study builds a functioning, testable model in less than 100 lines of code. It brings underused macroeconomic data into the machine learning space, benchmarks the neural network against traditional models like ARIMA and lagged MEV regressions, and tests the lag required for macroeconomic information to be absorbed by the market.

4 LITERATURE AND HYPOTHESIS

4.1 Introduction

This section covers three primary areas: The Efficient Market Hypothesis, Behavioural Finance, and Machine Learning. I will review seminal papers to critically evaluate, compare themes and methodology, and draw conclusions to position my work amongst other literature and existing theories.

4.2 Market Efficiency and The Random Walk Theory

In 1970, Eugene Fama formalised the "Efficient Market Hypothesis", proposing that an asset price fully reflects all the available information. He distinguished between 3 nested categories: "Weak", "Semi-Strong", and "Strong" form efficiency, depending on the type of information incorporated. Under this framework, a given player in the market cannot adopt a trading strategy based on past or public data and consistently earn a return in excess of the market. Fama's empirical review revealed limited evidence against weak and semi-strong efficiency, with anomalies often explained by other factors such as data mining and ignored transaction costs and taxes. Admittedly, critics scrutinise the questionable assumptions around perfect information, rational behaviour, and insider trading, which reduce real-world applications. Despite this, the EMH presents a compelling and widely accepted challenge to predictive modelling. If the markets are semi-strong, as suggested, then MEVs in my model should not contain enough information to provide explanatory power, earning constant excess returns. Testing this theory underpins this paper's contribution to predictive modelling.

Further to his work in 1970, Fama established and empirically tested the random walk hypothesis in 1965, finding that stock price volatility is "independent" and "identically distributed", rendering my attempt to draw technical patterns useless. Despite long-tailed distributions, his results suggested that all past and present public data has been incorporated into prices. Of course, this challenges my approach and positions my work as a potential disruptor.

In response to Fama (1965), Malkiel (2003) broadly supports EMH, suggesting that anomalies in the market are small, unstable, and often disappear once the model factors in accounting costs. However, he acknowledges consistent patterns such as mean reversion and momentum, though he doubts how they could be used as components in profitable trading strategies. Furthermore, Fama and French (1992) challenge the Capital Asset Pricing Model, concluding that a size factor and book-to-market values have greater explanatory power than a beta in evaluating returns. Both papers highlight tensions between the EMH and opposing academics.

Furthermore, Schwert (2003) reviews anomalies, finding that they often weaken or disappear over time, likely due to arbitrage. He argues that many anomalies provide no value when out of sample, making them useless. This is a direct contradiction to my approach to out-of-sample testing.

In summary, these seminal papers form the foundation of work in the field. However, much of the literature around the EMH identifies anomalies with no solution. They also do not refer to MEVs in the same way that my study is empowering them. On the other hand, BASE provides a repeatable, direct test of the predictive value found in observed inefficiencies. If my model can successfully predict FTSE100 Index prices using MEVs, the EMH assumptions discussed in this section have been sufficiently challenged, most notably in the semi-strong form.

4.3 Behavioural Finance and Market Anomalies

De Bondt and Thaler, in their 1985 paper, find that stocks with extremely poor past performance tend to yield statistically significant "reversal" performance when compared to previous winners over the following 3 years, evidence of investor overreaction. Furthermore, this effect is most common in January and persists up to 5 years after formation, encouraging my research and contradicting the weak form of the EMH. This study, therefore, provides a profitable trading strategy. Findings suggest players overweight recent news and underweight older data, creating predictable mispricing. This supports my lagged models, reflecting the same behavioural delay in players' market information processing.

Jegadeesh and Titman, in their 1993 paper, provide evidence of momentum; stocks that have performed well over the past three to twelve months tend to continue outperforming in the short term. The study highlights market underreaction to "firm-specific" events, leading to momentum profits. This finding directly contradicts the weak form of EMH. Furthermore, my reasoning is based on "delayed stock price reaction" to information released, supporting my lagged models.

Kahneman and Tversky proposed the Prospect Theory in their 1979 paper as an alternative to the expected utility theory. This core model suggests that investors are "risk averse in... sure gains and... risk seeking in... sure losses". Furthermore, players disproportionately weigh losses over equivalent gains, a phenomenon known as loss aversion. This effect causes mispricing and explains over and underreaction. As a result, this paper supports the idea that market prices regularly and systematically deviate from their fundamental values and, therefore, justifies my model in attempting to exploit these patterns. Although a seminal paper, Kahneman and Tversky oversimplify real-world scenarios and offer no incorporation of MEVs or a predictive mechanism.

Barberis et al. (1998) propose a behavioural model, concluding that investor sentiment causes underreaction in the short term and overreaction in the long term to new data. This stems from psychological biases such as "representativeness" and "conservatism", seeing false patterns in random data, and updating beliefs too slowly. However, a fundamental flaw in this paper is the assumption that psychological bias is persistent, where the "investor never changes the model". Furthermore, this approach is stock-specific with no MEV inclusion, while my study is market-wide.

Furthermore, Barber and Odean (2001) use gender as a proxy to show that overconfidence is often a precursor to overtrading, reducing returns, with men trading 45% more and underperforming compared to women. It is worth noting that there is no reference to why people trade, perhaps for enjoyment. The data is also from a single "large discount brokerage", which may not incorporate the true demographic within the markets. These findings support my model's focus on delayed market reactions and justify my set of lagged models; sentiment and overconfidence introduce consistent, learnable, and exploitable structures into the data.

In conclusion, we can deduce that players are irrational, leading to slow-reacting markets and pricing errors. The lag caused by this delay in the digestion of information validates my model and directly supports the use of lagged duplicates in analysis. My neural network will be built with this in mind, combining lagged MEVs to capture behavioural insights.

4.4 Forecasting with Traditional Methods

Traditional models, such as ARIMA, the CAPM, Arbitrage Pricing Theory, and Discounted Cash Flow, are common. However, these models rely on linearity, rational investor behaviour, and market equilibrium assumptions.

ARIMA is a strong indicator for short-term univariate time series. Mahadik et al. (2024) compare ARIMA to LSTM modelling, including similar preprocessing, feature scaling, and time series splitting to my approach. Furthermore, Mahadik et al. observed a 90% accuracy, close to that of L1M. Mahadik et al. suggest that ARIMA performs well in short-term horizons but fails to adapt to longer-term data.

Furthermore, Fama and French (1992) scrutinise the CAPM's limitations, suggesting that size and value have greater explanatory power than beta, which "does not seem to... explain... average stock returns" (Fama and French, 1992). In addition, the DCF and DDM models are prevalent in valuation but rely on stable and accurate assumptions, most notably in discounting (Froidevaux, 2004).

These alternative models have several weaknesses. Firstly, they struggle with non-linearity and given the complex relationships between MEVs and the FTSE100 Index price, my model is positioned in a gap in the existing infrastructure. Linking back to behavioural topics discussed in 4.3, these anomalies are non-linear, so exploiting them using the model would require a method that is more sophisticated than existing models. Furthermore, these models are static in structure and poorly perform when adapting to macroeconomic shifts and lagged economic reactions. My model aims to improve upon their performance. In support of this, Froidevaux (2004) suggests that traditional models yield inconsistent predictions under volatility, implying that valuation "is not a science". Furthermore, Mahadik et al. (2024) indicate that macro-based forecasting models lack flexibility, observing that performance degrades in dynamic markets.

Therefore, existing models offer a benchmark to compare my model to, specifically ARIMA, which will be selected as a control variable in this study. If my neural network can outperform ARIMA, it implies that nonlinear relationships are present and have explanatory power in predicting the market, yet to be unlocked.

4.5 Machine Learning and Neural Networks

The core premise of this study is that Machine Learning methods, specifically neural networks, can learn from patterns in data and overcome the limitations of traditional linear methods discussed in the previous section.

Lam (2004) successfully integrates MEVs into a neural network. The study concludes that neural networks have an edge in prediction but are not as good as the top money managers in the market. And, despite including MEVs, the paper deduces that they had little impact as they "do not outperform the... benchmark in this study".

Murekachiro (2020) built a hybrid model, similarly integrating MEVs. This paper introduces complex adaptive systems theory, linking AI to behavioural finance. Although the study offers a thorough methodology, limitations include high computational demands and limited generalisation, focusing on specific stock indices, raising concerns about adaptability; for example, finding an anomaly that "African stock markets relate differently to macroeconomic variables", demonstrating "varying market ecologies".

Nti et al. (2020) systematically reviewed 122 Machine Learning studies over 11 years, finding that MEVs are "not commonly used well" and that most studies focus too heavily on technical indicators, where only 7% used MEVs. The paper concludes by recommending that I combine multiple data sources for improved accuracy and that MEVs are underused, especially as the sole input for prediction. This supports my approach, which fills a gap by using only macroeconomic variables in a neural network, a combination rarely tested in existing literature.

Leigh et al. (2001) applied a neural network to predict the NYSE Composite Index. They employed an out-of-sample testing window to validate performance, supporting my

methodology and providing generalised and significant results. Although this study only includes neural networks as a component of its complex approach, it confirms that using neural networks for market prediction is feasible.

In addition, Sermpinis et al. (2019) compare neural network architecture in trading and find that Recurrent Neural Networks statistically outperform benchmarks, presenting "stronger profitability" compared to other types of neural networks, but exhibit high rates of overfitting and bring hefty computational costs. Reconciling these weaknesses with an otherwise strong paper, Sermpinis et al. emphasise the importance of robust testing and validation, a suggestion I can directly incorporate into my model.

As such, my approach to using MEVs in a neural network is valid and encouraged and has been somewhat done before. My model has a distinct macroeconomic focus, transparent architecture, and lagged variable structure, all supported by existing literature.

4.6 Synthesis and Positioning

This literature review reveals a clash between three dominant schools of thought. Firstly, the Efficient Market Hypothesis (Fama, 1970; Malkiel, 2003) suggest that prices fully reflect information and cannot be consistently predicted. Secondly, behavioural finance (De Bondt and Thaler, 1985) and (Barberis, 1998) demonstrate that lagging and irrational players in the market have delayed responses to the news. Finally, Machine Learning approaches (Leigh et al., 2001), (Sermpinis et al., 2019), and (Murekachiro, 2020) challenge both by modelling complex, nonlinear relationships in the economy and markets.

Most existing literature identifies weaknesses in theory and anomalies, with no effort to construct viable, predictive solutions. MEVs are theoretically relevant and are underused in Machine Learning research, with few papers successfully combining MEVs, behavioural lags and neural networks into a single transparent and accurate predictive framework. My research bridges the gap between theory and application by testing semi-strong forms of the EMH (Fama, 1970).

4.7 Hypotheses

As a result of this literature review, two hypotheses have been established:

Hypothesis 1	Neural networks can accurately price the FTSE100 Index at a given time interval, using MEVs as independent variables
Hypothesis 2	Neural networks can accurately predict the direction of the FTSE100 Index price change, using MEVs as independent variables

5 DATA AND METHODOLOGY

5.1 Independent Variable: MEVs

Central to building a neural network capable of predicting stock prices is a thorough and purposeful selection of Macroeconomic indicators to use as explanatory variables. Literature around the explanatory power of variables compared to one another in their impact on the economy is at a consensus, with key indicators including GDP, Unemployment, Inflation, Interest Rates, and Budget Deficit (Bloomenthal, 2023).

5.2 Dependent Variable: FTSE100 Index Price

The dependent variable is the output of the model, the FTSE100 Index price.

5.3 Control Variable (Hypothesis 1): The Lagged Fundamental Model

The first hypothesis can be controlled using a lagged model. To investigate whether past macroeconomic conditions contain explanatory power for current FTSE100 Index prices, a second dataset was created, which is equal in all aspects but lagged by several months. This version uses MEVs at time t-x (Eq. 2) to predict the FTSE100 Index price at time t.

$$\text{Base Model: } Y_t = f(X_t) + \varepsilon_t \tag{1}$$

$$\text{Lagged Model (n months): } Y_t = f(X_{t-n}) + \varepsilon_t \tag{2}$$

where Y_t is the FTSE100 Index price at time t
where X_t is a single factor combining all 26 MEVs at time t

This approach was inspired by the intuition of lagged models, testing if past values contain helpful information for forecasting future outcomes. A stronger performance when using lagged MEVs would suggest that the markets are inefficient and that it takes an interval of time (the lag) for the information to be incorporated into market prices.

5.4 Control Variable (Hypothesis 2): The Random Walk Model

The second hypothesis can be controlled for using a Random Walk Model, following Lo and McKinlay (1988). Firstly, we need to difference the FTSE100 Index (Eq. 3).

$$e_t = Y_t - Y_{t-1} \tag{3}$$

Secondly, we should establish a null hypothesis (Eq. 4).

$$H_0: \text{Cov}(e_t, e_{t-k}) = 0, \text{ where } \forall\ k > 0 \tag{4}$$

Next, we apply the Durbin Watson Formula (Eq. 5) to measure first-order autocorrelation, if any, between e_t and e_{t-1}.

$$\text{Durbin Watson} = \frac{\sum_{t=2}^{T} (e_t, e_{t-k})^2}{\sum_{t=2}^{T} e_t^2} \tag{5}$$

where T is the total number of time intervals
where e is the residual at time t

Finally, we apply the Ljung-Box Q Statistic (Eq. 6), testing for autocorrelation across multiple lags. This process checks whether autocorrelation within the dataset is jointly equal to 0, indicating whether residuals behave like white noise.

$$Q = n(n+2) \sum_{k=1}^{h} \frac{\hat{\rho}_k^2}{n-k} \tag{6}$$

where Q is the Ljung-Box Q Statistic
where n is the number of observations
where h is the number of lags tested
where $\hat{\rho}_k$ *is the estimated autocorrelation at lag k*

5.5 MEV Collection

The FTSE100 Index was introduced in January 1984, at 1000 (GBP.00), and MEVs span from January 1984 to December 2024 (London Stock Exchange, 2025) from 9 sources; Office of National Statistics (12), Bloomberg (4), FRED (3), Investing.com (3), Bank of England (1), Gold.org (1), World Bank (1), IFS.org (1), and UK Land Registry (1).

The Office for National Statistics is the UK's largest independent producer of official statistics for the United Kingdom; one of the most reputable and cited by academics. The ONS follows a standardised, rigorous methodology, meeting international standards such as the OECD. As suggested by 12 out of 26 of my datasets being sourced from the ONS, its website contains an enormous volume and range. The core weakness of the ONS is the lack of data from before 1997. Perhaps this was an institutional update across the whole ONS, supporting my suspicion regarding the coincidence of July 1997 with the birth of the internet; if this were not the case, I would have collected all my data from the ONS, which would have led to standard and consistent data, a more reliable model output, and easier data processing.

The Bank of England (BoE) is the central bank of the United Kingdom, responsible for monetary policy, financial stability, and the Great British Pound. Maintaining extensive datasets, the BoE is a credible, government-backed institution with data widely used by academics. Data is published in a regular, standardised, and consistent format across datasets. However, there was limited depth in some time series datasets, and some datasets are annual or quarterly, so they are less useful to this study.

Bloomberg is a premium, professional-grade data platform offering real-time data on the financial markets and the economy. The terminal has extensive coverage, collecting data at high frequencies and packaging it into clean, pre-processed, exportable tables. Furthermore, the terminal collates data from a range of sources. Despite these strengths, the terminal does not offer insight into how its data is collected.

A range of other financial data websites contributed to my final dataset. Providing free, accessible, downloadable data made them ideal for supplementing the final dataset with a few niche time series I could not find elsewhere. However, these websites have lower reliability, with data potentially lacking rigorous validation or containing inaccuracies and no indication of source or methodology used to collect it.

Given the nature of my model and advice from my supervisor, frequency was a key aspect in building an accurate model. Holding my research to the highest academic standard, I only included datasets with monthly or quarterly data available, with no effort to spline annual data inwards 12 times, artificially generating potentially misleading data, and by extension, a misleading model.

5.6 MEV Processing

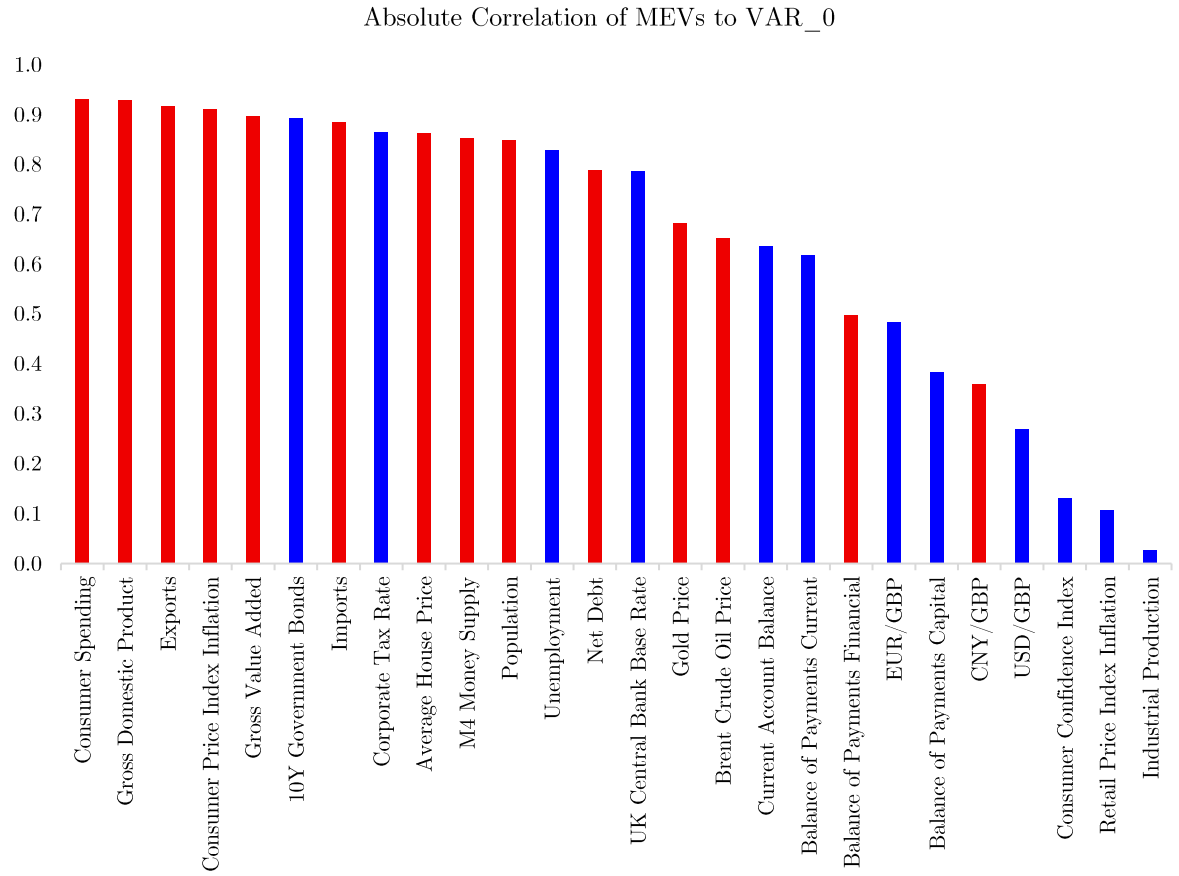
Once collected, all the data was processed into an Excel data frame organised by variable and in monthly columns for 480 intervals, automatically generating real-time summary statistics. Furthermore, Excel saves files as .xlsx extensions, allowing easy import into Stata and Excel. Some variables had to be dropped due to the absence of usable data, such as insolvencies, foreign direct investment in and out, and M&A acquisitions and disposals. I ensured that the economic explanatory power of these variables was captured by at least one of the other variables. Another important aspect of data cleaning was around units. The neural network holds no significance regarding the unit – it simply considers patterns between numbers, so unit disparity between variables theoretically should bear no impact. Concerning currency, I ensured that USD, CNY, and EUR were included as their own distinct variables to capture any impact they would have as the largest trade partners of the United Kingdom (UK Government, 2025). Where data was provided quarterly, I used a Python script to spline the data. Please refer to 10.58 and 10.59. Extreme variables were not dropped. The neural network must be accustomed to and trained on real-world data to capture and emulate real-world scenarios. This study compares AI to human data processing, so to cut data I deem extreme contradicts the purpose of this investigation.

5.7 Correlation: MEV to The FTSE100 Index

The correlation analysis in the appendix can individually reveal the relationship between MEVs and the FTSE100 Index, in Appendix 10.2. In the case of my neural network, positive or negative values have no bearing on model performance. Instead, the model considers relationships between variables only, so negative and positive values are normalised by taking the absolute value of all the correlations before analysis.

MEVs with the highest absolute correlation to VAR_0 (r values) are Consumer Spending (0.9311), GDP (0.9301), Exports (0.9162), CPI Inflation (0.9106), and Gross Value Added (0.8979). These variables have a high predictive power. Consumer spending and GDP directly relate to economic health and trajectory, while esports and GVA capture the productive capacity and trade balance, influencing investor sentiment and market confidence.

MEVs with the lowest absolute correlation to VAR_0 (r values) are Industrial Production (0.0227), RPI Inflation (0.1065), Consumer Confidence (0.1306), USD/GBP (0.2696), and CNY/GBP (0.3600). These variables have a low predictive power. This is most likely due to their indirect impact on equity markets or noise in the dataset. The range in r values, as seen below (red positive and blue negative), highlights the importance of feature selection in modelling and supports the theoretical argument that not all MEVs equally influence or even influence the equity markets with enough weight to be worthy of inclusion in my model.



5.8 Correlation: MEV to MEV

Please refer to 10.4 in the appendix. The relationships between MEVs highlight how complex and interrelated the dataset is.

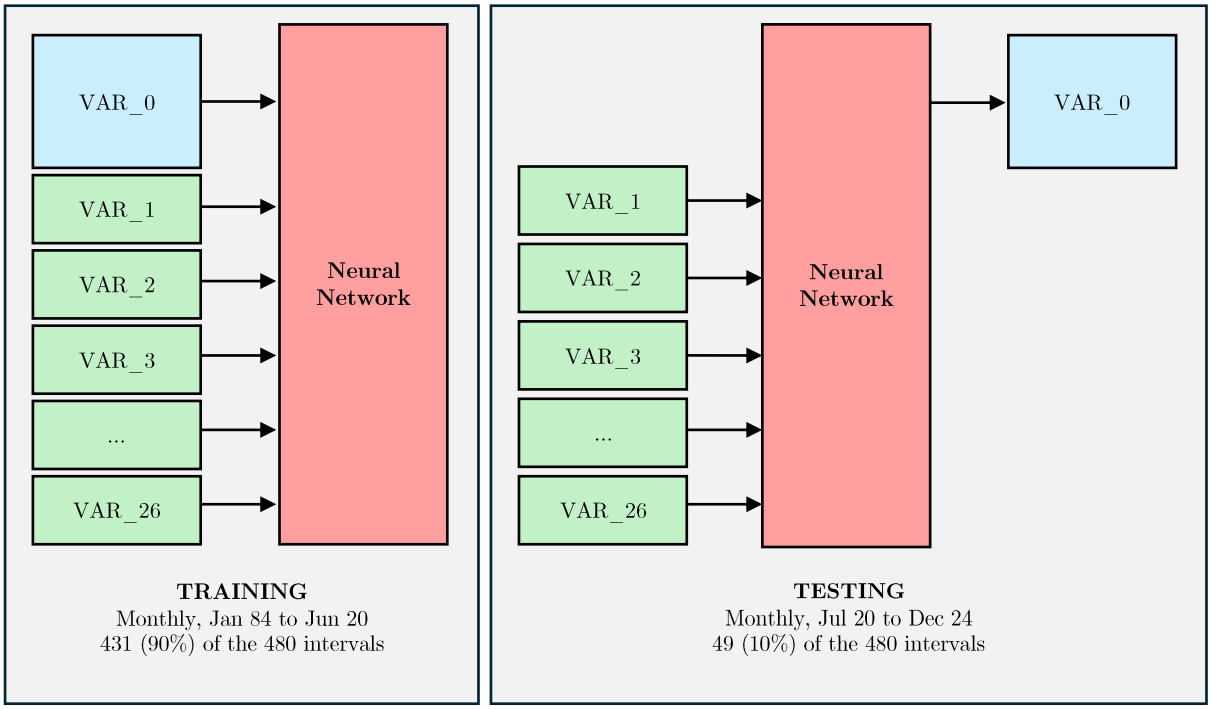
MEVs with the highest absolute correlation to each other (r values) are GDP & Consumer Spending (0.9979), Balance of Payments Current & UK Current Account Balance (0.9948), CPI Inflation & Gross Value Added (0.9918), Exports & Imports (0.9853), and Gross Value Added & Average House Price (0.9843). Such high correlations, as expected given the narrow focus of the dataset, suggest that many of the variables move in tandem, capturing similar economic activities. For example, GDP and consumer spending are inherently linked, as consumer spending is a key determinant of GDP. Furthermore, GVA and CPI inflation both reflect underlying economic output and price changes, which are related.

MEVs with the lowest absolute correlation to each other (r values) are Unemployment & RPI Inflation (0.0012), RPI Inflation & EUR/GBP (0.0023), Balance of Payments Current & CNY/GBP (0.0034), UK Current Account Balance & CNY/GBP (0.0046), and CNY/GBP & EUR/GBP (0.0046). These lower correlations may indicate that the associated MEVs provide unique, non-overlapping information to the model, useful in reducing multicollinearity and enhancing the explanatory breadth of the neural network.

Inclusion of both correlated and non-correlated MEVs supports a more thoughtful selection of inputs, a balance between capturing core economic drivers of the FTSE100 Index price, while avoiding overfitting and a model with too narrow a focus on the broader economy. The average absolute correlation between MEVs is 0.5351, with a standard deviation of 0.3173.

5.9 Training and Testing Windows

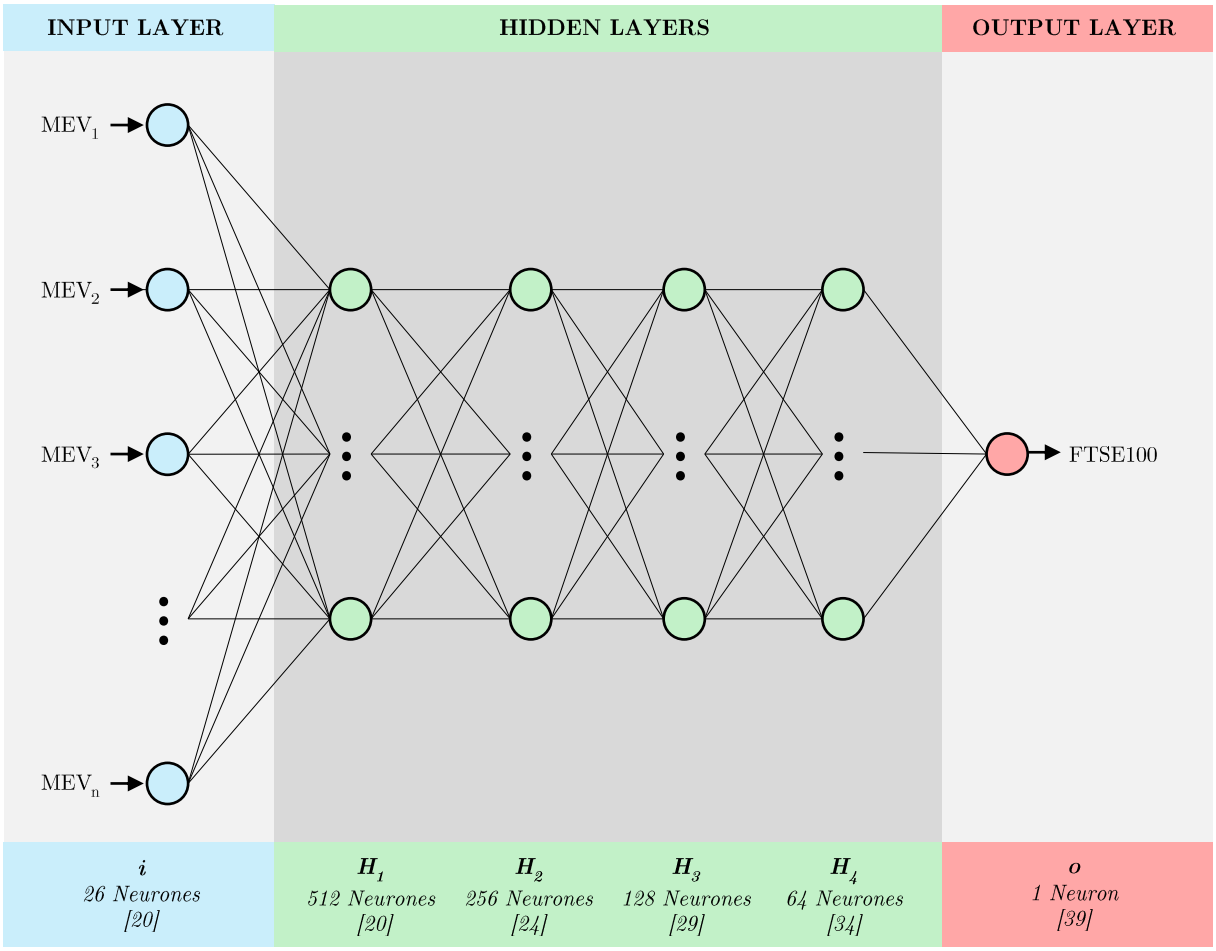
The two phases of model deployment are shown below. Firstly, the model will be trained on MEVs and the FTSE100 Index to learn relationships between the variables. Secondly, the newly trained model will be provided MEVs from the holdout period of data I kept untouched, so that I could "test" the model's accuracy.



5.10 Context of Data

The data covers January 1984 to December 2024, 40 years of monthly macroeconomic data. This timeframe includes key events such as Black Monday (1987), the Dot-com bubble (1990s), the Great Financial Crisis (2008), Brexit (2016), and COVID-19 (2020). This ensures that data spans multiple market cycles: booms, recessions, and recovery, allowing the model to learn under various economic conditions and produce more generalised results. Data is focused only on the United Kingdom. The FTSE100 Index was chosen as a proxy of the equity market in the UK, including the largest 100 companies by market capitalisation. Many FTSE100 Index firms operate globally; however, it can be assumed that each company's core operations are in the United Kingdom, based on their London Stock Exchange listing.

5.11 BASE as an AI Model



This model can be categorised as an Artificial Intelligence model, within Deep Learning, a subsection of Machine Learning. The Model is a Feedforward Neural Network, specifically a Fully Connected Dense Network, as each neuron in layer n is connected to each neuron in layers n±1. For a line-by-line explanation of the code, please refer to 10.45.

5.12 Using BASE

1. The model should be trained on data between January 1984 and June 2020 (training windows in section 5.9).
2. The model should then be used to predict the FTSE100 Index price for July 2020 to December 2024. This should be done 40 times, as seen in section 6.1, each referred to as an "iteration".
3. Since July 2020 to December 2024 are past values, this serves as a holdout period, allowing us to identify the most accurate model versions. The output of each of the 40 iterations should be evaluated using the hold-out period:
   A. The iteration with the lowest Mean Squared Error to the actual data from July 2020 to December 2024 should be noted. This model has the highest accuracy in pricing the FTSE100 Index at a single interval, given the MEV values for that interval.

B. The iteration with the highest correlation to the actual data from July 2020 to December 2024 should be noted. This model has the highest accuracy in determining movement in the FTSE100 Index, albeit without considering magnitude.

C. The remaining 38 versions of the model should be discarded.

4. The user has two versions to use, in 2 different cases:

A. The user wants to price the FTSE100 Index right now, to calculate whether it is over or underpriced. They should use the first. The individual will take a long or short position, accordingly. This investor supports the efficient market hypothesis.

B. The user wants to invest in the FTSE100 Index right now, to profit from an eventual price increase or decrease based on movement. They should use the second. The individual will take a long or short position, accordingly. This investor rejects the efficient market hypothesis.

5.13 Testing BASE

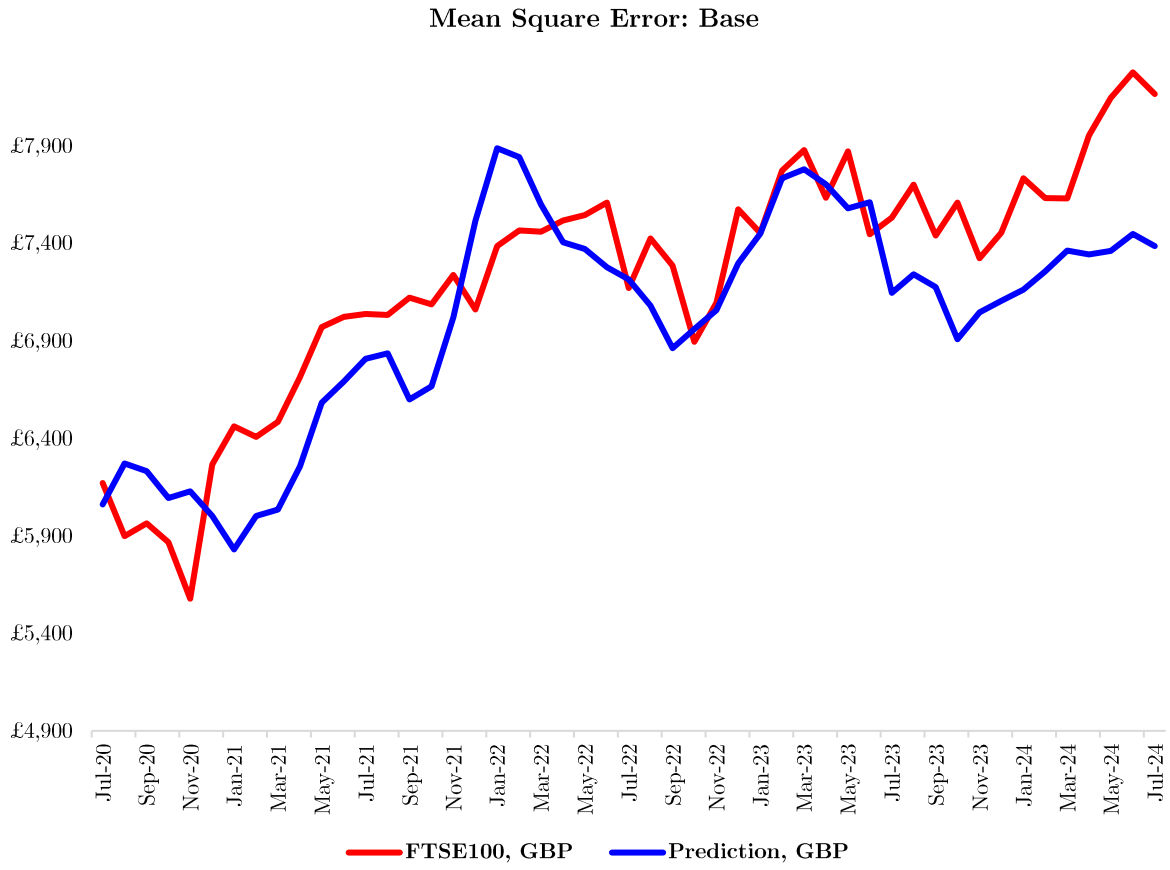
Neural network performance will be compared to 2 other models: the Lagged Model and the ARIMA Model.

6 EMPIRICAL ANALYSIS

6.1 The Best of 40

BASE is relatively small, with only 480 training intervals and 26 MEVs. Despite this, the "best of 40" approach is still 95.28% (MSE) and 88.25% (correlation) accurate for BASE; therefore, the most feasible way of predicting the FTSE100 Index price, while also meeting the requirements given the constraints of a dissertation.

6.2 Accuracy (MSE): BASE

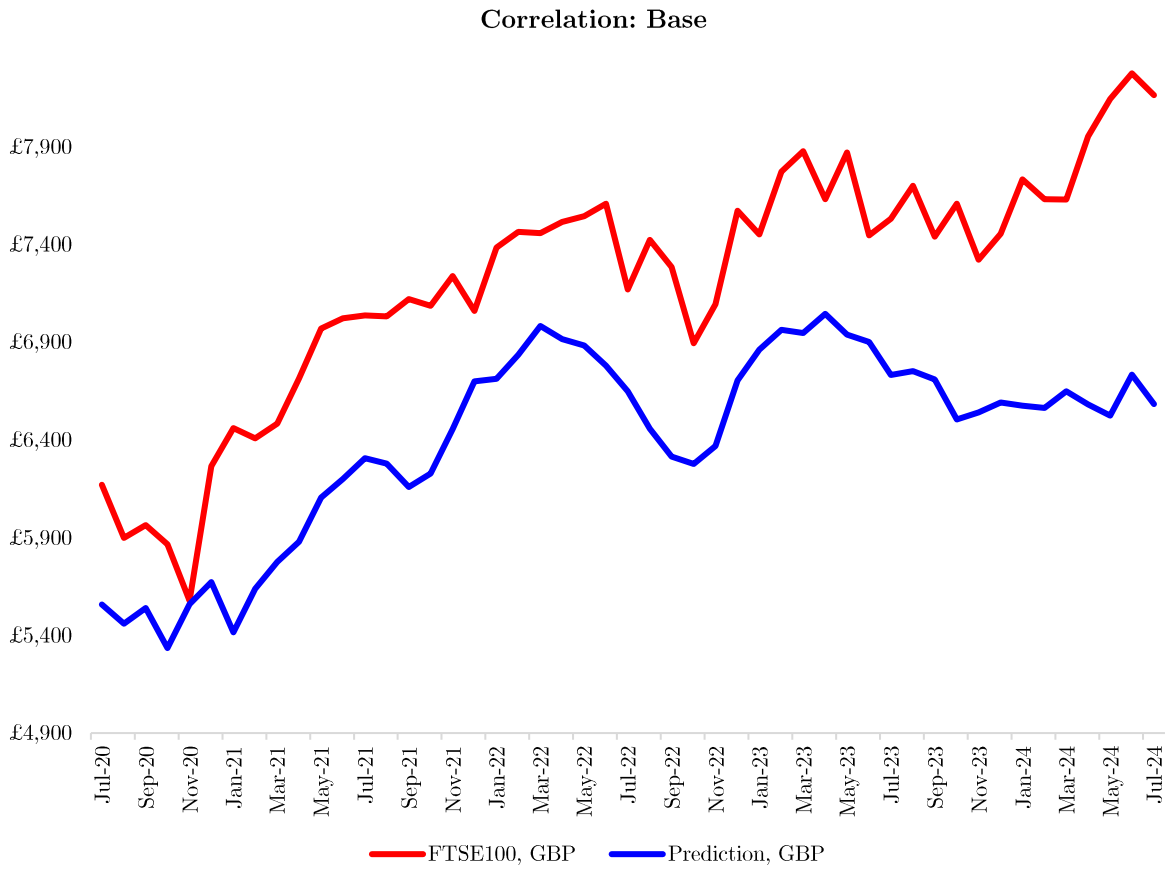


	Error	*Absolute Error*	*% Error*
Average	-207.70	339.73	4.72%
Minimum	-828.78	2.67	0.04%
Maximum	550.03	828.78	10.01%
Range	1378.81	826.11	9.98%

Accuracy	95.28%

In the case of current price prediction (Hypothesis 1), the best of 40 outputs an average absolute error of 339.73, or a percentage error of 4.72%. Therefore, the model can predict the FTSE100 Index price with 95.28% MSE accuracy. At best, the model was 0.04% out (99.96% accurate), and at worst, 10.01% out (89.99% accurate).

6.3 Accuracy (Correlation): BASE



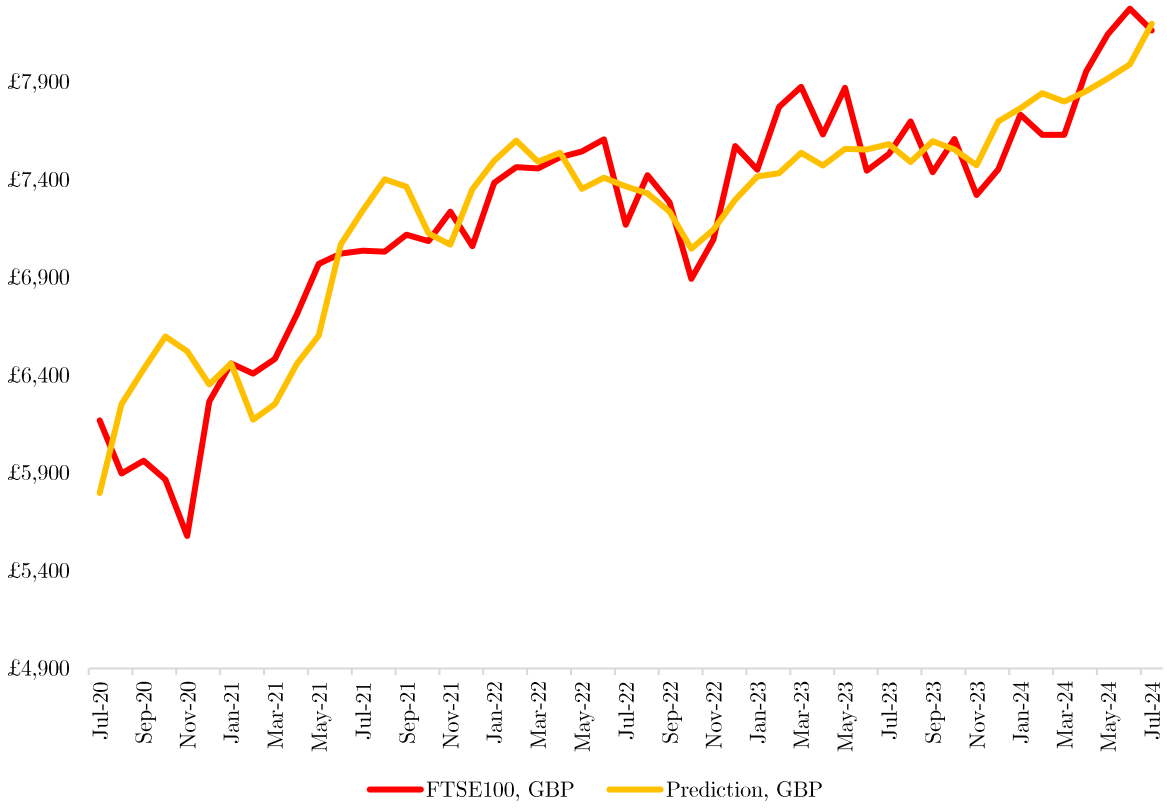
	Error	*Absolute Error*	*% Error*
Average	-807.69	807.69	11.04%
Minimum	-1619.18	17.10	0.31%
Maximum	-17.10	1619.18	19.88%
Range	1602.08	1602.08	19.57%

Accuracy	88.25%

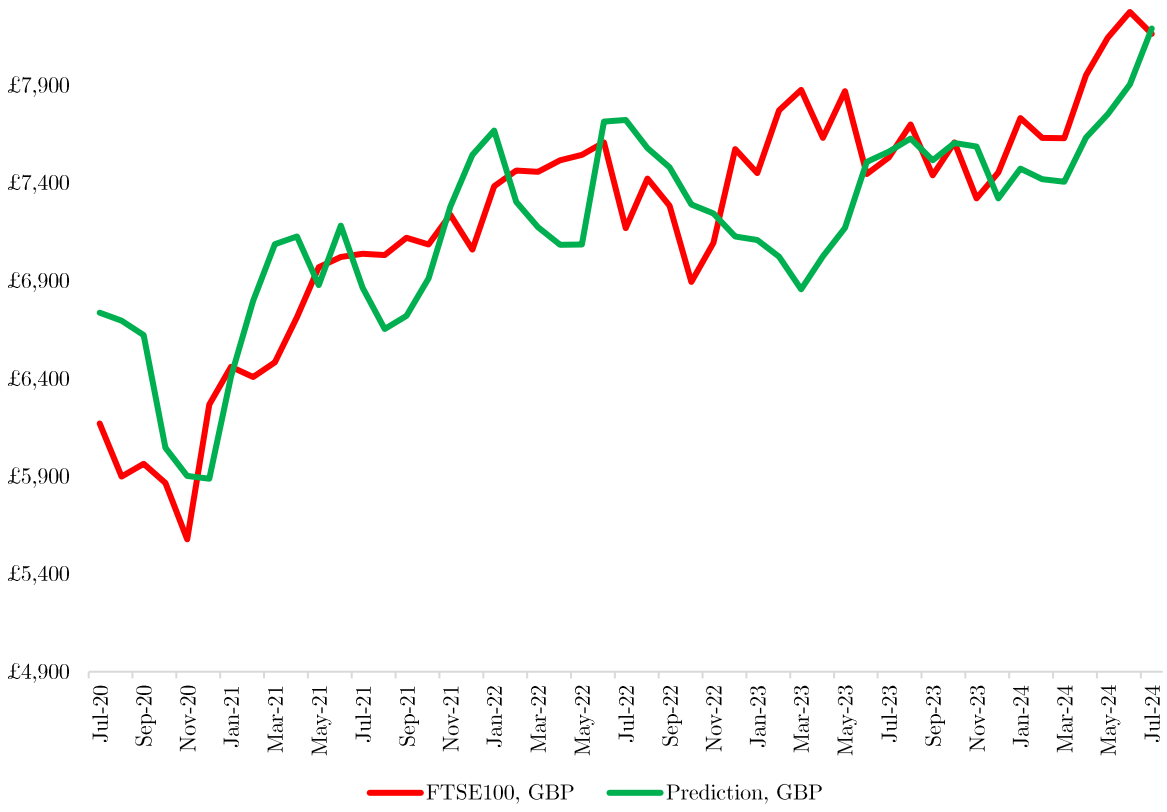
For future price movement prediction (Hypothesis 2), the best of 40 outputs an average absolute error of 807.69, or a percentage error of 11.04%. Therefore, the model can predict the FTSE100 Index price with 88.25% correlation accuracy. At best, the model was 0.31% out (99.69% accurate), and at worst, 19.88% out (80.12% accurate).

6.4 Accuracy (MSE and Correlation): L1M, L6M, L9M, L12M

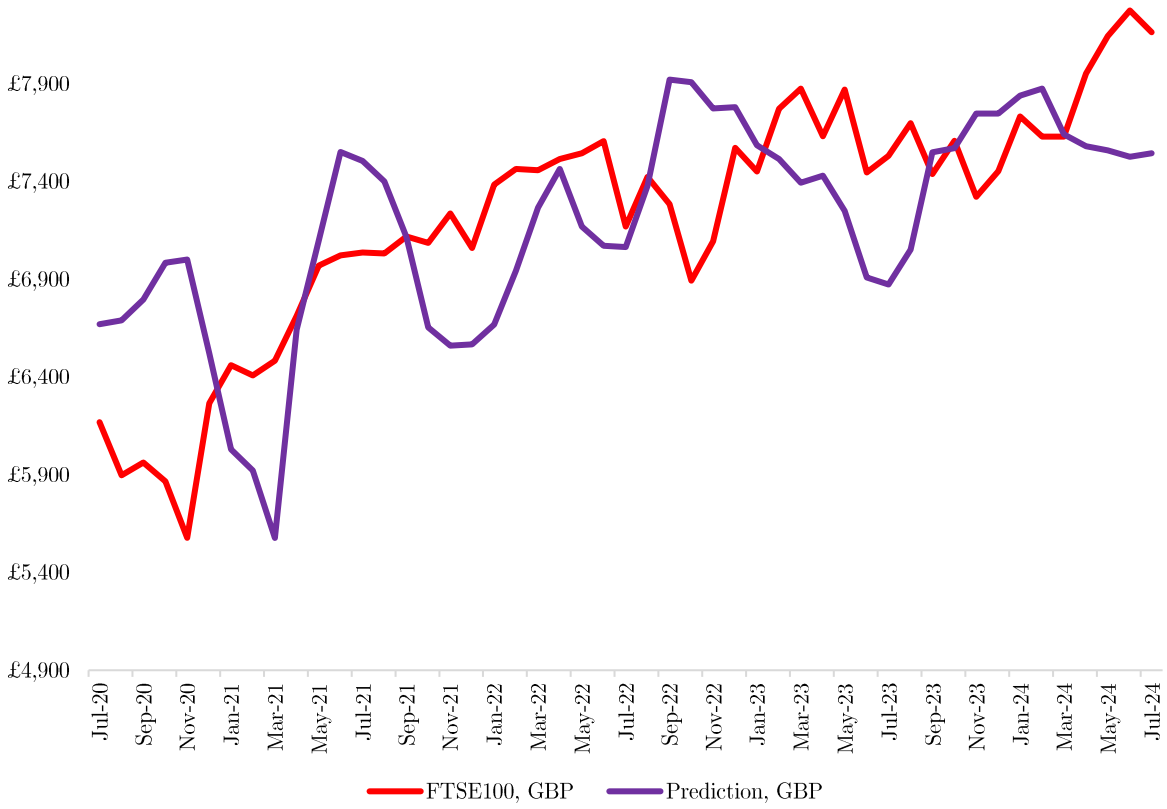
Mean Square Error: L1M



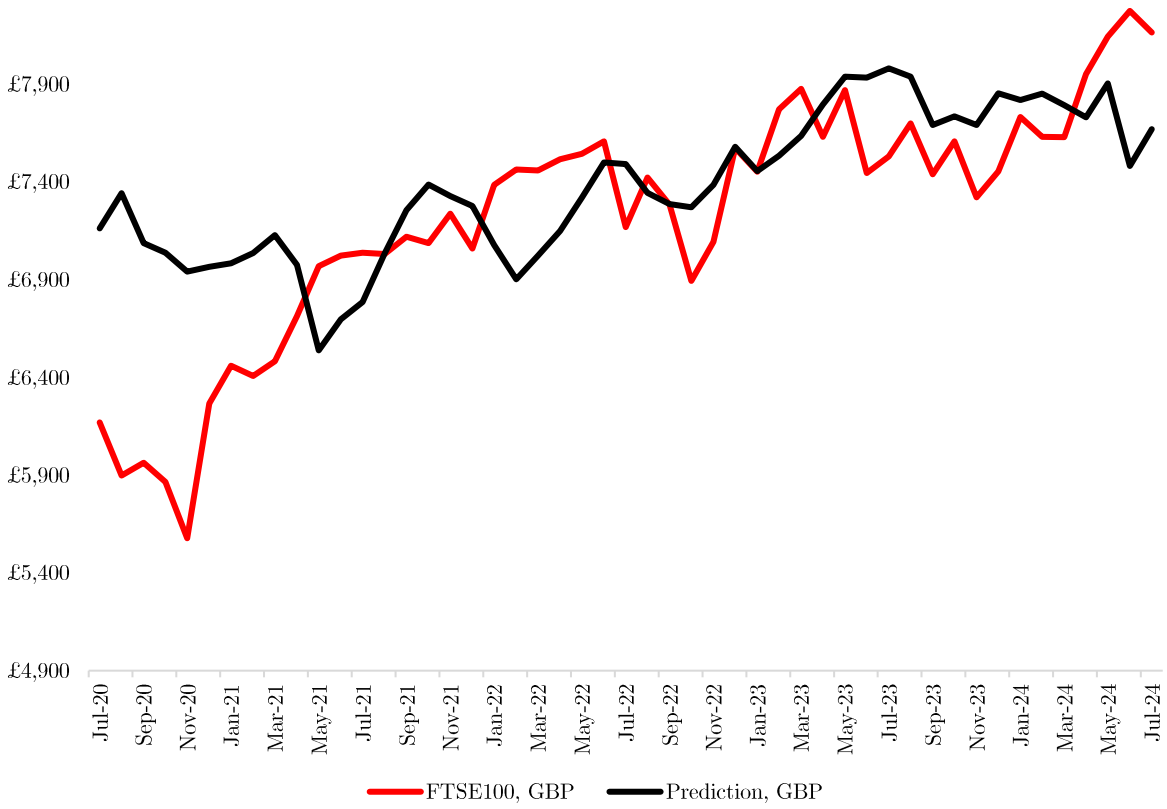
Mean Square Error: L6M



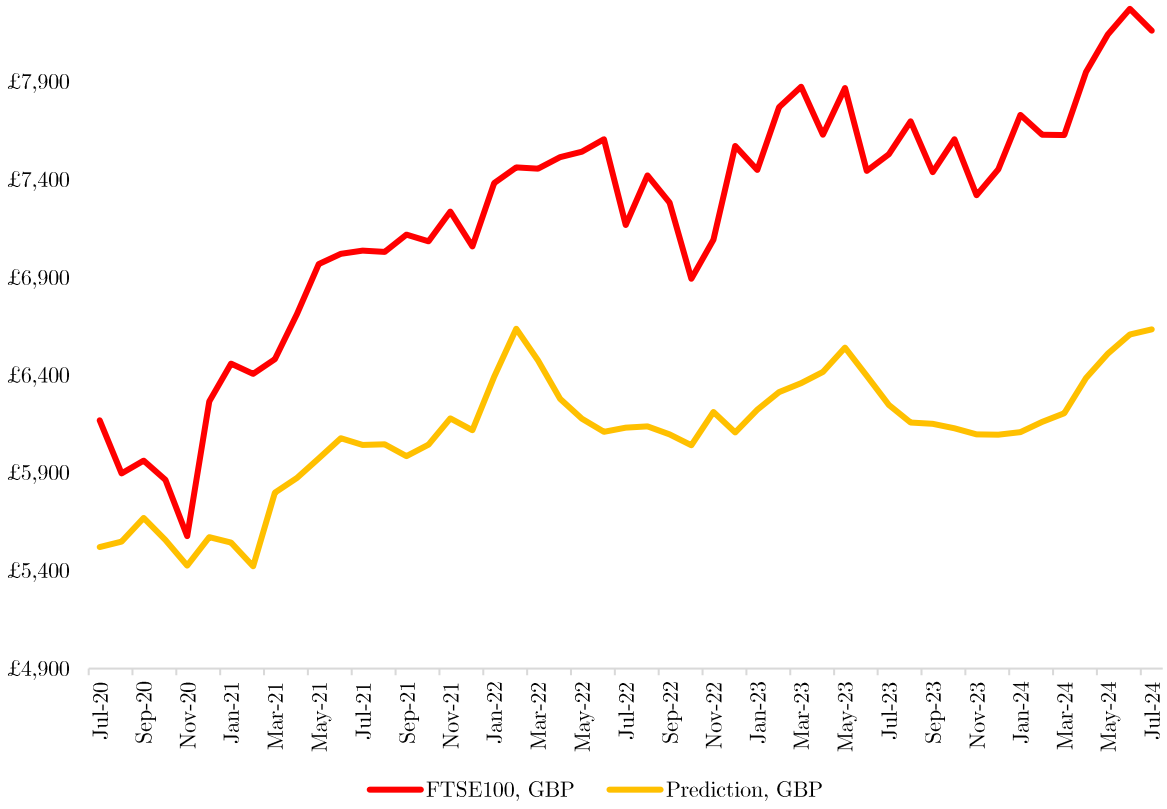
Mean Square Error: L9M



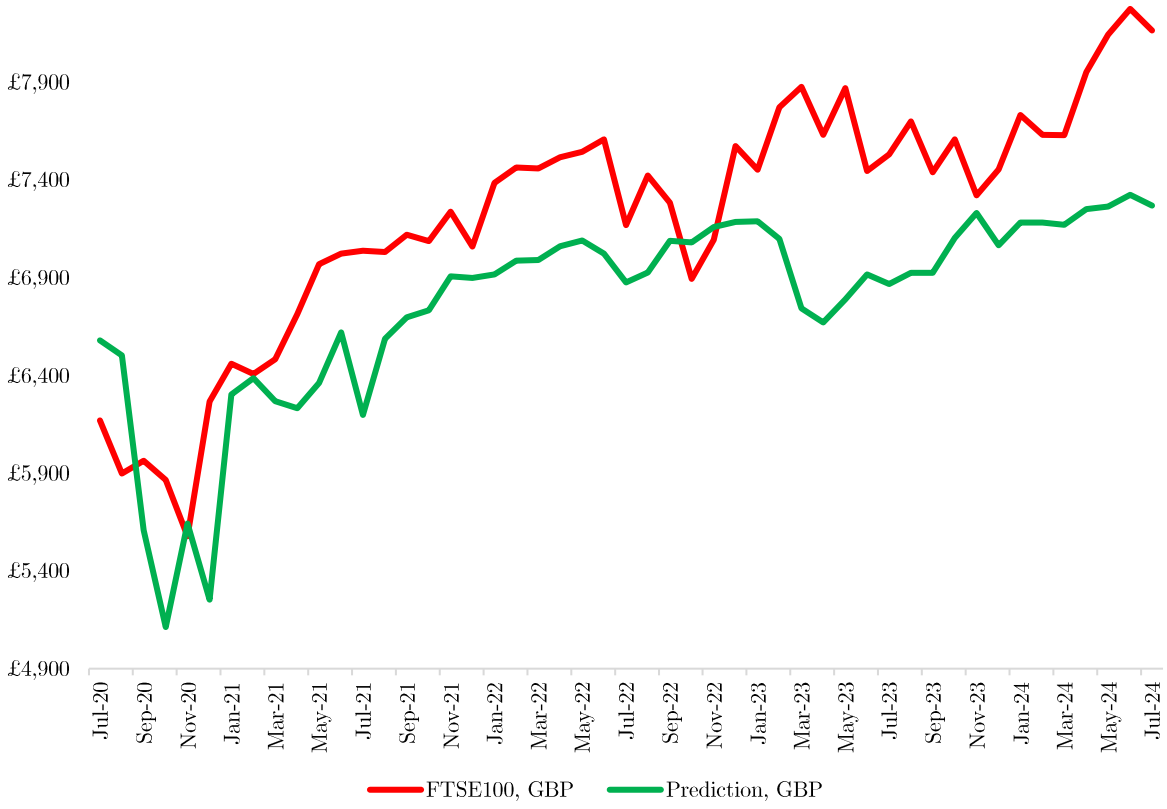
Mean Square Error: L12M



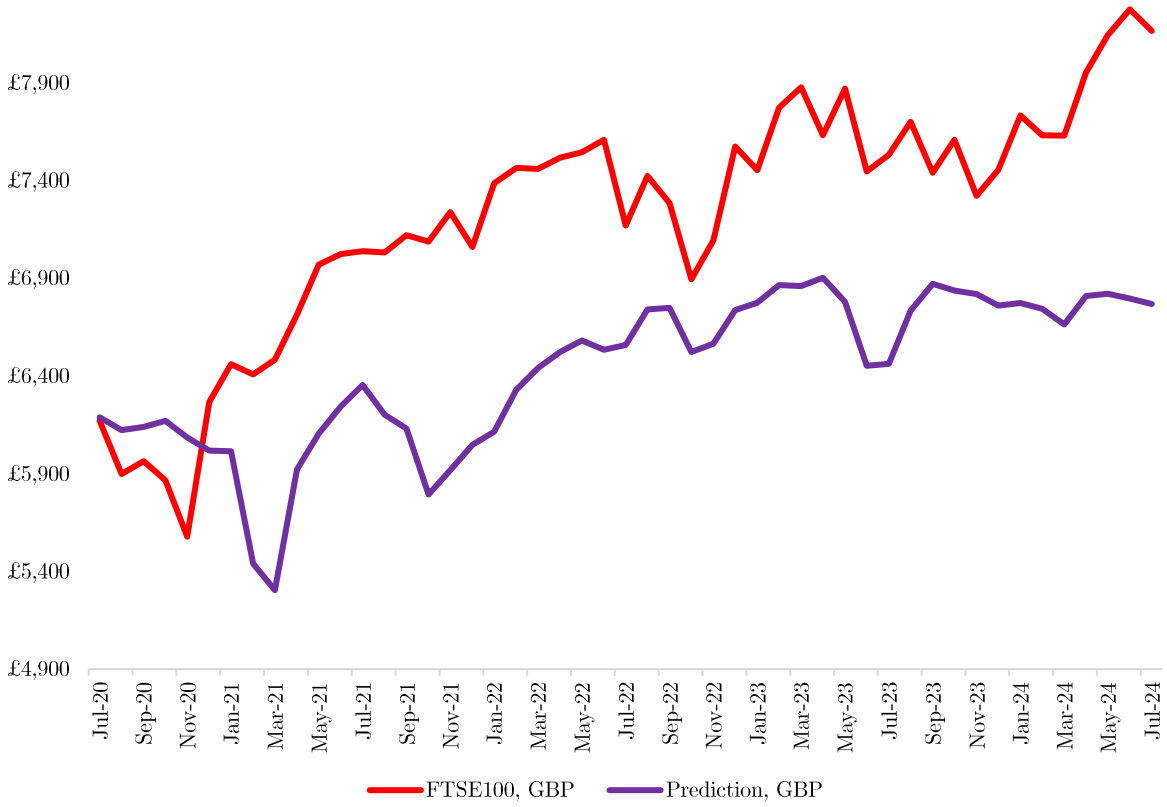
Correlation: L1M



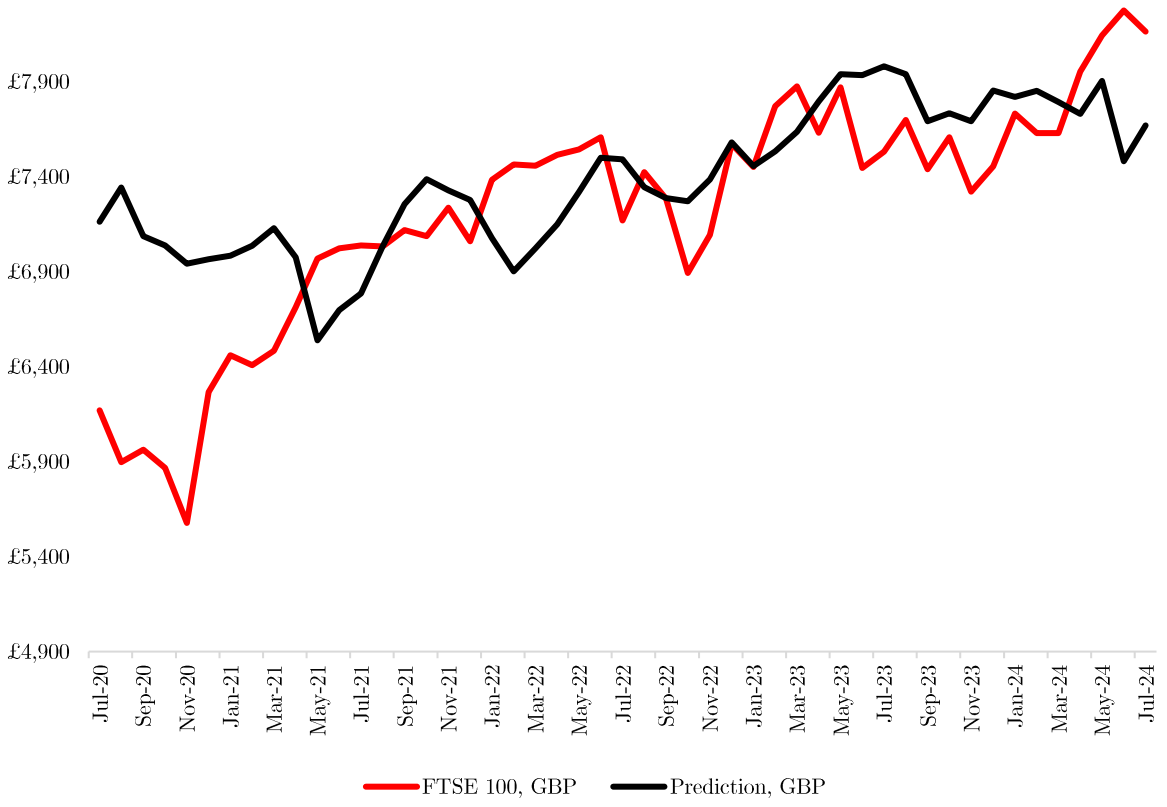
Correlation: L6M



Correlation: L9M

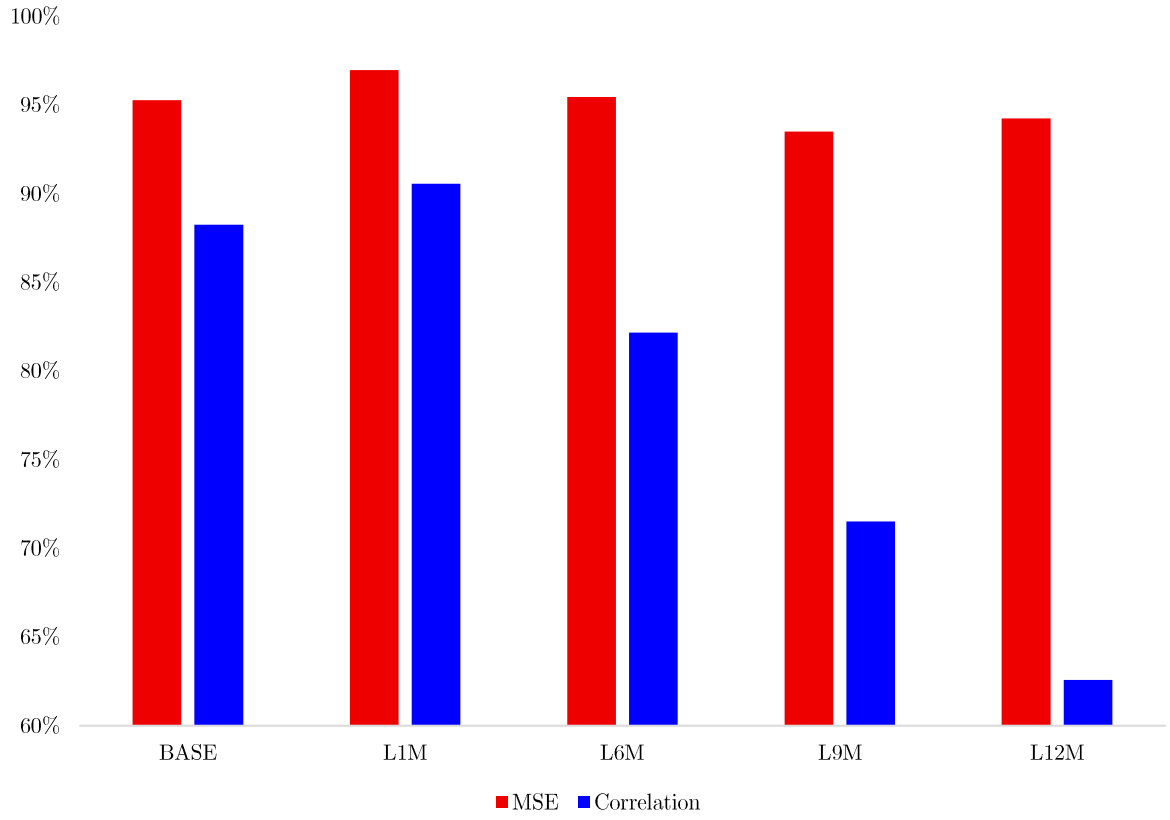


Correlation: L12M



We now come to our lagged models, comparing BASE, L1M, L6M, L9M, and L12M. Across the board, MSE results were strong, ranging from 93.51% in L9M to 96.96% in L1M. Correlation results were less centralised, ranging from 62.58% in L12M to 90.58% in L1M. Ultimately, L1M had the highest accuracy across MSE and correlation. There is a short delay period for macroeconomic absorption, where players in the market take time to digest news, analyst activity, and monetary policy signalling. L1M provides better predictive power than BASE to support this, so we can conclude that a lag of around 4 weeks provides incremental information gain into market prices. More practically, L1M aligns with reporting cycles, aligning a sensible market reaction period with realistic information availability timeframes.

Chart Comparison: BASE, L1M, L6M, L9M, L12M



	MSE	*Rank*	*Correlation*	*Rank*
BASE	95.28%	3	88.25%	2
L1M	96.97%	1	90.58%	1
L6M	95.46%	2	82.17%	3
L9M	93.51%	5	71.52%	4
L12M	94.25%	4	62.58%	5

We observe a decline in accuracy across MSE and correlation as lag increases. Signalling becomes outdated, and more recent and impactful news may be released, leading to a more "random" price movement. Furthermore, 6-12 months is sufficient for a significant change in economic regime or cycle, leading to signals breaking off from their impact in the market, or becoming outdated and already priced into the market. This is consistent with economic intuition and the decay of value in past information with time.

Strong L1M results challenge strong form efficiency, suggesting prices do not immediately adjust to new macroeconomic data. Furthermore, declining accuracy after 6 months indicates a version of semi-strong efficiency, where information is incorporated eventually, but not

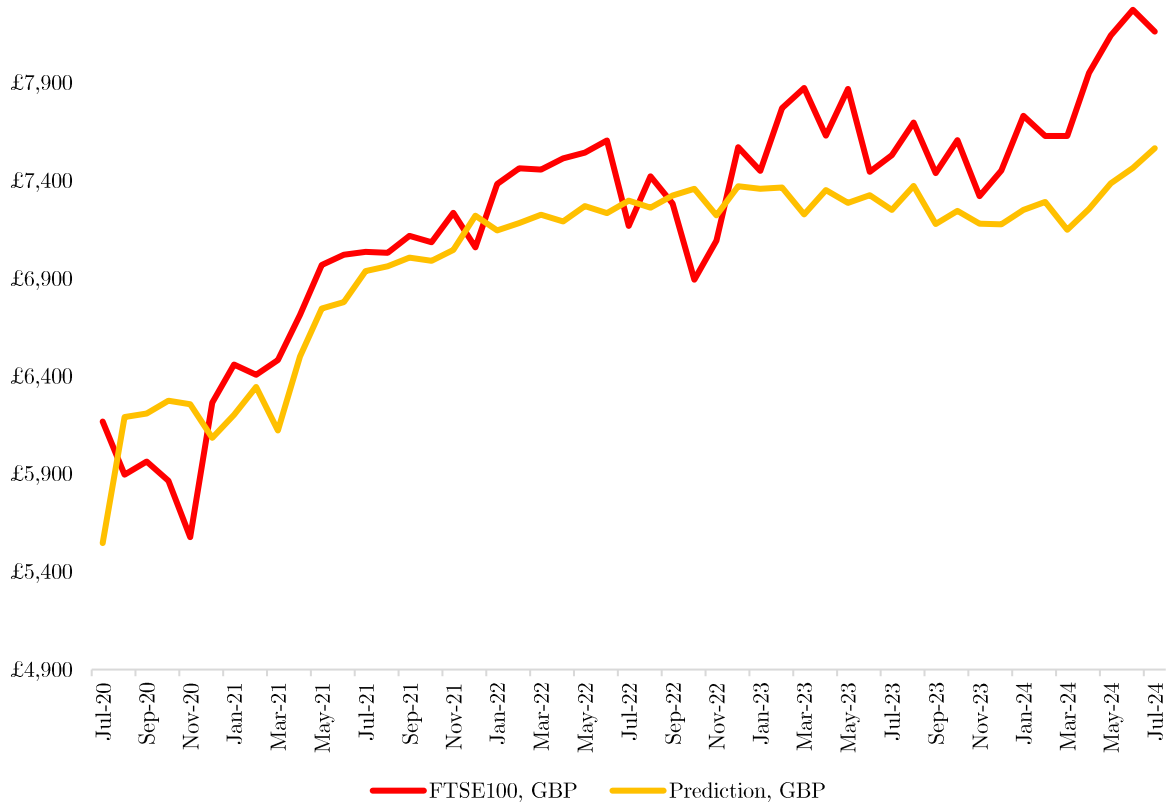
immediately. In the short term, where L1M exceeds the accuracy of BASE, the evidence undermines the weak form of the Efficient Market Hypothesis. The results imply that the markets are inefficient in processing recent macroeconomic data and that the long-term loss of correlation restores some EMH credibility. This directly supports Lo and MacKinlay's (1988) findings and Jegadeesh and Titman (1993), where momentum and mean reversion effects may exist in the short term. Furthermore, these papers find that predictability fades as the market corrects itself, hence the drop in accuracy (correlation) with L6M, L9M, and L12M.

It is also worth noting that the same architecture and identical modelling have been used with L1M, L6M, L9M, and L12M compared to BASE. This approach promotes comparability and follows the scientific method as a control variable to improve the reliability of inferences.

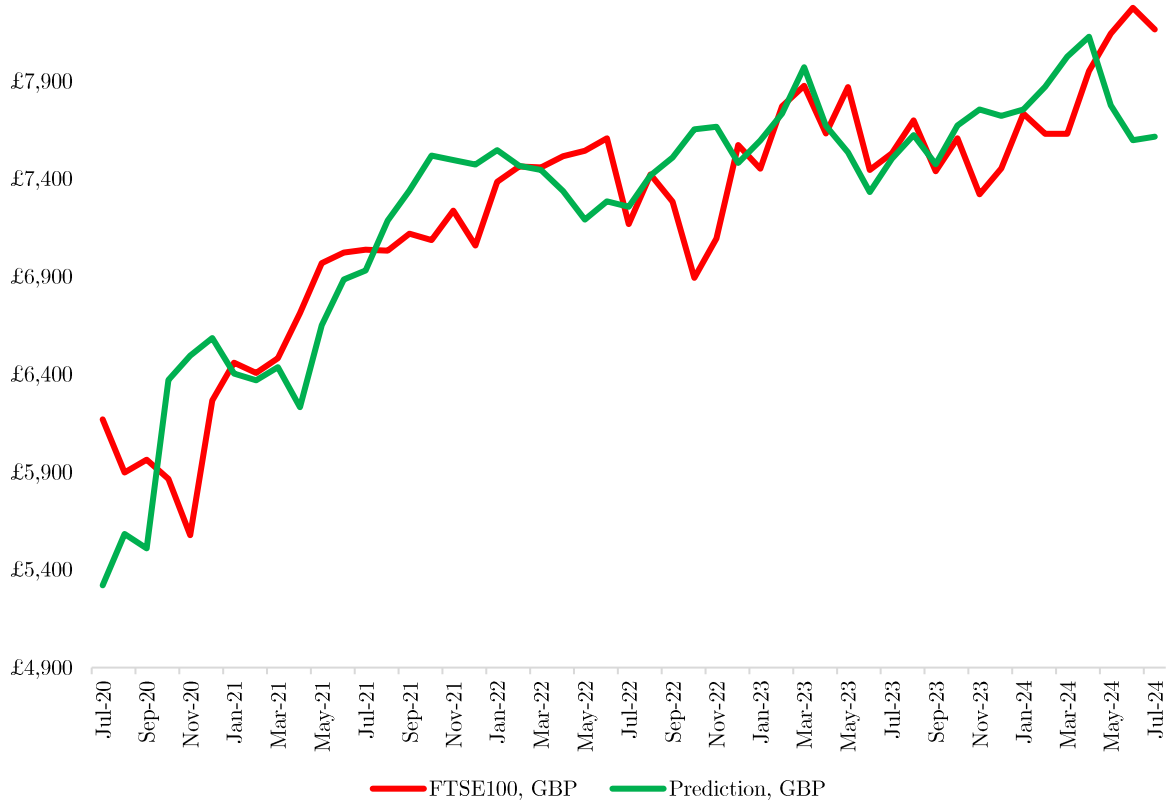
In conclusion, the lagged model shows that past MEV data can increase the pricing accuracy of the neural network, supporting both hypotheses 1 and 2. However, limiting the lag length helps avoid adding irrelevant market noise. L1M exhibits the golden 1-month lag, capturing short-term investor underreaction and delayed MEV signalling absorption. L1M outperformed BASE, so there is evidence that the markets are inefficient, as they can be predicted using past prices.

6.5 Accuracy (MSE and Correlation): L2M, L3M, L4M, L5M

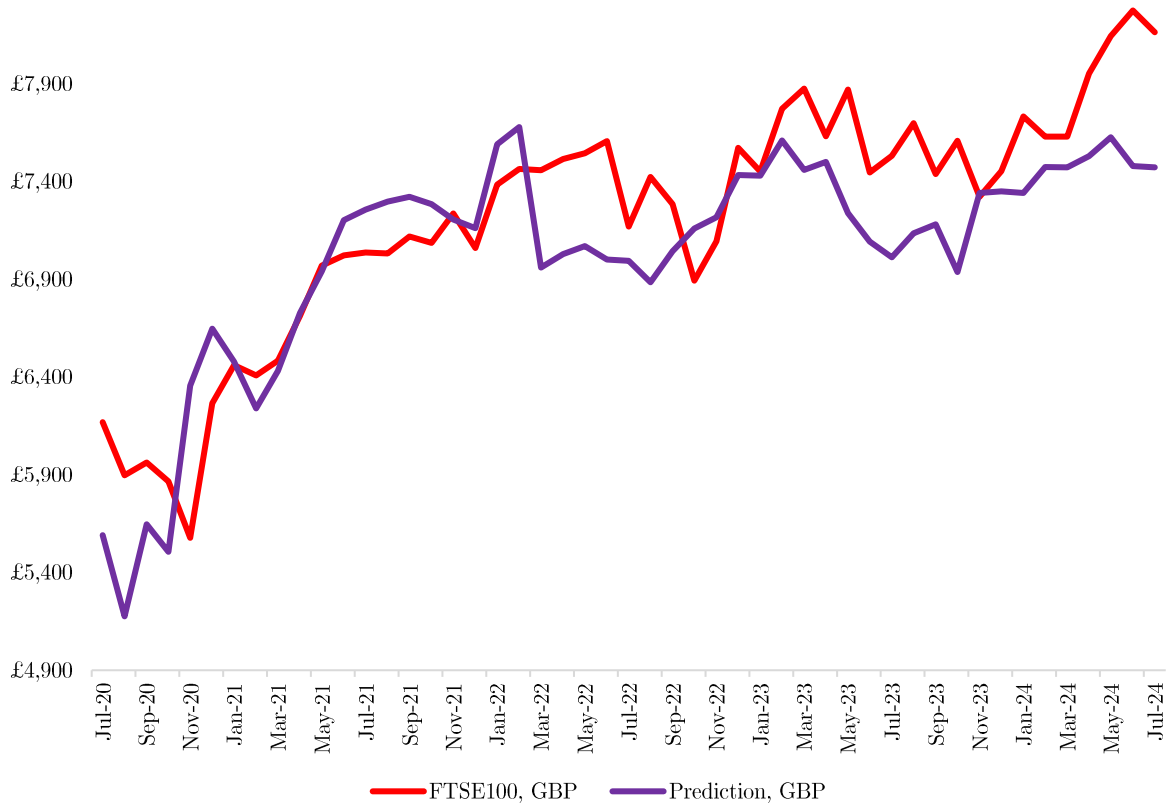
Mean Square Error: L2M



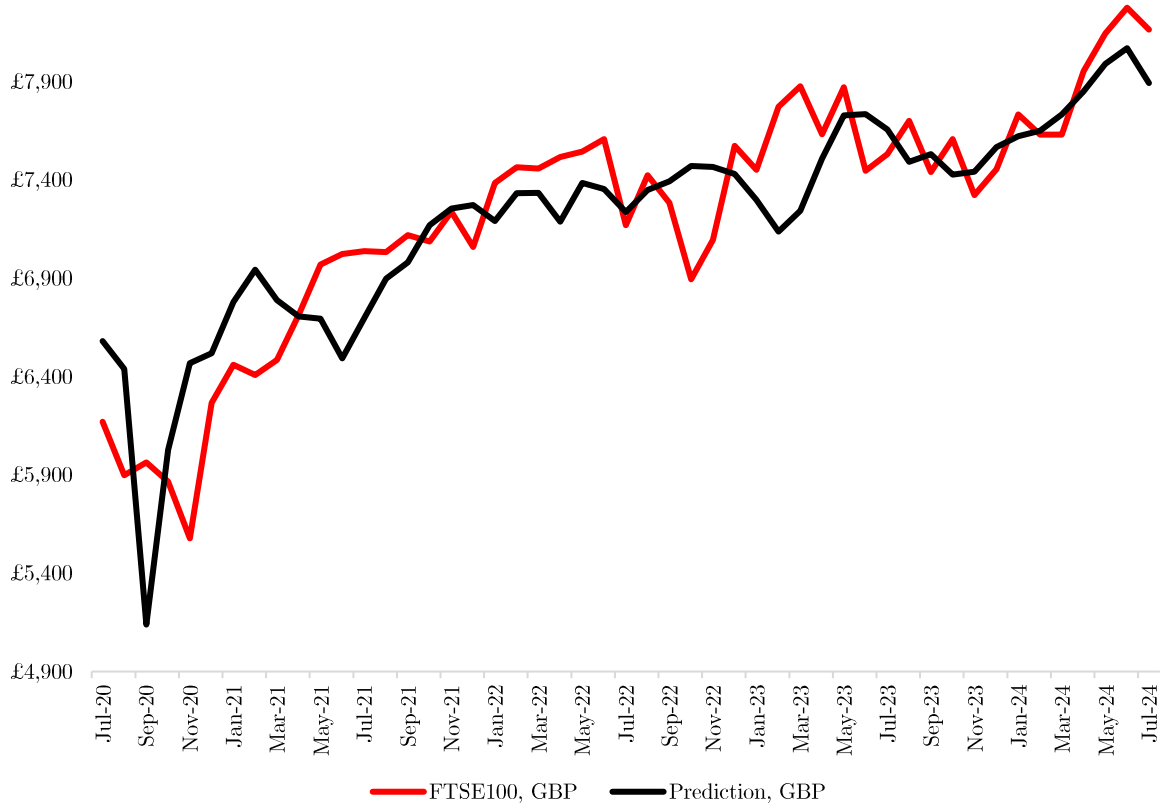
Mean Square Error: L3M



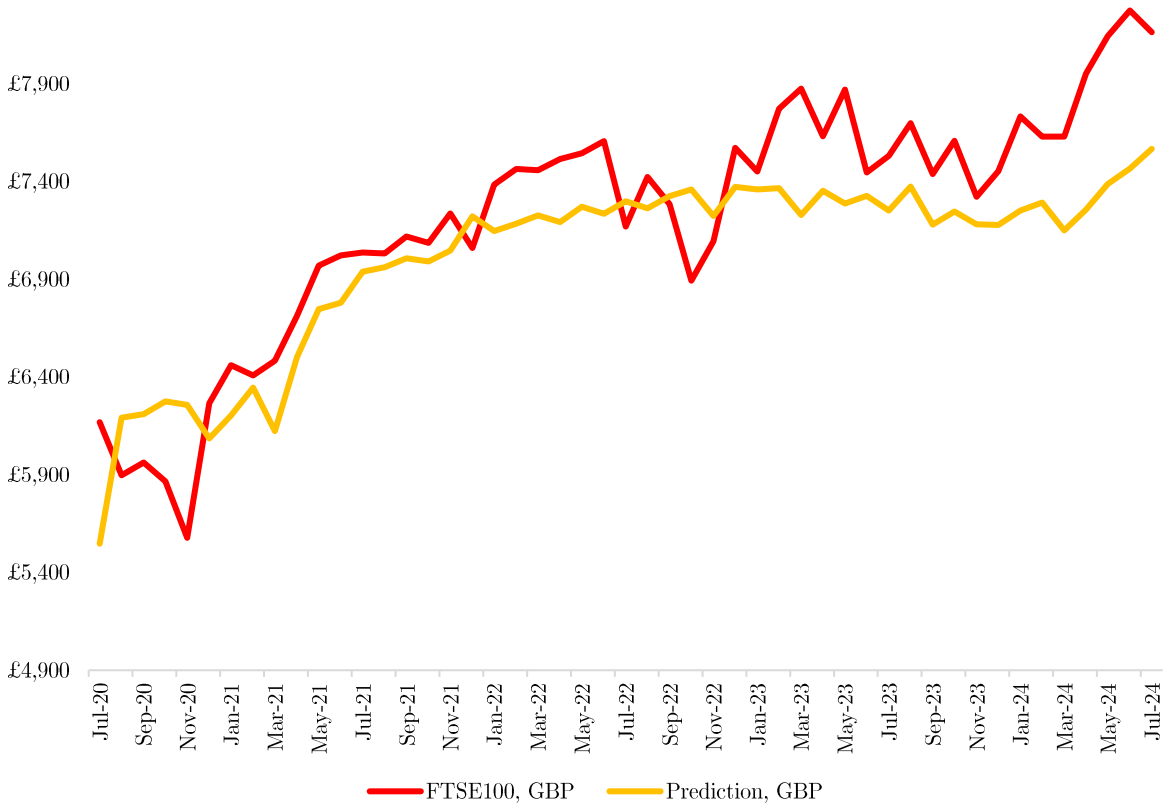
Mean Square Error: L4M



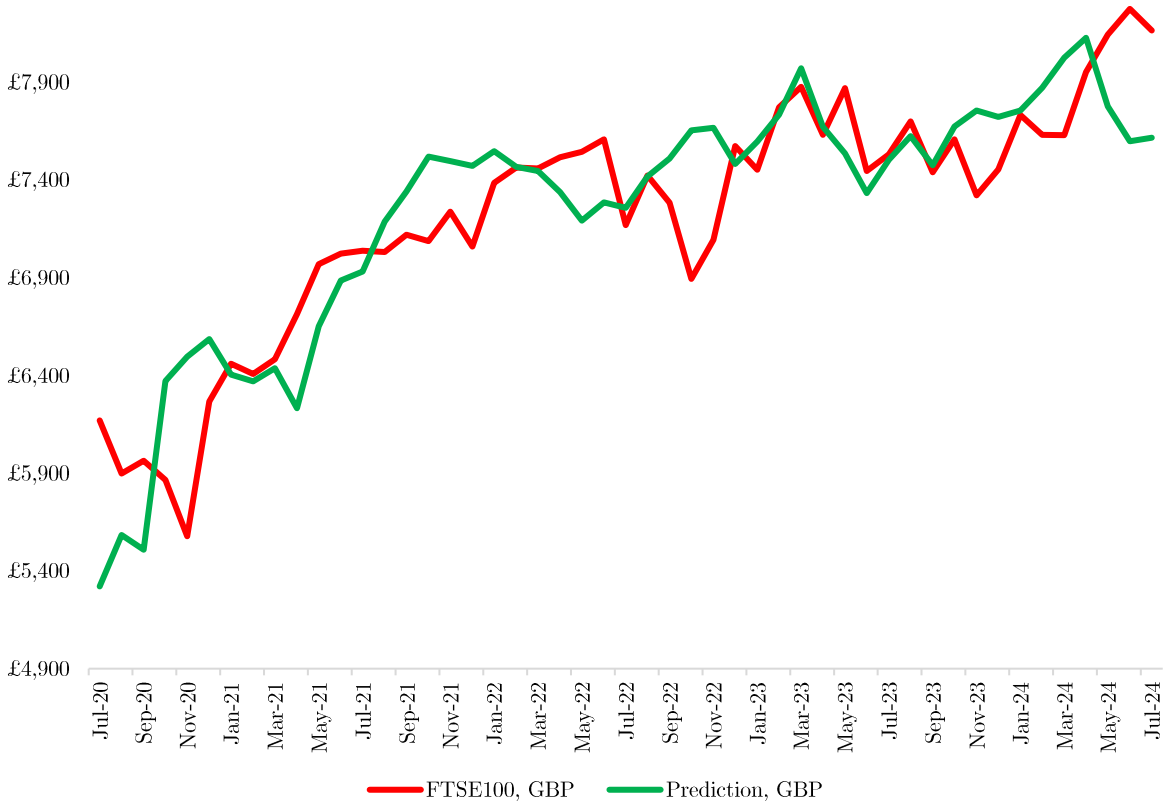
Mean Square Error: L5M



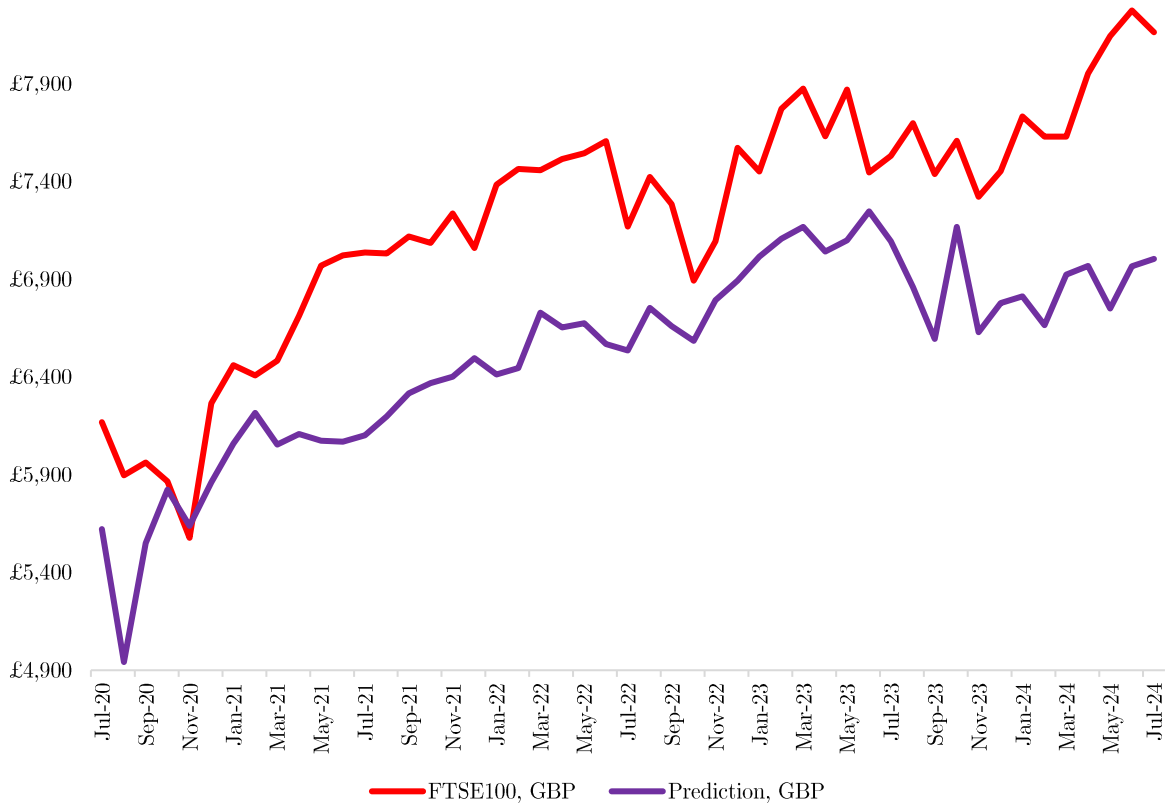
Correlation: L2M



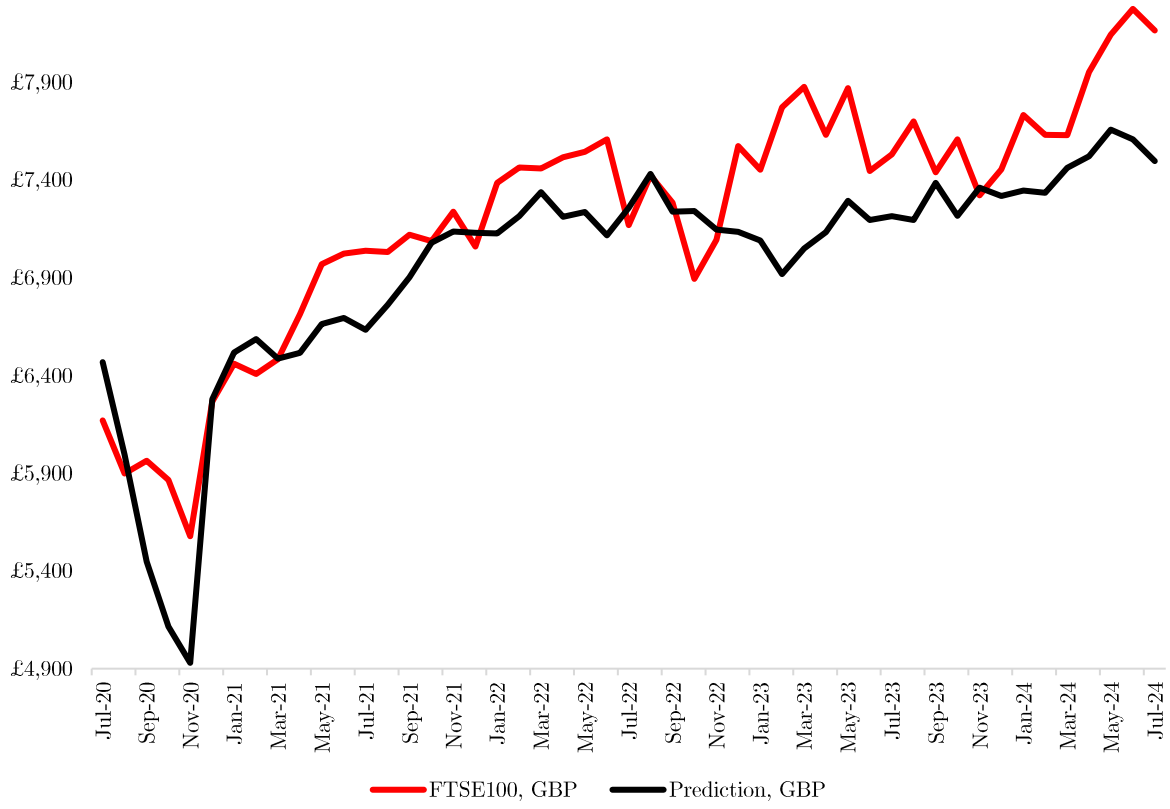
Correlation: L3M



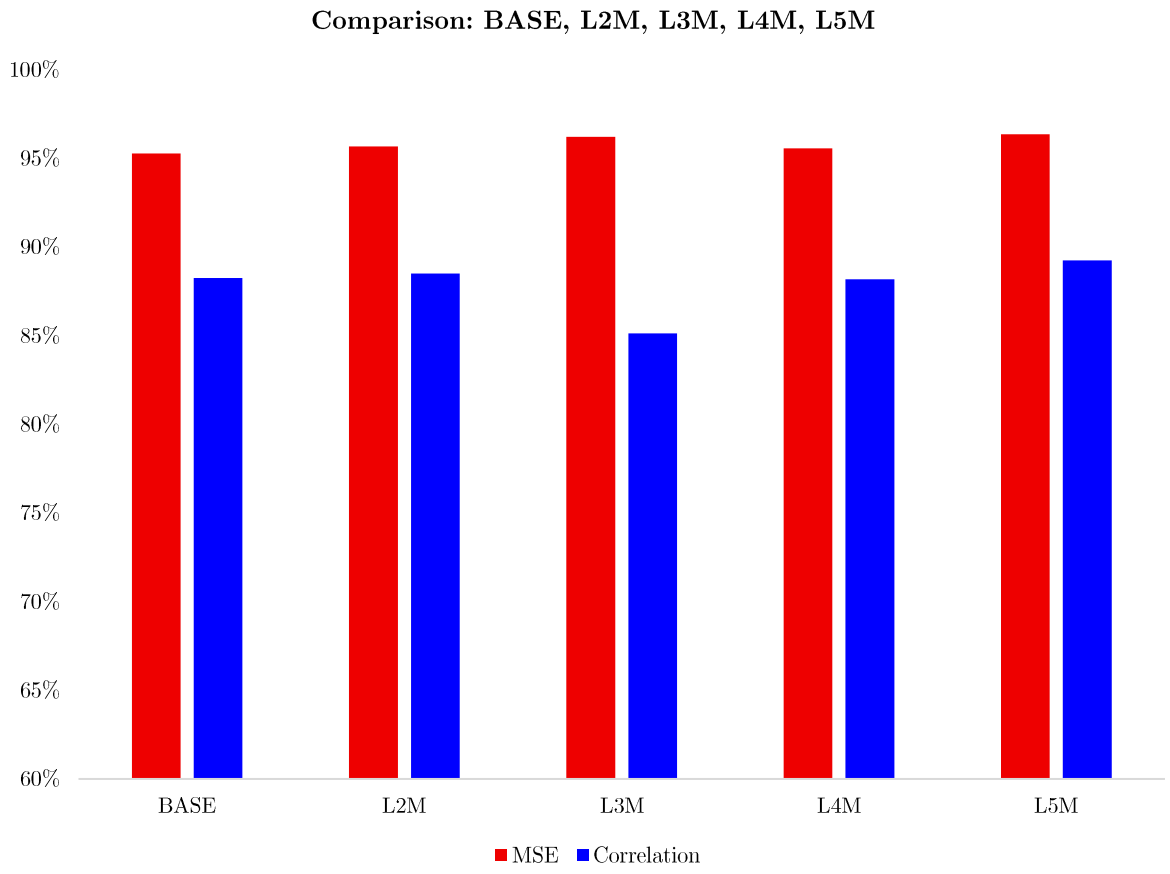
Correlation: L4M



Correlation: L5M



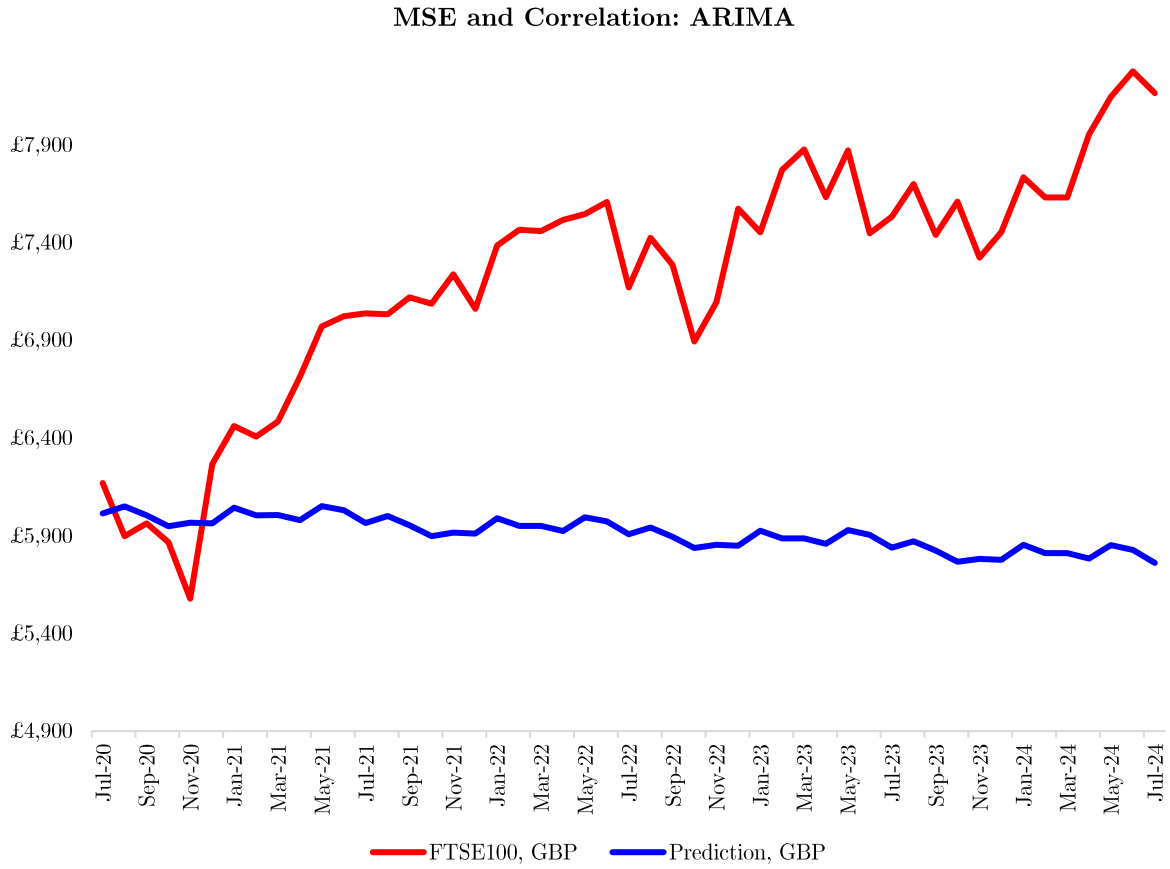
Although we have established that L1M is more accurate than L6M, there may be an intermediate lag of 2, 3, 4, or 5 months, yielding even better results than L1M. This warrants further exploration. Below is a summary of L2M, L3M, L4M, and L5M models to compare their performance.



	MSE	*Rank*	*Correlation*	*Rank*
BASE	95.28%	5	88.25%	3
L2M	95.68%	3	88.51%	2
L3M	96.21%	2	85.13%	5
L4M	95.57%	4	88.17%	4
L5M	96.35%	1	89.24%	1

L1M remains the strongest performer, confirming it as the optimal lag length. Performance drops off either side of L1M, with L2M and L5M coming closest, but still trailing on both metrics. Furthermore, there is evident decay after L5M, with L6M and beyond showing sharp drops in correlation and inconsistency in MSE.

6.6 Accuracy (MSE and Correlation): ARIMA



	Error	*Absolute Error*	*% Error*
Average	-1300.41	1327.45	17.77%
Minimum	-2448.00	40.05	0.67%
Maximum	389.12	2448.00	29.58%
Range	2837.12	2407.95	28.91%

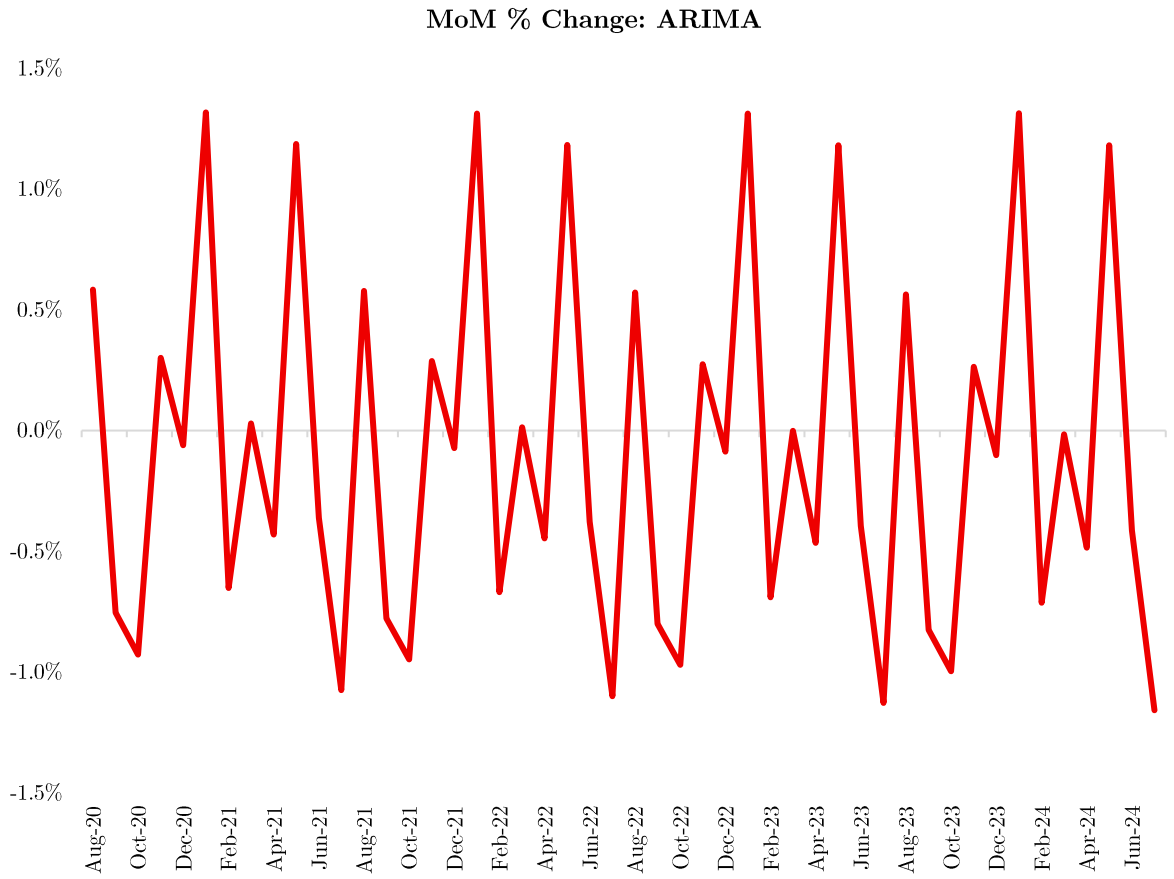
Accuracy	82.23%

Correlation	-63.90%

While the Durbin Watson test outputted ≈ 2, indicating no significant first order autocorrelation, our Ljung-Box outputted a Q statistic of ≈ 21.09 at p $\approx 99.40\%$ and df $= 40$. Given that this is beyond conventional thresholds (less than 1%), we fail to reject the null hypothesis of no autocorrelation. Therefore, residuals from ARIMA(1,1,1) are not autocorrelated and are therefore statistically indistinguishable from white noise.

Combining this with the knowledge that L1M outperformed both BASE and ARIMA implies that exploitable structures exist and that the traditional nonlinear models fail to capture them successfully. Refer to 5.4 for further information, and for the accompanying model script and commentary, please refer to appendix 10.48 and 10.49.

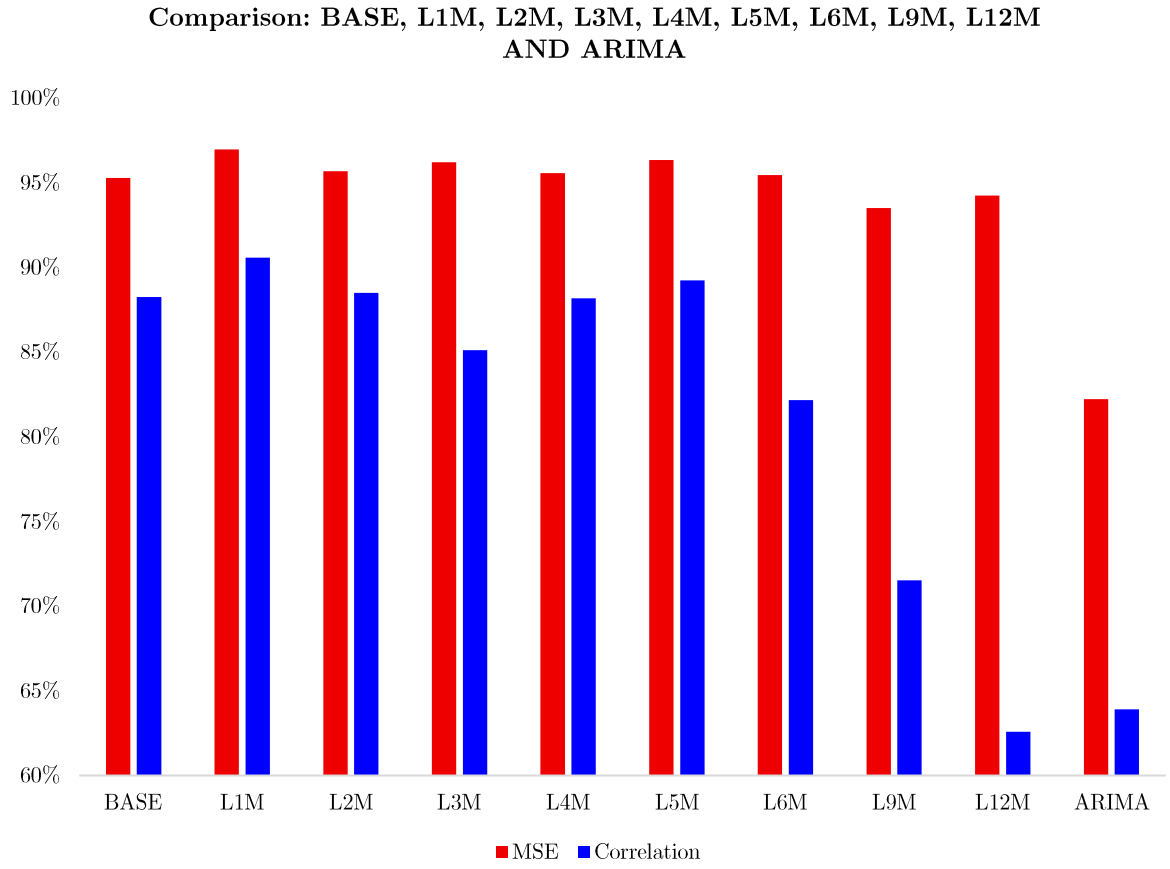
This study applies ARIMA as a control variable to BASE. ARIMA inherently assumes linearity and stationarity, which are poor fits for equity indices, so if BASE consistently outperformed ARIMA, it challenges the assumption of market efficiency and highlights the potential of AI in harnessing complex non-linear relationships. While ARIMA was not coded to capture nonlinear relationships, BASE was designed to handle high-dimensional data, identifying complex, nonlinear relationships, perfect for real-world MEV data. ARIMA has the lowest correlation and MSE accuracy scores of all models, falling short of BASE by 24.35% (correlation) and 13.05% (MSE). This shows that past prices do not contain enough information to predict future prices. Furthermore, the low correlation value suggests that the model has missed key trends, and the MSE value suggests substantial mispricing regularly. This suggests that the UK Equity market is influenced by more than just its historical values, and external variables such as MEVs hold explanatory power.



These findings reflect the conclusions of Lo and MacKinlay (1998), who rejected the random walk using similar autocorrelation tests. Furthermore, this evidence aligns with Poterba and Summer (1988) and their argument that price movements exhibit mean reversion over time. This aligns with the above figure, showing cyclical predictions. Although ARIMA is designed well for this, it does not capture the nuanced and complex time series of FTSE100 Index prices. Hypothesis two, therefore, is reinforced, as ARIMA cannot match the MSE accuracy of BASE. Hypothesis two is also supported, as ARIMA's weak correlation confirms the superiority of MEV-fed neural networks.

6.7 Overall Model Comparison

Finally, we can compare all the models at once. L1M is our winner.



	MSE	*Rank*	*Correlation*	*Rank*
BASE	95.28%	7	88.25%	4
L1M	96.97%	1	90.58%	1
L2M	95.68%	4	88.51%	3
L3M	96.21%	3	85.13%	6
L4M	95.57%	5	88.17%	5
L5M	96.35%	2	89.24%	2
L6M	95.46%	6	82.17%	7
L9M	93.51%	9	71.52%	8
L12M	94.25%	8	62.58%	10
ARIMA	82.23%	10	-63.90%	9

6.8 Limits to My Approach

FTSE100 Index predictions outputted by the model were predicted for July 2020 to December 2024, compared to the actual historical figures. As such, when modelling the "best of 40" approach, the 40 iterations were compared against the actual FTSE100 Index price for July 2020 to December 2024, with no question raised as to whether the FTSE100 Index was priced correctly during this time. Therefore, no valid conclusion can be made about price derivation using this model. Although this seems to undermine the entire investigation, this paper aimed to provide an accurate prediction of the FTSE100 Index price, not an accurate prediction of what the FTSE100 Index price should have been. One could argue that accurately predicting the price, despite inherent FTSE100 Index market mispricing, has captured a nuance in the market that investors have not yet observed or understood.

Secondly, monthly data was sometimes unavailable, so quarterly splined data had to be used. Furthermore, lots of data came with a cost. Where funding and time are less restricting, I would have completed an analysis on perhaps 10-15 key MEVs and then spent time and money developing daily datasets. Making the dataset roughly 30 times larger would have captured more intricate and complex relationships, making modelling more accurate. Next, we come to parameters. Although all fine-tuned, the script was 87 lines long, so parameter tuning was elementary.

Furthermore, I could not automatically optimise tuning, so I had to manually apply trial and error to see which combinations had the best outcomes. In addition, correlation analysis was completed after the model had been used. In retrospect, I should have correlated the MEVs with VAR_0 and based on research in the field, only included MEVs with a correlation to VAR_0 above a certain threshold, for example, 75%. Finally, it is worth considering "fit". Over or under-fitting refers to how well a model responds to its training data. A model may "overfit" the training dataset, too focused on repeated patterns from the original training dataset, unable to adapt and apply the relationships between MEVs and FTSE100 Index prices to new scenarios.

7 DISCUSSION

7.1 Approach Successes

This study has been a successful attempt at pricing the FTSE100 Index. L1M has real-world and practical applications, with repeatable methodology, in less than 100 lines of code at no cost. Academically, this study contributes a simple Feed Forward Neural Network that challenges the Efficient Market Hypothesis, achieving an accuracy of 96.97% based on MSE, and 90.58% with respect to correlation. Addressing a gap in the literature, unlike Malkiel (2003) and Fama (1970), this paper offers a robust and tested tool for market prediction. Upon empirical performance evaluation, behavioural inefficiencies are revealed in the FTSE100 Index price, suggesting rejection of the semi-strong form of efficiency. Ultimately, both hypotheses were affirmed; price and price change direction can be predicted using L1M.

7.2 Approach Weaknesses

A key weakness is the assumption that the market absorbs all information with the same intensity and speed. A one-month lag appears optimal, but this assumes homogeneity in MEV absorption. In reality, it may take varying amounts of time for each MEV's "value" to be "absorbed" by the players in the market, leading to a lag in price adjustment, depending on the outlet, market conditions, ease of understanding, and education levels in the population. Despite this, 1 month does seem a sensible estimate. Therefore, a simple uniform lagged model may not capture the correct lag periods for every MEV. Secondly, model error is discussed in 6.8, where the user may compare each epoch to an inherently mispriced market. In addition, data was limited to 480 monthly intervals between January 1984 and December 2024, restricting generalisability and robustness. This reduces the reliability and generalisability of results and restricts the scope for predictions. As discussed in 6.8, there is a risk of overfitting, where the model may have learned noise or overfitted to its training data. A key ambition of this investigation, filling a gap in the literature where no actionable trading strategies were suggested, is using BASE as a trading model. Firstly, the study assumes that the user can access real-time MEVs. Implementing this into a real trading system would require extensive automation, testing, and connection of data sources into Python, including using APIs. Even once integrated, most consumer trading platforms will not support algorithmic trading and may restrict trades based on margins, capital requirements, limits to leverage, and shorting constraints. Finally, we must consider that this research aims not to price the FTSE100 Index fairly, but accurately. In conjunction, another core assumption that underpins the model is the accuracy of prediction in the past, and the assumption that accuracy will continue. Visually inspecting results from the model is enough evidence to deduce that for the hold-out (testing) period of July 2020 to Jul 2024, some periods are more accurately predicted than others.

7.3 Comparing BASE, LxM's, and ARIMA

BASE secured MSE accuracy of 95.28% and a correlation accuracy of 88.25%, L1M secured MSE accuracy of 96.97% and a correlation accuracy of 90.58%, and ARIMA seriously underperformed; L1M outperforms BASE, which outperforms ARIMA. Weak ARIMA performance reflects traditional linear models' limitations, as Tsay (2010) and Schwert (2003) covered. Comparing L1M to the EMH challenges market efficiency. If the markets were efficient, past data would not have predictive power. Of course, we have proven that it does. The two hypotheses in this paper consider accuracy and direction of price prediction. MSE evaluates how close two values are, and as a model optimisation tool, is focused on achieving predictions that are as close to reality as possible. Correlation measures movement direction,

and as a model optimisation tool, it is focused on directional movement and accuracy in predicting directional price changes.

7.4 Hypotheses

H1 is supported because L1M could predict the FTSE100 Index Price with an MSE accuracy of 96.97%. H2 is supported because L1M could predict the FTSE100 Index Price with a correlation accuracy of 90.58%.

Suppose the markets had been efficient; by the time L1M had outputted a price, the price would have already adjusted if it had been correct. Given that L1M could accurately predict the market, the hypothesis has been affirmed, and we can reject a semi-strong form of the market. Linking back to Barberis et al. (1998) and Adaptive Market Theory proposed by Neely et al. (2007), we can argue that the markets adjust over time, but lag-based models can still exploit temporary inefficiency.

7.5 Improvements

Introducing data at daily intervals instead of monthly would give the model more data to infer statistical relationships, becoming more accurate with more data to train on. In addition, section 6.7 highlighted the improved accuracy when lagging MEVs. To guide future research, this could be optimised, perhaps in combination with daily data, to identify the lag exactly to the day for each MEV, which supports the most accurate prediction. Given more time and money, I could have introduced more advanced architecture, such as LSTM or CNN modelling, which would have captured temporal relationships and spatial dependencies between MEVs; ultimately, making for a more accurate and reliable model. Furthermore, a more powerful computer would have accelerated model development, as the machine used in this study took roughly 10 minutes per set of 40 iterations, and a specialised and more powerful computer may have produced results faster. Additional time spent on fine-tuning and more structured analysis of the combination of hyperparameter tuning would improve accuracy and reliability. Real-time API integration would have made the entire model public, open source, and free of charge for anyone to use. From an academic perspective, repeatability is a key consideration that this approach would satisfy. Finally, dynamic lagging of MEVs based on historical absorption rates would optimise modelling and take complex nonlinear relationship discovery by the neural network, one step further.

7.6 Guiding Future Research

Exploration of other markets and indices would test the generalisability beyond the United Kingdom and the FTSE100 Index. Furthermore, investigating sector-specific predictions and selecting MEVs with the sector in mind could lead to more accurate models tailored to specific industries. Although quite a large extension on my existing work, textual sentiment analysis could be applied using Python. One could scrape news articles off the internet, use AI processing to score them across a range of metrics, and then convert qualitative data into values to incorporate into the model, and with the proper tooling, this could all be automated. Regarding testing, the ultimate test of generalisability would be training the model in one region or market and testing it on another. This would support research and development to ensure the model is robust and applicable. Future research could also lead to the development of a real-time trading algorithm. Although not the core focus of this paper, with API and data scraping from the internet, as well as linkage to a broker which permits algorithmic trading, this model could be used to predict prices and trade accordingly, especially regarding

hypothesis 1. Finally, the model only includes 480 intervals, so testing it during extreme economic conditions could become an early form of a stress test.

8 CONCLUSION

8.1 Summary

This paper explores the explanatory potential of MEVs in predicting the FTSE100 Index price, using them to train a neural network. The model was trained on 26 macroeconomic variables and tested over 40 iterations, with selection based on best MSE and correlation accuracy. BASE was the original model. LxM variations offer lags between one and twelve months as a control variable. An ARIMA model was used as a second control variable. Upon comparison of all models, L1M outperforms, securing MSE accuracy of 96.97% and a correlation accuracy of 90.58%. This result suggests the value of lagging MEV data for accurate prediction.

8.2 Motivation

Since the 1970s, academics have been challenged to reconcile the EMH with real-world data. While one should not be able to predict the market using past or present information if the markets are semi-strong efficient, persistent anomalies say otherwise. Combining this literature with research on AI and Machine Learning, I was presented with an opportunity to challenge traditional finance theory with a practical, dynamic, and flexible model to accurately and consistently predict the stock market. This mode applied MEVs and tested whether data could systematically predict stock prices at a given interval. This study incorporates MEVs, underused in Machine Learning, as the driving force of a simple neural network to attempt to predict the market and therefore challenge efficiency. And not only does this paper discuss efficiency, but it also goes one step further in testing a replicable model with real-world data. Behavioural theory and anomalies cause lags, and the study considers how a neural network could explore the nonlinear relationships between the information released and the market price inflicted as a result.

8.3 Data and Setting

26 MEVs cover data across macroeconomic variables such as GDP, Unemployment, and Inflation between January 1984 and December 2024, in 480 monthly intervals. The data was sourced from reputable databases, providing reliability in results and predictions. This dissertation approaches the neural network using two training and testing windows, as seen in section 5.9. Admittedly, this paper utilises a small dataset of only 12,480 data points. Despite this, the study yields significant results.

8.4 Findings

BASE secured MSE accuracy of 95.28% and a correlation accuracy of 88.25%. L1M secured MSE accuracy of 96.97% and a correlation accuracy of 90.58%. ARIMA did not perform nearly as well, given its linear approach to time series modelling and the nonlinearity and noise in MEV data. As a result, both Hypothesis 1 and Hypothesis 2 were strongly supported. Therefore, we can conclude that MEVs as input variables for neural networks can be used to predict FTSE100 Index prices.

8.5 Implications

Based on the evidence in this study, the semi-strong form of the EMH should be rejected. To at least some extent, past and present data in the form of MEVs hold predictive power, most

explanatory when lagged by 1 month. A basic neural network predicting the markets with such accuracy makes its use as a trading model viable. With low computational and financial costs, the model is replicable, dynamic, and scalable, as shown in previous discussions about API and brokerage integrations. Academically, I have bridged a gap in the literature by showing that a simple, transparent, and customisable neural network can outperform traditional models. This contributes to the broader debate and excitement around AI and its market predictive use.

8.6 Stakeholders

Of course, the most financially motivated amongst readers of this paper will be interested in using the model to trade. Although AI modelling has a long way to go, it has the potential to drive trading models to earn consistent profits through explainable decision-making and suggesting a position to take in the market. For academics, I have provided a basis for future studies combining financial and behavioural theory, with MEV data and Machine Learning. Finally, this study offers insight into the response lag for policymakers, identifying systematic risks and delays in policy transmission within the population.

8.7 Final Thoughts

This study confirms that FTSE100 Index prices are partially predictable using MEVs, demonstrating that simple, interpretable neural networks can capture meaningful patterns missed by traditional models. A simple, transparent neural network in less than 100 lines of code outperformed the legacy model by almost 20%. If anything can be taken from this paper, it should be the excitement and justification for recent AI-driven financial research momentum. Complex insight can be unlocked from clean, structured data powered by intelligent modelling, without deep computation infrastructure. As such, this dissertation contributes to Finance academia and Machine Learning technology, supporting broader excitement and anticipation within the Artificial Intelligence space.

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10 SUPPLEMENTARY APPENDIX

10.1 Summary Statistics

Variable	Obs	Mean	Standard Deviation	Min	Max
FTSE100 Index Price	487	4858.19	1992.168	1000	8275.38
Gross Domestic Product	487	471644.1	105994.7	282605	639452
Population	487	60400000	3319229	56300000	66800000
Unemployment	487	6.816016	2.295825	3.6	11.9
UK Central Bank Base Rate	487	4.977734	4.120133	0.25	14.875
Consumer Price Index Inflation	487	80.46221	22.97717	39.42698	133
Retail Price Index Inflation	487	3.798768	2.52479	-1.6	14.2
Industrial Production	487	0.0887064	1.519874	-16.2	11.4
M4 Money Supply	487	1379.58	903.3762	174.3	3113.3
Consumer Confidence Index	487	-11.02464	11.98436	-49	10
Gold Price	487	17.75955	14.19213	5.135029	60.00225
10Y Government Bonds	487	5.352752	3.321641	0.105	12.795
Exports	487	17500000000	7610000000	5370000000	33500000000
Imports	487	23600000000	12500000000	5870000000	58900000000
Balance of Payments Current	487	-8908.187	8751.938	-39615	11056
Balance of Payments Capital	487	-265.7855	760.7962	-8435	659.9945
Balance of Payments Financial	487	9657.614	11748.32	-16088	53841
Brent Crude Oil Price	487	46.69513	31.15751	9.616667	132.8252
Net Debt	487	788.7838	717.8176	132.5	2610.3
Current Account Balance	487	-8893.953	8458.66	-39615	11056
Gross Value Added	487	307175.6	144629.5	84008	640429
USD/GBP	487	1.558909	0.2048615	1.079	2.0815
CNY/GBP	487	10.29056	3.053071	2.9131	15.9108
EUR/GBP	487	1.34728	0.1787325	1.047307	1.791152
Consumer Spending	487	283454.3	71583.63	151361	389410
Corporate Tax Rate	487	0.2875154	0.0645417	0.19	0.5
Average House Price	487	123463.4	71751.39	23572	265727

10.2 Correlation of the FTSE100 Index to MEVs

		Correlation
VAR_24	Consumer Spending	0.9311
VAR_1	Gross Domestic Product	0.9301
VAR_12	Exports	0.9162
VAR_5	Consumer Price Index Inflation	0.9106
VAR_20	Gross Value Added	0.8979
VAR_11	10Y Government Bonds	-0.8923

VAR_13	Imports	0.8853
VAR_25	Corporate Tax Rate	-0.8648
VAR_26	Average House Price	0.8630
VAR_8	M4 Money Supply	0.8527
VAR_2	Population	0.8497
VAR_3	Unemployment	-0.8291
VAR_18	Net Debt	0.7882
VAR_4	UK Central Bank Base Rate	-0.7865
VAR_10	Gold Price	0.6828
VAR_17	Brent Crude Oil Price	0.6525
VAR_19	Current Account Balance	-0.6373
VAR_14	Balance of Payments Current	-0.6183
VAR_16	Balance of Payments Financial	0.4990
VAR_23	EUR/GBP	-0.4834
VAR_15	Balance of Payments Capital	-0.3849
VAR_22	CNY/GBP	0.3600
VAR_21	USD/GBP	-0.2696
VAR_9	Consumer Confidence Index	-0.1306
VAR_6	Retail Price Index Inflation	-0.1065
VAR_7	Industrial Production	-0.0277

10.3 Macroeconomic Variables (MEVs)

Number	Time Series	Description	Units
0	FTSE100 Index Price	Value of the FTSE100 Index, representing the performance of the largest 100 companies listed on the London Stock Exchange	£, pence
1	Gross Domestic Product	The total monetary value of all the goods and services produced in the United Kingdom	£, millions
2	Population	Total population of the United Kingdom	Integer
3	Unemployment	Percentage of the labour force who are jobless and actively seeking employment	%
4	UK Central Bank Base Rate	Interest rate set by the Bank of England for lending and borrowing	%
5	Consumer Price Index Inflation	The measure of the changes in the prices consumers pay for goods and services.	Index, 2015=100
6	Retail Price Index Inflation	Inflation measure, which includes mortgage payments as well as the CPI	£
7	Industrial Production	Total output of the mining, manufacturing, and utilities sectors in the United Kingdom	% Change
8	M4 Money Supply	Broad measure of the money supply in the United Kingdom, including cash, bank deposits, and other liquid assets	£
9	Consumer Confidence Index	The measure of optimism regarding the future of finances in the United Kingdom	£
10	Gold Price	Market price of Gold	£/g
11	10Y Government Bonds	Interest yield on United Kingdom government bonds, with 10 years to maturity	%
12	Exports	Total value of all goods and services consumed in the United Kingdom, purchased from parties overseas	Integer
13	Imports	Total value of all goods and services produced in the United Kingdom, sold to parties overseas.	Integer
14	Balance of Payments Current	Net flow of trade in goods and services, income, and current transfers into and out of the United Kingdom	£, millions
15	Balance of Payments Capital	Net transfer of capital into and out of the United Kingdom	£, millions
16	Balance of Payments Financial	Net flow of foreign direct investment into and out of the United Kingdom	£, millions

17	Brent Crude Oil Price	Market price of Crude Oil extracted in the North Sea	$/bbl
18	Net Debt	Total value of all government debt less government assets	£, billions
19	Current Account Balance	Net value of trade balance, net income, and direct financial transfers	£, millions
20	Gross Value Added	The measure of the value of goods and services produced in the United Kingdom, less taxes	£, millions
21	USD/GBP	Market price of GBP in United States Dollar terms	£
22	CNY/GBP	Market price of GBP in Chinese Yuan terms	£
23	EUR/GBP	Market price of GBP in Euro terms	£
24	Consumer Spending	Total expenditure made by households in the United Kingdom	£, millions
25	Corporate Tax Rate	Percentage of earnings before tax, that is deducted as tax, for companies in the United Kingdom	%
26	Average House Price	Average price of residential property in the United Kingdom	£

10.4 Correlation of MEVs to MEVs

	VAR_1	**VAR_2**	**VAR_3**	**VAR_4**	**VAR_5**	**VAR_6**	**VAR_7**	**VAR_8**	**VAR_9**	**VAR_10**	**VAR_11**	**VAR_12**	**VAR_13**
VAR_1	1.0000												
VAR_2	0.9365	1.0000											
VAR_3	-0.8283	-0.6983	1.0000										
VAR_4	-0.8298	-0.7813	0.5542	1.0000									
VAR_5	0.9636	0.9688	-0.7598	-0.8077	1.0000								
VAR_6	-0.0397	0.0478	0.0012	0.3954	0.0074	1.0000							
VAR_7	-0.0432	-0.0510	0.0388	0.0322	-0.0550	-0.0661	1.0000						
VAR_8	0.9496	0.9820	-0.6819	-0.8078	0.9719	0.0535	-0.0664	1.0000					
VAR_9	-0.2723	-0.3840	0.0428	0.1370	-0.3771	-0.4781	0.0855	-0.4491	1.0000				
VAR_10	0.7921	0.9370	-0.4772	-0.6406	0.8780	0.1955	-0.0435	0.9233	-0.5332	1.0000			
VAR_11	-0.9193	-0.8628	0.7362	0.9373	-0.8839	0.2999	0.0261	-0.8727	0.1634	-0.7022	1.0000		
VAR_12	0.9712	0.9554	-0.7431	-0.8330	0.9747	0.0228	-0.0359	0.9636	-0.3648	0.8566	-0.8982	1.0000	

VAR_13	0.9627	0.9745	-0.7280	-0.7858	0.9742	0.1122	-0.0506	0.9776	-0.4080	0.9011	-0.8613	0.9853	1.0000
VAR_14	-0.6987	-0.6689	0.5109	0.5877	-0.6428	0.1296	0.0428	-0.6618	0.1141	-0.5527	0.6540	-0.6585	-0.6717
VAR_15	-0.4257	-0.4771	0.3240	0.2965	-0.4470	-0.1354	-0.0380	-0.4458	0.1856	-0.4721	0.3511	-0.4524	-0.4688
VAR_16	0.5517	0.5057	-0.4228	-0.4863	0.4864	-0.1520	-0.0212	0.5007	-0.0317	0.3862	-0.5313	0.5189	0.5175
VAR_17	0.7491	0.7108	-0.3877	-0.6402	0.7235	0.1570	-0.0666	0.7882	-0.5013	0.6993	-0.6459	0.7810	0.7808
VAR_18	0.8675	0.9781	-0.6197	-0.6820	0.9301	0.1516	-0.0460	0.9420	-0.3870	0.9642	-0.7658	0.9084	0.9450
VAR_19	-0.7206	-0.6895	0.5275	0.6082	-0.6620	0.1335	0.0315	-0.6822	0.1207	-0.5672	0.6756	-0.6790	-0.6908
VAR_20	0.9745	0.9794	-0.7712	-0.7830	0.9918	0.0501	-0.0535	0.9805	-0.3739	0.8898	-0.8712	0.9750	0.9850
VAR_21	-0.2877	-0.4935	0.1266	0.3043	-0.3948	-0.0661	-0.0228	-0.4103	0.2037	-0.5748	0.3053	-0.3943	-0.4274
VAR_22	0.2457	-0.0242	-0.4443	-0.2763	0.1807	-0.3344	-0.0079	0.0403	0.2010	-0.2464	-0.2795	0.1625	0.0684
VAR_23	-0.5830	-0.6862	0.2734	0.6066	-0.6789	0.0023	0.0351	-0.6990	0.5423	-0.7052	0.5542	-0.6655	-0.6624
VAR_24	0.9979	0.9216	-0.8337	-0.8389	0.9552	-0.0677	-0.0405	0.9357	-0.2506	0.7645	-0.9247	0.9649	0.9507
VAR_25	-0.9058	-0.9124	0.7528	0.8195	-0.9084	0.1033	0.0285	-0.8895	0.2593	-0.7748	0.9063	-0.9018	-0.8927
VAR_26	0.9718	0.9677	-0.7732	-0.7648	0.9623	0.0729	-0.0674	0.9765	-0.3829	0.8727	-0.8619	0.9531	0.9715

	VAR_14	**VAR_15**	**VAR_16**	**VAR_17**	**VAR_18**	**VAR_19**	**VAR_20**	**VAR_21**	**VAR_22**	**VAR_23**	**VAR_24**	**VAR_25**	**VAR_26**
VAR_14	1.0000												
VAR_15	0.3105	1.0000											
VAR_16	-0.7538	-0.2731	1.0000										
VAR_17	-0.5663	-0.2569	0.4175	1.0000									
VAR_18	-0.6203	-0.4952	0.4495	0.6651	1.0000								
VAR_19	0.9948	0.3193	-0.7512	-0.5856	-0.6383	1.0000							
VAR_20	-0.6621	-0.4635	0.5016	0.7370	0.9431	-0.6818	1.0000						
VAR_21	0.1565	0.3334	-0.1249	-0.0736	-0.5748	0.1611	-0.4013	1.0000					
VAR_22	-0.0034	0.1164	0.0459	0.0585	-0.1539	-0.0046	0.1267	0.5347	1.0000				
VAR_23	0.4076	0.3106	-0.3002	-0.5329	-0.6616	0.4209	-0.6519	0.2858	-0.0046	1.0000			
VAR_24	-0.7056	-0.4159	0.5605	0.7372	0.8459	-0.7278	0.9637	-0.2702	0.2737	-0.5756	1.0000		
VAR_25	0.6732	0.4190	-0.5374	-0.6002	-0.8522	0.6956	-0.8971	0.3355	-0.1565	0.6838	-0.9057	1.0000	
VAR_26	-0.6765	-0.4576	0.5200	0.7556	0.9186	-0.6975	0.9853	-0.3443	0.1047	-0.6142	0.9600	-0.8790	1.0000

10.5 Regression of FTSE100 Index to MEVs

VAR_0	Coefficient	Standard Error	t-value	P>|t|	95% Confidence Interval		R^2
VAR_1	0.0175	0.0003	55.7500	0.0000	0.0169	0.0181	0.8650
_cons	-3386.3210	151.5740	-22.3400	0.0000	-3684.1440	-3088.4980	
VAR_2	0.0005	0.0000	35.4900	0.0000	0.0005	0.0005	0.7220
_cons	-25962.2700	869.7645	-29.8500	0.0000	-27671.2400	-24253.2900	
VAR_3	-719.4602	22.0282	-32.6600	0.0000	-762.7426	-676.1778	0.6874
_cons	9762.0430	158.4161	61.6200	0.0000	9450.7760	10073.3100	
VAR_4	-380.3056	13.5585	-28.0500	0.0000	-406.9464	-353.6649	0.6186
_cons	6751.2510	87.5744	77.0900	0.0000	6579.1790	6923.3230	
VAR_5	78.9471	1.6275	48.5100	0.0000	75.7493	82.1448	0.8291
cons	-1494.0640	136.1744	-10.9700	0.0000	-1761.6290	-1226.4990	
VAR_6	-84.0004	35.6250	-2.3600	0.0190	-153.9987	-14.0020	0.0113
_cons	5177.2880	162.4441	31.8700	0.0000	4858.1070	5496.4690	
VAR_7	-36.2966	59.4951	-0.6100	0.5420	-153.1965	80.6033	0.0008
_cons	4861.4100	90.4862	53.7300	0.0000	4683.6170	5039.2030	
VAR_8	1.8805	0.0523	35.9500	0.0000	1.7777	1.9833	0.7272
_cons	2263.8680	86.2234	26.2600	0.0000	2094.4500	2433.2850	
VAR_9	-21.7068	7.4835	-2.9000	0.0040	-36.4110	-7.0027	0.0171
cons	4618.8800	121.7935	37.9200	0.0000	4379.5720	4858.1880	
VAR_10	95.8451	4.6568	20.5800	0.0000	86.6950	104.9952	0.4662
_cons	3156.0240	105.8247	29.8200	0.0000	2948.0930	3363.9560	
VAR_11	-535.1588	12.2945	-43.5300	0.0000	-559.3158	-511.0017	0.7962
_cons	7722.7620	77.4286	99.7400	0.0000	7570.6250	7874.8990	
VAR_12	0.0000	0.0000	50.3600	0.0000	0.0000	0.0000	0.8395
_cons	670.8062	90.6905	7.4000	0.0000	492.6114	849.0011	
VAR_13	0.0000	0.0000	41.9200	0.0000	0.0000	0.0000	0.7837
_cons	1541.4060	89.5834	17.2100	0.0000	1365.3870	1717.4260	
VAR_14	-0.1407	0.0081	-17.3300	0.0000	-0.1567	-0.1248	0.3823
_cons	3604.4430	101.3954	35.5500	0.0000	3405.2150	3803.6720	
VAR_15	-1.0078	0.1097	-9.1800	0.0000	-1.2235	-0.7922	0.1481
_cons	4590.3240	88.3584	51.9500	0.0000	4416.7110	4763.9360	

VAR_16	0.0846	0.0067	12.6800	0.0000	0.0715	0.0977	0.2490
_cons	4041.0780	101.4207	39.8400	0.0000	3841.8000	4240.3570	
VAR_17	41.7230	2.2000	18.9700	0.0000	37.4003	46.0456	0.4258
_cons	2909.9320	123.4581	23.5700	0.0000	2667.3530	3152.5100	
VAR_18	2.1874	0.0776	28.2000	0.0000	2.0350	2.3398	0.6212
cons	3132.7980	82.6808	37.8900	0.0000	2970.3410	3295.2550	
VAR_19	-0.1501	0.0082	-18.2100	0.0000	-0.1663	-0.1339	0.4062
_cons	3523.1710	101.0996	34.8500	0.0000	3324.5240	3721.8190	
VAR_20	0.0124	0.0003	44.9100	0.0000	0.0118	0.0129	0.8062
_cons	1059.2280	93.4785	11.3300	0.0000	875.5551	1242.9010	
VAR_21	-2621.6900	425.2153	-6.1700	0.0000	-3457.1820	-1786.1990	0.0727
_cons	8945.1670	668.5595	13.3800	0.0000	7631.5360	10258.8000	
VAR_22	234.9288	27.6421	8.5000	0.0000	180.6157	289.2418	0.1296
_cons	2440.6410	296.6833	8.2300	0.0000	1857.6970	3023.5840	
VAR_23	-5387.7830	443.0612	-12.1600	0.0000	-6258.3400	-4517.2270	0.2337
_cons	12117.0400	602.1466	20.1200	0.0000	10933.9000	13300.1800	
VAR_24	0.0259	0.0005	56.2300	0.0000	0.0250	0.0268	0.8670
_cons	-2486.9820	134.7308	-18.4600	0.0000	-2751.7100	-2222.2540	
VAR_25	-26691.9900	703.8506	-37.9200	0.0000	-28074.9700	-25309.0200	0.7478
_cons	12532.5500	207.3938	60.4300	0.0000	12125.0500	12940.0500	
VAR_26	0.0240	0.0006	37.6100	0.0000	0.0227	0.0252	0.7447
_cons	1900.0120	90.9416	20.8900	0.0000	1721.3230	2078.7000	

10.6 BASE: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	-807.69	807.69	11.04%
Minimum	-1619.18	17.10	0.31%
Maximum	-17.10	1619.18	19.88%
Range	1602.08	1602.08	19.57%
Correlation			**88.25%**

10.7 L1M: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	-1110.79	1110.79	15.08%
Minimum	-1666.67	151.26	2.71%
Maximum	-151.26	1666.67	21.00%
Range	1515.41	1515.41	18.29%
Correlation			**90.58%**

10.8 L2M: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	-207.71	312.09	4.32%
Minimum	-809.94	40.50	0.56%
Maximum	679.65	809.94	12.19%
Range	1489.59	769.44	11.63%
Correlation			**88.51%**

10.9 L3M: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	20.51	263.09	3.79%
Minimum	-848.95	2.20	0.03%
Maximum	918.73	918.73	16.47%
Range	1767.68	916.54	16.44%
Correlation			**85.13%**

10.10 L4M: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	-691.99	694.51	9.47%
Minimum	-1392.85	42.73	0.73%
Maximum	61.75	1392.85	17.10%
Range	1454.60	1350.11	16.37%
Correlation			**88.17%**

10.11 L5M: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	-256.78	307.86	4.25%
Minimum	-852.52	3.25	0.05%

Maximum	348.11	852.52	12.78%
Range	1200.64	849.28	12.73%
Correlation			**89.24%**

10.12 L6M: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	-448.36	502.71	6.89%
Minimum	-1133.35	21.45	0.33%
Maximum	604.65	1133.35	16.17%
Range	1738.00	1111.90	15.84%
Correlation			**82.17%**

10.13 L9M: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	-782.61	832.84	11.34%
Minimum	-1480.06	19.13	0.31%
Maximum	506.93	1480.06	18.25%
Range	1986.99	1460.94	17.94%
Correlation			**71.52%**

10.14 L12M: Performance (Best Correlation of 40)

	Error	Absolute Error	% Error
Average	169.92	387.02	5.75%
Minimum	-793.98	3.18	0.05%
Maximum	1445.47	1445.47	24.51%
Range	2239.44	1442.28	24.46%
Correlation			**62.58%**

10.15 BASE: Performance (Lowest MSE of 40)

	Error	Absolute Error	% Error
Average	-207.70	339.73	4.72%
Minimum	-828.78	2.67	0.04%
Maximum	550.03	828.78	10.01%
Range	1378.81	826.11	9.98%
Accuracy (MSE)			**95.28%**

10.16 L1M: Performance (Lowest MSE of 40)

	Error	Absolute Error	% Error
Average	23.85	206.98	3.03%
Minimum	-372.54	0.02	0.00%
Maximum	946.33	946.33	16.97%
Range	1318.87	946.31	16.97%
Accuracy (MSE)			**96.97%**

10.17 L2M: Performance (Lowest MSE of 40)

	Error	Absolute Error	% Error
Average	-207.71	312.09	4.32%
Minimum	-809.94	40.50	0.56%
Maximum	679.65	809.94	12.19%
Range	1489.59	769.44	11.63%
Accuracy (MSE)			**95.68%**

10.18 L3M: Performance (Lowest MSE of 40)

	Error	Absolute Error	% Error
Average	20.51	263.09	3.79%
Minimum	-848.95	2.20	0.03%
Maximum	918.73	918.73	16.47%
Range	1767.68	916.54	16.44%
Accuracy (MSE)			**96.21%**

10.19 L4M: Performance (Lowest MSE of 40)

	Error	Absolute Error	% Error
Average	-187.58	317.71	4.43%
Minimum	-795.98	13.68	0.20%
Maximum	777.74	795.98	13.94%
Range	1573.72	782.30	13.74%
Accuracy (MSE)			**95.57%**

10.20 L5M: Performance (Lowest MSE of 40)

	Error	Absolute Error	% Error
Average	-17.95	250.92	3.65%
Minimum	-824.70	7.69	0.11%
Maximum	890.24	890.24	15.96%
Range	1714.94	882.56	15.85%
Accuracy (MSE)			**96.35%**

10.21 L6M: Performance (Lowest MSE of 40)

	Error	Absolute Error	% Error
Average	-38.85	321.57	4.54%
Minimum	-1019.48	3.01	0.04%
Maximum	797.71	1019.48	13.53%
Range	1817.19	1016.47	13.49%
Accuracy (MSE)			**95.46%**

10.22 L9M: Performance (Lowest MSE of 40)

	Error	Absolute Error	% Error
Average	-30.98	450.78	6.49%
Minimum	-906.83	3.77	0.05%

Maximum	1424.08	1424.08	25.53%
Range	2330.91	1420.31	25.48%
Accuracy (MSE)			**93.51%**

10.23 L12M: Performance (Lowest MSE of 40)

	Error	**Absolute Error**	**% Error**
Average	169.92	387.02	5.75%
Minimum	-793.98	3.18	0.05%
Maximum	1445.47	1445.47	24.51%
Range	2239.44	1442.28	24.46%
Accuracy (MSE)			**94.25%**

10.24 ARIMA: Performance (Best Correlation and Lowest MSE of 40)

	Error	**Absolute Error**	**% Error**
Average	-1300.41	1327.45	17.77%
Minimum	-2448.00	40.05	0.67%
Maximum	389.12	2448.00	29.58%
Range	2837.12	2407.95	28.91%
Accuracy (MSE)			**82.23%**
Correlation			**-63.90%**

10.25 BASE: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5557.54	-612.20	612.20	9.92%
Aug-20	5897.76	5459.28	-438.48	438.48	7.43%
Sep-20	5963.57	5539.46	-424.11	424.11	7.11%
Oct-20	5866.10	5334.81	-531.29	531.29	9.06%
Nov-20	5577.27	5560.17	-17.10	17.10	0.31%
Dec-20	6266.19	5673.17	-593.03	593.03	9.46%
Jan-21	6460.52	5415.20	-1045.32	1045.32	16.18%
Feb-21	6407.46	5637.36	-770.10	770.10	12.02%
Mar-21	6483.43	5776.18	-707.25	707.25	10.91%
Apr-21	6713.63	5879.31	-834.32	834.32	12.43%
May-21	6969.81	6104.91	-864.91	864.91	12.41%
Jun-21	7022.61	6200.00	-822.61	822.61	11.71%
Jul-21	7037.47	6306.44	-731.03	731.03	10.39%
Aug-21	7032.30	6279.11	-753.19	753.19	10.71%
Sep-21	7119.70	6158.14	-961.56	961.56	13.51%
Oct-21	7086.42	6227.21	-859.21	859.21	12.12%
Nov-21	7237.57	6455.20	-782.37	782.37	10.81%
Dec-21	7059.45	6700.10	-359.35	359.35	5.09%
Jan-22	7384.54	6712.64	-671.91	671.91	9.10%
Feb-22	7464.37	6835.06	-629.31	629.31	8.43%
Mar-22	7458.25	6983.56	-474.69	474.69	6.36%
Apr-22	7515.68	6915.64	-600.04	600.04	7.98%
May-22	7544.55	6883.65	-660.90	660.90	8.76%
Jun-22	7607.66	6779.17	-828.49	828.49	10.89%
Jul-22	7169.28	6648.75	-520.53	520.53	7.26%
Aug-22	7423.43	6457.83	-965.60	965.60	13.01%
Sep-22	7284.15	6314.67	-969.48	969.48	13.31%
Oct-22	6893.81	6276.16	-617.65	617.65	8.96%
Nov-22	7094.53	6368.03	-726.50	726.50	10.24%
Dec-22	7573.05	6704.92	-868.13	868.13	11.46%
Jan-23	7451.74	6862.23	-589.51	589.51	7.91%
Feb-23	7771.70	6962.50	-809.20	809.20	10.41%
Mar-23	7876.28	6946.64	-929.64	929.64	11.80%
Apr-23	7631.74	7044.91	-586.83	586.83	7.69%
May-23	7870.57	6938.87	-931.70	931.70	11.84%
Jun-23	7446.14	6901.54	-544.60	544.60	7.31%
Jul-23	7531.53	6731.99	-799.54	799.54	10.62%
Aug-23	7699.41	6752.63	-946.78	946.78	12.30%
Sep-23	7439.13	6708.66	-730.47	730.47	9.82%
Oct-23	7608.08	6504.12	-1103.96	1103.96	14.51%
Nov-23	7321.72	6541.30	-780.42	780.42	10.66%
Dec-23	7453.75	6591.82	-861.93	861.93	11.56%
Jan-24	7733.24	6574.30	-1158.94	1158.94	14.99%
Feb-24	7630.57	6562.86	-1067.71	1067.71	13.99%
Mar-24	7630.02	6648.55	-981.47	981.47	12.86%
Apr-24	7952.62	6580.80	-1371.82	1371.82	17.25%
May-24	8144.13	6524.95	-1619.18	1619.18	19.88%
Jun-24	8275.38	6734.62	-1540.76	1540.76	18.62%
Jul-24	8164.12	6582.64	-1581.48	1581.48	19.37%

10.26 L1M: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5521.20	-648.55	648.55	10.51%
Aug-20	5897.76	5549.89	-347.87	347.87	5.90%
Sep-20	5963.57	5671.43	-292.14	292.14	4.90%
Oct-20	5866.10	5558.15	-307.95	307.95	5.25%
Nov-20	5577.27	5426.01	-151.26	151.26	2.71%
Dec-20	6266.19	5573.11	-693.08	693.08	11.06%
Jan-21	6460.52	5544.64	-915.88	915.88	14.18%
Feb-21	6407.46	5423.30	-984.16	984.16	15.36%
Mar-21	6483.43	5799.29	-684.14	684.14	10.55%
Apr-21	6713.63	5875.58	-838.05	838.05	12.48%
May-21	6969.81	5975.23	-994.58	994.58	14.27%
Jun-21	7022.61	6078.77	-943.84	943.84	13.44%
Jul-21	7037.47	6043.29	-994.18	994.18	14.13%
Aug-21	7032.30	6047.21	-985.09	985.09	14.01%
Sep-21	7119.70	5986.47	-1133.23	1133.23	15.92%
Oct-21	7086.42	6045.18	-1041.24	1041.24	14.69%
Nov-21	7237.57	6180.84	-1056.73	1056.73	14.60%
Dec-21	7059.45	6119.17	-940.28	940.28	13.32%
Jan-22	7384.54	6392.26	-992.28	992.28	13.44%
Feb-22	7464.37	6638.63	-825.74	825.74	11.06%
Mar-22	7458.25	6476.28	-981.97	981.97	13.17%
Apr-22	7515.68	6280.15	-1235.53	1235.53	16.44%
May-22	7544.55	6178.84	-1365.71	1365.71	18.10%
Jun-22	7607.66	6110.78	-1496.88	1496.88	19.68%
Jul-22	7169.28	6131.83	-1037.45	1037.45	14.47%
Aug-22	7423.43	6138.94	-1284.49	1284.49	17.30%
Sep-22	7284.15	6098.85	-1185.30	1185.30	16.27%
Oct-22	6893.81	6042.79	-851.02	851.02	12.34%
Nov-22	7094.53	6212.87	-881.66	881.66	12.43%
Dec-22	7573.05	6108.75	-1464.30	1464.30	19.34%
Jan-23	7451.74	6224.61	-1227.13	1227.13	16.47%
Feb-23	7771.70	6315.15	-1456.55	1456.55	18.74%
Mar-23	7876.28	6360.58	-1515.70	1515.70	19.24%
Apr-23	7631.74	6417.02	-1214.72	1214.72	15.92%
May-23	7870.57	6541.57	-1329.00	1329.00	16.89%
Jun-23	7446.14	6398.15	-1047.99	1047.99	14.07%
Jul-23	7531.53	6248.73	-1282.80	1282.80	17.03%
Aug-23	7699.41	6158.31	-1541.10	1541.10	20.02%
Sep-23	7439.13	6152.45	-1286.68	1286.68	17.30%
Oct-23	7608.08	6128.61	-1479.47	1479.47	19.45%
Nov-23	7321.72	6097.61	-1224.11	1224.11	16.72%
Dec-23	7453.75	6096.55	-1357.20	1357.20	18.21%
Jan-24	7733.24	6109.16	-1624.08	1624.08	21.00%
Feb-24	7630.57	6162.50	-1468.07	1468.07	19.24%
Mar-24	7630.02	6205.85	-1424.17	1424.17	18.67%
Apr-24	7952.62	6386.07	-1566.55	1566.55	19.70%
May-24	8144.13	6510.28	-1633.85	1633.85	20.06%
Jun-24	8275.38	6608.71	-1666.67	1666.67	20.14%
Jul-24	8164.12	6635.83	-1528.29	1528.29	18.72%

10.27 L2M: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5547.05	-622.69	622.69	10.09%
Aug-20	5897.76	6192.05	294.29	294.29	4.99%
Sep-20	5963.57	6210.55	246.98	246.98	4.14%
Oct-20	5866.10	6274.96	408.86	408.86	6.97%
Nov-20	5577.27	6256.92	679.65	679.65	12.19%
Dec-20	6266.19	6084.79	-181.40	181.40	2.89%
Jan-21	6460.52	6203.20	-257.32	257.32	3.98%
Feb-21	6407.46	6345.87	-61.59	61.59	0.96%
Mar-21	6483.43	6122.50	-360.93	360.93	5.57%
Apr-21	6713.63	6501.89	-211.74	211.74	3.15%
May-21	6969.81	6747.42	-222.39	222.39	3.19%
Jun-21	7022.61	6780.06	-242.55	242.55	3.45%
Jul-21	7037.47	6939.34	-98.13	98.13	1.39%
Aug-21	7032.30	6962.54	-69.76	69.76	0.99%
Sep-21	7119.70	7007.49	-112.21	112.21	1.58%
Oct-21	7086.42	6991.07	-95.35	95.35	1.35%
Nov-21	7237.57	7046.80	-190.77	190.77	2.64%
Dec-21	7059.45	7222.05	162.60	162.60	2.30%
Jan-22	7384.54	7147.42	-237.12	237.12	3.21%
Feb-22	7464.37	7183.96	-280.41	280.41	3.76%
Mar-22	7458.25	7226.49	-231.76	231.76	3.11%
Apr-22	7515.68	7192.08	-323.60	323.60	4.31%
May-22	7544.55	7271.35	-273.20	273.20	3.62%
Jun-22	7607.66	7234.72	-372.94	372.94	4.90%
Jul-22	7169.28	7298.96	129.68	129.68	1.81%
Aug-22	7423.43	7262.48	-160.95	160.95	2.17%
Sep-22	7284.15	7324.65	40.50	40.50	0.56%
Oct-22	6893.81	7359.66	465.85	465.85	6.76%
Nov-22	7094.53	7223.45	128.92	128.92	1.82%
Dec-22	7573.05	7372.66	-200.39	200.39	2.65%
Jan-23	7451.74	7359.17	-92.57	92.57	1.24%
Feb-23	7771.70	7365.51	-406.19	406.19	5.23%
Mar-23	7876.28	7229.44	-646.84	646.84	8.21%
Apr-23	7631.74	7353.19	-278.55	278.55	3.65%
May-23	7870.57	7288.14	-582.43	582.43	7.40%
Jun-23	7446.14	7326.83	-119.31	119.31	1.60%
Jul-23	7531.53	7251.48	-280.05	280.05	3.72%
Aug-23	7699.41	7374.35	-325.06	325.06	4.22%
Sep-23	7439.13	7178.97	-260.16	260.16	3.50%
Oct-23	7608.08	7246.58	-361.51	361.51	4.75%
Nov-23	7321.72	7181.03	-140.69	140.69	1.92%
Dec-23	7453.75	7177.57	-276.18	276.18	3.71%
Jan-24	7733.24	7251.99	-481.25	481.25	6.22%
Feb-24	7630.57	7293.29	-337.28	337.28	4.42%
Mar-24	7630.02	7150.01	-480.01	480.01	6.29%
Apr-24	7952.62	7256.54	-696.08	696.08	8.75%
May-24	8144.13	7387.19	-756.94	756.94	9.29%
Jun-24	8275.38	7465.44	-809.94	809.94	9.79%
Jul-24	8164.12	7567.29	-596.83	596.83	7.31%

10.28 L3M: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5320.79	-848.95	848.95	13.76%
Aug-20	5897.76	5583.12	-314.64	314.64	5.33%
Sep-20	5963.57	5509.20	-454.37	454.37	7.62%
Oct-20	5866.10	6371.06	504.96	504.96	8.61%
Nov-20	5577.27	6496.00	918.73	918.73	16.47%
Dec-20	6266.19	6585.39	319.20	319.20	5.09%
Jan-21	6460.52	6403.71	-56.81	56.81	0.88%
Feb-21	6407.46	6369.09	-38.37	38.37	0.60%
Mar-21	6483.43	6437.54	-45.89	45.89	0.71%
Apr-21	6713.63	6232.14	-481.49	481.49	7.17%
May-21	6969.81	6650.82	-318.99	318.99	4.58%
Jun-21	7022.61	6885.79	-136.82	136.82	1.95%
Jul-21	7037.47	6931.94	-105.54	105.54	1.50%
Aug-21	7032.30	7186.53	154.23	154.23	2.19%
Sep-21	7119.70	7341.14	221.44	221.44	3.11%
Oct-21	7086.42	7519.02	432.60	432.60	6.10%
Nov-21	7237.57	7495.86	258.29	258.29	3.57%
Dec-21	7059.45	7472.86	413.41	413.41	5.86%
Jan-22	7384.54	7547.26	162.72	162.72	2.20%
Feb-22	7464.37	7466.57	2.20	2.20	0.03%
Mar-22	7458.25	7445.68	-12.57	12.57	0.17%
Apr-22	7515.68	7337.88	-177.80	177.80	2.37%
May-22	7544.55	7191.47	-353.08	353.08	4.68%
Jun-22	7607.66	7285.45	-322.21	322.21	4.24%
Jul-22	7169.28	7257.05	87.77	87.77	1.22%
Aug-22	7423.43	7418.71	-4.72	4.72	0.06%
Sep-22	7284.15	7509.30	225.15	225.15	3.09%
Oct-22	6893.81	7653.25	759.44	759.44	11.02%
Nov-22	7094.53	7666.60	572.07	572.07	8.06%
Dec-22	7573.05	7481.91	-91.14	91.14	1.20%
Jan-23	7451.74	7595.97	144.23	144.23	1.94%
Feb-23	7771.70	7734.48	-37.22	37.22	0.48%
Mar-23	7876.28	7971.65	95.37	95.37	1.21%
Apr-23	7631.74	7671.18	39.44	39.44	0.52%
May-23	7870.57	7536.37	-334.20	334.20	4.25%
Jun-23	7446.14	7332.42	-113.72	113.72	1.53%
Jul-23	7531.53	7503.64	-27.89	27.89	0.37%
Aug-23	7699.41	7624.71	-74.70	74.70	0.97%
Sep-23	7439.13	7475.03	35.90	35.90	0.48%
Oct-23	7608.08	7672.75	64.67	64.67	0.85%
Nov-23	7321.72	7755.04	433.32	433.32	5.92%
Dec-23	7453.75	7723.04	269.29	269.29	3.61%
Jan-24	7733.24	7755.53	22.28	22.28	0.29%
Feb-24	7630.57	7871.14	240.57	240.57	3.15%
Mar-24	7630.02	8026.04	396.02	396.02	5.19%
Apr-24	7952.62	8127.46	174.84	174.84	2.20%
May-24	8144.13	7776.66	-367.47	367.47	4.51%
Jun-24	8275.38	7598.75	-676.63	676.63	8.18%
Jul-24	8164.12	7616.15	-547.97	547.97	6.71%

10.29 L4M: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5621.77	-547.97	547.97	8.88%
Aug-20	5897.76	4941.81	-955.95	955.95	16.21%
Sep-20	5963.57	5549.70	-413.87	413.87	6.94%
Oct-20	5866.10	5823.37	-42.73	42.73	0.73%
Nov-20	5577.27	5639.02	61.75	61.75	1.11%
Dec-20	6266.19	5861.46	-404.73	404.73	6.46%
Jan-21	6460.52	6059.50	-401.02	401.02	6.21%
Feb-21	6407.46	6215.86	-191.60	191.60	2.99%
Mar-21	6483.43	6054.79	-428.64	428.64	6.61%
Apr-21	6713.63	6109.31	-604.32	604.32	9.00%
May-21	6969.81	6074.82	-894.99	894.99	12.84%
Jun-21	7022.61	6069.47	-953.14	953.14	13.57%
Jul-21	7037.47	6102.08	-935.39	935.39	13.29%
Aug-21	7032.30	6198.44	-833.86	833.86	11.86%
Sep-21	7119.70	6316.93	-802.77	802.77	11.28%
Oct-21	7086.42	6368.58	-717.84	717.84	10.13%
Nov-21	7237.57	6402.31	-835.26	835.26	11.54%
Dec-21	7059.45	6496.06	-563.39	563.39	7.98%
Jan-22	7384.54	6413.59	-970.95	970.95	13.15%
Feb-22	7464.37	6446.07	-1018.30	1018.30	13.64%
Mar-22	7458.25	6728.71	-729.54	729.54	9.78%
Apr-22	7515.68	6654.37	-861.31	861.31	11.46%
May-22	7544.55	6675.44	-869.11	869.11	11.52%
Jun-22	7607.66	6568.79	-1038.87	1038.87	13.66%
Jul-22	7169.28	6536.06	-633.22	633.22	8.83%
Aug-22	7423.43	6753.02	-670.41	670.41	9.03%
Sep-22	7284.15	6660.16	-623.99	623.99	8.57%
Oct-22	6893.81	6585.90	-307.91	307.91	4.47%
Nov-22	7094.53	6792.49	-302.04	302.04	4.26%
Dec-22	7573.05	6893.26	-679.79	679.79	8.98%
Jan-23	7451.74	7015.86	-435.89	435.89	5.85%
Feb-23	7771.70	7107.03	-664.67	664.67	8.55%
Mar-23	7876.28	7168.35	-707.93	707.93	8.99%
Apr-23	7631.74	7041.50	-590.24	590.24	7.73%
May-23	7870.57	7098.59	-771.98	771.98	9.81%
Jun-23	7446.14	7247.53	-198.61	198.61	2.67%
Jul-23	7531.53	7094.95	-436.58	436.58	5.80%
Aug-23	7699.41	6860.19	-839.22	839.22	10.90%
Sep-23	7439.13	6594.36	-844.77	844.77	11.36%
Oct-23	7608.08	7168.20	-439.89	439.89	5.78%
Nov-23	7321.72	6629.07	-692.65	692.65	9.46%
Dec-23	7453.75	6778.07	-675.68	675.68	9.06%
Jan-24	7733.24	6812.87	-920.38	920.38	11.90%
Feb-24	7630.57	6664.98	-965.59	965.59	12.65%
Mar-24	7630.02	6924.33	-705.69	705.69	9.25%
Apr-24	7952.62	6967.87	-984.75	984.75	12.38%
May-24	8144.13	6751.28	-1392.85	1392.85	17.10%
Jun-24	8275.38	6966.67	-1308.71	1308.71	15.81%
Jul-24	8164.12	7004.01	-1160.11	1160.11	14.21%

10.30 L5M: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	6467.71	297.97	297.97	4.83%
Aug-20	5897.76	5995.57	97.81	97.81	1.66%
Sep-20	5963.57	5447.80	-515.77	515.77	8.65%
Oct-20	5866.10	5116.16	-749.94	749.94	12.78%
Nov-20	5577.27	4929.56	-647.71	647.71	11.61%
Dec-20	6266.19	6278.06	11.87	11.87	0.19%
Jan-21	6460.52	6517.42	56.90	56.90	0.88%
Feb-21	6407.46	6586.30	178.84	178.84	2.79%
Mar-21	6483.43	6486.68	3.25	3.25	0.05%
Apr-21	6713.63	6515.63	-198.00	198.00	2.95%
May-21	6969.81	6663.72	-306.10	306.10	4.39%
Jun-21	7022.61	6694.95	-327.66	327.66	4.67%
Jul-21	7037.47	6633.63	-403.84	403.84	5.74%
Aug-21	7032.30	6761.53	-270.77	270.77	3.85%
Sep-21	7119.70	6902.53	-217.17	217.17	3.05%
Oct-21	7086.42	7079.65	-6.77	6.77	0.10%
Nov-21	7237.57	7136.67	-100.90	100.90	1.39%
Dec-21	7059.45	7130.33	70.88	70.88	1.00%
Jan-22	7384.54	7126.73	-257.81	257.81	3.49%
Feb-22	7464.37	7214.23	-250.14	250.14	3.35%
Mar-22	7458.25	7338.02	-120.23	120.23	1.61%
Apr-22	7515.68	7211.38	-304.30	304.30	4.05%
May-22	7544.55	7236.56	-307.99	307.99	4.08%
Jun-22	7607.66	7117.07	-490.59	490.59	6.45%
Jul-22	7169.28	7257.23	87.95	87.95	1.23%
Aug-22	7423.43	7431.43	8.00	8.00	0.11%
Sep-22	7284.15	7237.30	-46.85	46.85	0.64%
Oct-22	6893.81	7241.92	348.11	348.11	5.05%
Nov-22	7094.53	7145.69	51.16	51.16	0.72%
Dec-22	7573.05	7135.03	-438.02	438.02	5.78%
Jan-23	7451.74	7090.30	-361.44	361.44	4.85%
Feb-23	7771.70	6919.18	-852.52	852.52	10.97%
Mar-23	7876.28	7048.80	-827.48	827.48	10.51%
Apr-23	7631.74	7133.66	-498.08	498.08	6.53%
May-23	7870.57	7294.37	-576.21	576.21	7.32%
Jun-23	7446.14	7194.86	-251.28	251.28	3.37%
Jul-23	7531.53	7214.36	-317.17	317.17	4.21%
Aug-23	7699.41	7195.76	-503.65	503.65	6.54%
Sep-23	7439.13	7385.29	-53.84	53.84	0.72%
Oct-23	7608.08	7216.29	-391.79	391.79	5.15%
Nov-23	7321.72	7360.39	38.67	38.67	0.53%
Dec-23	7453.75	7318.01	-135.74	135.74	1.82%
Jan-24	7733.24	7345.70	-387.54	387.54	5.01%
Feb-24	7630.57	7334.37	-296.20	296.20	3.88%
Mar-24	7630.02	7462.88	-167.14	167.14	2.19%
Apr-24	7952.62	7521.82	-430.80	430.80	5.42%
May-24	8144.13	7657.62	-486.51	486.51	5.97%
Jun-24	8275.38	7607.26	-668.12	668.12	8.07%
Jul-24	8164.12	7496.54	-667.58	667.58	8.18%

10.31 L6M: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	6579.58	409.84	409.84	6.64%
Aug-20	5897.76	6502.41	604.65	604.65	10.25%
Sep-20	5963.57	5608.39	-355.18	355.18	5.96%
Oct-20	5866.10	5112.75	-753.35	753.35	12.84%
Nov-20	5577.27	5642.94	65.67	65.67	1.18%
Dec-20	6266.19	5252.81	-1013.38	1013.38	16.17%
Jan-21	6460.52	6303.53	-156.99	156.99	2.43%
Feb-21	6407.46	6386.01	-21.45	21.45	0.33%
Mar-21	6483.43	6268.75	-214.68	214.68	3.31%
Apr-21	6713.63	6232.25	-481.38	481.38	7.17%
May-21	6969.81	6362.14	-607.67	607.67	8.72%
Jun-21	7022.61	6620.02	-402.59	402.59	5.73%
Jul-21	7037.47	6197.11	-840.36	840.36	11.94%
Aug-21	7032.30	6587.01	-445.30	445.30	6.33%
Sep-21	7119.70	6696.91	-422.79	422.79	5.94%
Oct-21	7086.42	6734.13	-352.29	352.29	4.97%
Nov-21	7237.57	6906.82	-330.75	330.75	4.57%
Dec-21	7059.45	6899.00	-160.46	160.46	2.27%
Jan-22	7384.54	6917.27	-467.27	467.27	6.33%
Feb-22	7464.37	6987.97	-476.40	476.40	6.38%
Mar-22	7458.25	6990.95	-467.30	467.30	6.27%
Apr-22	7515.68	7061.27	-454.41	454.41	6.05%
May-22	7544.55	7089.98	-454.57	454.57	6.03%
Jun-22	7607.66	7023.26	-584.40	584.40	7.68%
Jul-22	7169.28	6875.75	-293.53	293.53	4.09%
Aug-22	7423.43	6926.42	-497.01	497.01	6.70%
Sep-22	7284.15	7089.09	-195.07	195.07	2.68%
Oct-22	6893.81	7079.89	186.08	186.08	2.70%
Nov-22	7094.53	7159.85	65.32	65.32	0.92%
Dec-22	7573.05	7185.09	-387.97	387.97	5.12%
Jan-23	7451.74	7188.58	-263.16	263.16	3.53%
Feb-23	7771.70	7099.01	-672.69	672.69	8.66%
Mar-23	7876.28	6742.93	-1133.35	1133.35	14.39%
Apr-23	7631.74	6671.46	-960.28	960.28	12.58%
May-23	7870.57	6788.46	-1082.11	1082.11	13.75%
Jun-23	7446.14	6916.08	-530.06	530.06	7.12%
Jul-23	7531.53	6868.40	-663.13	663.13	8.80%
Aug-23	7699.41	6925.11	-774.30	774.30	10.06%
Sep-23	7439.13	6924.32	-514.81	514.81	6.92%
Oct-23	7608.08	7103.06	-505.02	505.02	6.64%
Nov-23	7321.72	7230.74	-90.98	90.98	1.24%
Dec-23	7453.75	7066.42	-387.33	387.33	5.20%
Jan-24	7733.24	7182.13	-551.12	551.12	7.13%
Feb-24	7630.57	7182.06	-448.51	448.51	5.88%
Mar-24	7630.02	7169.95	-460.07	460.07	6.03%
Apr-24	7952.62	7250.93	-701.69	701.69	8.82%
May-24	8144.13	7263.60	-880.53	880.53	10.81%
Jun-24	8275.38	7324.19	-951.19	951.19	11.49%
Jul-24	8164.12	7269.72	-894.40	894.40	10.96%

10.32 L9M: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	6188.87	19.13	19.13	0.31%
Aug-20	5897.76	6122.27	224.51	224.51	3.81%
Sep-20	5963.57	6139.09	175.52	175.52	2.94%
Oct-20	5866.10	6170.72	304.62	304.62	5.19%
Nov-20	5577.27	6084.20	506.93	506.93	9.09%
Dec-20	6266.19	6017.00	-249.19	249.19	3.98%
Jan-21	6460.52	6015.21	-445.31	445.31	6.89%
Feb-21	6407.46	5439.83	-967.63	967.63	15.10%
Mar-21	6483.43	5303.58	-1179.85	1179.85	18.20%
Apr-21	6713.63	5921.34	-792.29	792.29	11.80%
May-21	6969.81	6105.37	-864.44	864.44	12.40%
Jun-21	7022.61	6243.01	-779.60	779.60	11.10%
Jul-21	7037.47	6353.88	-683.59	683.59	9.71%
Aug-21	7032.30	6200.96	-831.34	831.34	11.82%
Sep-21	7119.70	6130.75	-988.95	988.95	13.89%
Oct-21	7086.42	5793.44	-1292.98	1292.98	18.25%
Nov-21	7237.57	5918.30	-1319.27	1319.27	18.23%
Dec-21	7059.45	6048.02	-1011.43	1011.43	14.33%
Jan-22	7384.54	6115.18	-1269.36	1269.36	17.19%
Feb-22	7464.37	6329.32	-1135.06	1135.06	15.21%
Mar-22	7458.25	6439.97	-1018.28	1018.28	13.65%
Apr-22	7515.68	6521.57	-994.11	994.11	13.23%
May-22	7544.55	6580.76	-963.79	963.79	12.77%
Jun-22	7607.66	6533.33	-1074.33	1074.33	14.12%
Jul-22	7169.28	6558.74	-610.54	610.54	8.52%
Aug-22	7423.43	6740.00	-683.43	683.43	9.21%
Sep-22	7284.15	6748.66	-535.49	535.49	7.35%
Oct-22	6893.81	6522.34	-371.47	371.47	5.39%
Nov-22	7094.53	6564.51	-530.02	530.02	7.47%
Dec-22	7573.05	6736.14	-836.91	836.91	11.05%
Jan-23	7451.74	6774.51	-677.23	677.23	9.09%
Feb-23	7771.70	6865.05	-906.65	906.65	11.67%
Mar-23	7876.28	6859.28	-1017.00	1017.00	12.91%
Apr-23	7631.74	6902.04	-729.70	729.70	9.56%
May-23	7870.57	6778.84	-1091.73	1091.73	13.87%
Jun-23	7446.14	6451.05	-995.09	995.09	13.36%
Jul-23	7531.53	6460.79	-1070.75	1070.75	14.22%
Aug-23	7699.41	6734.02	-965.39	965.39	12.54%
Sep-23	7439.13	6870.95	-568.18	568.18	7.64%
Oct-23	7608.08	6836.96	-771.12	771.12	10.14%
Nov-23	7321.72	6819.20	-502.52	502.52	6.86%
Dec-23	7453.75	6759.44	-694.31	694.31	9.31%
Jan-24	7733.24	6772.17	-961.07	961.07	12.43%
Feb-24	7630.57	6743.19	-887.38	887.38	11.63%
Mar-24	7630.02	6662.65	-967.37	967.37	12.68%
Apr-24	7952.62	6808.65	-1143.97	1143.97	14.38%
May-24	8144.13	6820.42	-1323.71	1323.71	16.25%
Jun-24	8275.38	6795.32	-1480.06	1480.06	17.89%
Jul-24	8164.12	6767.45	-1396.67	1396.67	17.11%

10.33 L12M: Predictions (best correlation of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	7162.41	992.67	992.67	16.09%
Aug-20	5897.76	7343.23	1445.47	1445.47	24.51%
Sep-20	5963.57	7086.86	1123.29	1123.29	18.84%
Oct-20	5866.10	7038.28	1172.18	1172.18	19.98%
Nov-20	5577.27	6941.40	1364.13	1364.13	24.46%
Dec-20	6266.19	6966.31	700.12	700.12	11.17%
Jan-21	6460.52	6984.47	523.95	523.95	8.11%
Feb-21	6407.46	7035.67	628.21	628.21	9.80%
Mar-21	6483.43	7128.66	645.23	645.23	9.95%
Apr-21	6713.63	6976.66	263.03	263.03	3.92%
May-21	6969.81	6538.56	-431.25	431.25	6.19%
Jun-21	7022.61	6697.44	-325.17	325.17	4.63%
Jul-21	7037.47	6785.29	-252.18	252.18	3.58%
Aug-21	7032.30	7035.48	3.18	3.18	0.05%
Sep-21	7119.70	7255.53	135.83	135.83	1.91%
Oct-21	7086.42	7387.39	300.97	300.97	4.25%
Nov-21	7237.57	7328.56	90.99	90.99	1.26%
Dec-21	7059.45	7277.89	218.44	218.44	3.09%
Jan-22	7384.54	7076.97	-307.57	307.57	4.17%
Feb-22	7464.37	6901.78	-562.59	562.59	7.54%
Mar-22	7458.25	7023.06	-435.19	435.19	5.84%
Apr-22	7515.68	7149.73	-365.95	365.95	4.87%
May-22	7544.55	7319.86	-224.69	224.69	2.98%
Jun-22	7607.66	7499.79	-107.87	107.87	1.42%
Jul-22	7169.28	7491.22	321.94	321.94	4.49%
Aug-22	7423.43	7344.54	-78.89	78.89	1.06%
Sep-22	7284.15	7287.50	3.35	3.35	0.05%
Oct-22	6893.81	7271.11	377.30	377.30	5.47%
Nov-22	7094.53	7385.40	290.87	290.87	4.10%
Dec-22	7573.05	7580.96	7.91	7.91	0.10%
Jan-23	7451.74	7455.86	4.12	4.12	0.06%
Feb-23	7771.70	7534.12	-237.58	237.58	3.06%
Mar-23	7876.28	7636.12	-240.16	240.16	3.05%
Apr-23	7631.74	7795.85	164.11	164.11	2.15%
May-23	7870.57	7939.05	68.48	68.48	0.87%
Jun-23	7446.14	7934.29	488.15	488.15	6.56%
Jul-23	7531.53	7981.26	449.73	449.73	5.97%
Aug-23	7699.41	7939.09	239.68	239.68	3.11%
Sep-23	7439.13	7692.08	252.95	252.95	3.40%
Oct-23	7608.08	7735.10	127.02	127.02	1.67%
Nov-23	7321.72	7692.32	370.60	370.60	5.06%
Dec-23	7453.75	7853.33	399.58	399.58	5.36%
Jan-24	7733.24	7819.49	86.25	86.25	1.12%
Feb-24	7630.57	7852.16	221.59	221.59	2.90%
Mar-24	7630.02	7793.81	163.79	163.79	2.15%
Apr-24	7952.62	7730.26	-222.36	222.36	2.80%
May-24	8144.13	7904.95	-239.18	239.18	2.94%
Jun-24	8275.38	7481.40	-793.98	793.98	9.59%
Jul-24	8164.12	7669.75	-494.37	494.37	6.06%

10.34 BASE: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	6059.70	-110.04	110.04	1.78%
Aug-20	5897.76	6270.10	372.34	372.34	6.31%
Sep-20	5963.57	6231.93	268.36	268.36	4.50%
Oct-20	5866.10	6093.61	227.51	227.51	3.88%
Nov-20	5577.27	6127.30	550.03	550.03	9.86%
Dec-20	6266.19	6001.96	-264.23	264.23	4.22%
Jan-21	6460.52	5829.85	-630.67	630.67	9.76%
Feb-21	6407.46	6001.30	-406.16	406.16	6.34%
Mar-21	6483.43	6034.11	-449.32	449.32	6.93%
Apr-21	6713.63	6256.29	-457.34	457.34	6.81%
May-21	6969.81	6582.78	-387.03	387.03	5.55%
Jun-21	7022.61	6691.16	-331.45	331.45	4.72%
Jul-21	7037.47	6807.74	-229.73	229.73	3.26%
Aug-21	7032.30	6834.96	-197.35	197.35	2.81%
Sep-21	7119.70	6597.91	-521.79	521.79	7.33%
Oct-21	7086.42	6665.29	-421.13	421.13	5.94%
Nov-21	7237.57	7019.51	-218.06	218.06	3.01%
Dec-21	7059.45	7513.85	454.40	454.40	6.44%
Jan-22	7384.54	7886.35	501.81	501.81	6.80%
Feb-22	7464.37	7840.76	376.39	376.39	5.04%
Mar-22	7458.25	7597.91	139.66	139.66	1.87%
Apr-22	7515.68	7404.84	-110.84	110.84	1.47%
May-22	7544.55	7369.32	-175.23	175.23	2.32%
Jun-22	7607.66	7275.82	-331.84	331.84	4.36%
Jul-22	7169.28	7214.37	45.09	45.09	0.63%
Aug-22	7423.43	7079.30	-344.13	344.13	4.64%
Sep-22	7284.15	6861.84	-422.32	422.32	5.80%
Oct-22	6893.81	6959.82	66.01	66.01	0.96%
Nov-22	7094.53	7056.47	-38.06	38.06	0.54%
Dec-22	7573.05	7295.62	-277.43	277.43	3.66%
Jan-23	7451.74	7449.07	-2.67	2.67	0.04%
Feb-23	7771.70	7732.33	-39.37	39.37	0.51%
Mar-23	7876.28	7778.70	-97.58	97.58	1.24%
Apr-23	7631.74	7701.39	69.65	69.65	0.91%
May-23	7870.57	7578.11	-292.46	292.46	3.72%
Jun-23	7446.14	7609.57	163.43	163.43	2.19%
Jul-23	7531.53	7145.42	-386.11	386.11	5.13%
Aug-23	7699.41	7239.76	-459.65	459.65	5.97%
Sep-23	7439.13	7175.44	-263.69	263.69	3.54%
Oct-23	7608.08	6906.70	-701.38	701.38	9.22%
Nov-23	7321.72	7044.77	-276.95	276.95	3.78%
Dec-23	7453.75	7104.07	-349.68	349.68	4.69%
Jan-24	7733.24	7161.31	-571.93	571.93	7.40%
Feb-24	7630.57	7255.39	-375.18	375.18	4.92%
Mar-24	7630.02	7361.67	-268.35	268.35	3.52%
Apr-24	7952.62	7341.73	-610.89	610.89	7.68%
May-24	8144.13	7360.23	-783.90	783.90	9.63%
Jun-24	8275.38	7446.60	-828.78	828.78	10.01%
Jul-24	8164.12	7384.93	-779.19	779.19	9.54%

10.35 L1M: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5797.20	-372.54	372.54	6.04%
Aug-20	5897.76	6252.98	355.22	355.22	6.02%
Sep-20	5963.57	6431.31	467.74	467.74	7.84%
Oct-20	5866.10	6598.62	732.52	732.52	12.49%
Nov-20	5577.27	6523.60	946.33	946.33	16.97%
Dec-20	6266.19	6352.34	86.15	86.15	1.37%
Jan-21	6460.52	6460.54	0.02	0.02	0.00%
Feb-21	6407.46	6172.06	-235.40	235.40	3.67%
Mar-21	6483.43	6253.21	-230.22	230.22	3.55%
Apr-21	6713.63	6459.06	-254.57	254.57	3.79%
May-21	6969.81	6603.93	-365.88	365.88	5.25%
Jun-21	7022.61	7070.24	47.63	47.63	0.68%
Jul-21	7037.47	7242.57	205.10	205.10	2.91%
Aug-21	7032.30	7401.90	369.60	369.60	5.26%
Sep-21	7119.70	7364.24	244.54	244.54	3.43%
Oct-21	7086.42	7125.76	39.34	39.34	0.56%
Nov-21	7237.57	7068.19	-169.38	169.38	2.34%
Dec-21	7059.45	7351.77	292.32	292.32	4.14%
Jan-22	7384.54	7497.84	113.30	113.30	1.53%
Feb-22	7464.37	7600.22	135.85	135.85	1.82%
Mar-22	7458.25	7492.60	34.35	34.35	0.46%
Apr-22	7515.68	7538.60	22.92	22.92	0.30%
May-22	7544.55	7353.70	-190.85	190.85	2.53%
Jun-22	7607.66	7409.98	-197.68	197.68	2.60%
Jul-22	7169.28	7365.72	196.44	196.44	2.74%
Aug-22	7423.43	7329.78	-93.65	93.65	1.26%
Sep-22	7284.15	7237.38	-46.77	46.77	0.64%
Oct-22	6893.81	7046.26	152.45	152.45	2.21%
Nov-22	7094.53	7145.04	50.51	50.51	0.71%
Dec-22	7573.05	7297.36	-275.69	275.69	3.64%
Jan-23	7451.74	7417.86	-33.88	33.88	0.45%
Feb-23	7771.70	7433.15	-338.55	338.55	4.36%
Mar-23	7876.28	7537.83	-338.45	338.45	4.30%
Apr-23	7631.74	7473.47	-158.27	158.27	2.07%
May-23	7870.57	7557.43	-313.15	313.15	3.98%
Jun-23	7446.14	7554.34	108.20	108.20	1.45%
Jul-23	7531.53	7582.49	50.96	50.96	0.68%
Aug-23	7699.41	7491.59	-207.82	207.82	2.70%
Sep-23	7439.13	7597.61	158.48	158.48	2.13%
Oct-23	7608.08	7555.11	-52.97	52.97	0.70%
Nov-23	7321.72	7473.95	152.23	152.23	2.08%
Dec-23	7453.75	7698.41	244.66	244.66	3.28%
Jan-24	7733.24	7765.61	32.37	32.37	0.42%
Feb-24	7630.57	7842.60	212.03	212.03	2.78%
Mar-24	7630.02	7799.74	169.72	169.72	2.22%
Apr-24	7952.62	7853.47	-99.15	99.15	1.25%
May-24	8144.13	7917.58	-226.55	226.55	2.78%
Jun-24	8275.38	7990.03	-285.35	285.35	3.45%
Jul-24	8164.12	8198.40	34.28	34.28	0.42%

10.36 L2M: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5547.05	-622.69	622.69	10.09%
Aug-20	5897.76	6192.05	294.29	294.29	4.99%
Sep-20	5963.57	6210.55	246.98	246.98	4.14%
Oct-20	5866.10	6274.96	408.86	408.86	6.97%
Nov-20	5577.27	6256.92	679.65	679.65	12.19%
Dec-20	6266.19	6084.79	-181.40	181.40	2.89%
Jan-21	6460.52	6203.20	-257.32	257.32	3.98%
Feb-21	6407.46	6345.87	-61.59	61.59	0.96%
Mar-21	6483.43	6122.50	-360.93	360.93	5.57%
Apr-21	6713.63	6501.89	-211.74	211.74	3.15%
May-21	6969.81	6747.42	-222.39	222.39	3.19%
Jun-21	7022.61	6780.06	-242.55	242.55	3.45%
Jul-21	7037.47	6939.34	-98.13	98.13	1.39%
Aug-21	7032.30	6962.54	-69.76	69.76	0.99%
Sep-21	7119.70	7007.49	-112.21	112.21	1.58%
Oct-21	7086.42	6991.07	-95.35	95.35	1.35%
Nov-21	7237.57	7046.80	-190.77	190.77	2.64%
Dec-21	7059.45	7222.05	162.60	162.60	2.30%
Jan-22	7384.54	7147.42	-237.12	237.12	3.21%
Feb-22	7464.37	7183.96	-280.41	280.41	3.76%
Mar-22	7458.25	7226.49	-231.76	231.76	3.11%
Apr-22	7515.68	7192.08	-323.60	323.60	4.31%
May-22	7544.55	7271.35	-273.20	273.20	3.62%
Jun-22	7607.66	7234.72	-372.94	372.94	4.90%
Jul-22	7169.28	7298.96	129.68	129.68	1.81%
Aug-22	7423.43	7262.48	-160.95	160.95	2.17%
Sep-22	7284.15	7324.65	40.50	40.50	0.56%
Oct-22	6893.81	7359.66	465.85	465.85	6.76%
Nov-22	7094.53	7223.45	128.92	128.92	1.82%
Dec-22	7573.05	7372.66	-200.39	200.39	2.65%
Jan-23	7451.74	7359.17	-92.57	92.57	1.24%
Feb-23	7771.70	7365.51	-406.19	406.19	5.23%
Mar-23	7876.28	7229.44	-646.84	646.84	8.21%
Apr-23	7631.74	7353.19	-278.55	278.55	3.65%
May-23	7870.57	7288.14	-582.43	582.43	7.40%
Jun-23	7446.14	7326.83	-119.31	119.31	1.60%
Jul-23	7531.53	7251.48	-280.05	280.05	3.72%
Aug-23	7699.41	7374.35	-325.06	325.06	4.22%
Sep-23	7439.13	7178.97	-260.16	260.16	3.50%
Oct-23	7608.08	7246.58	-361.51	361.51	4.75%
Nov-23	7321.72	7181.03	-140.69	140.69	1.92%
Dec-23	7453.75	7177.57	-276.18	276.18	3.71%
Jan-24	7733.24	7251.99	-481.25	481.25	6.22%
Feb-24	7630.57	7293.29	-337.28	337.28	4.42%
Mar-24	7630.02	7150.01	-480.01	480.01	6.29%
Apr-24	7952.62	7256.54	-696.08	696.08	8.75%
May-24	8144.13	7387.19	-756.94	756.94	9.29%
Jun-24	8275.38	7465.44	-809.94	809.94	9.79%
Jul-24	8164.12	7567.29	-596.83	596.83	7.31%

10.37 L3M: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5320.79	-848.95	848.95	13.76%
Aug-20	5897.76	5583.12	-314.64	314.64	5.33%
Sep-20	5963.57	5509.20	-454.37	454.37	7.62%
Oct-20	5866.10	6371.06	504.96	504.96	8.61%
Nov-20	5577.27	6496.00	918.73	918.73	16.47%
Dec-20	6266.19	6585.39	319.20	319.20	5.09%
Jan-21	6460.52	6403.71	-56.81	56.81	0.88%
Feb-21	6407.46	6369.09	-38.37	38.37	0.60%
Mar-21	6483.43	6437.54	-45.89	45.89	0.71%
Apr-21	6713.63	6232.14	-481.49	481.49	7.17%
May-21	6969.81	6650.82	-318.99	318.99	4.58%
Jun-21	7022.61	6885.79	-136.82	136.82	1.95%
Jul-21	7037.47	6931.94	-105.54	105.54	1.50%
Aug-21	7032.30	7186.53	154.23	154.23	2.19%
Sep-21	7119.70	7341.14	221.44	221.44	3.11%
Oct-21	7086.42	7519.02	432.60	432.60	6.10%
Nov-21	7237.57	7495.86	258.29	258.29	3.57%
Dec-21	7059.45	7472.86	413.41	413.41	5.86%
Jan-22	7384.54	7547.26	162.72	162.72	2.20%
Feb-22	7464.37	7466.57	2.20	2.20	0.03%
Mar-22	7458.25	7445.68	-12.57	12.57	0.17%
Apr-22	7515.68	7337.88	-177.80	177.80	2.37%
May-22	7544.55	7191.47	-353.08	353.08	4.68%
Jun-22	7607.66	7285.45	-322.21	322.21	4.24%
Jul-22	7169.28	7257.05	87.77	87.77	1.22%
Aug-22	7423.43	7418.71	-4.72	4.72	0.06%
Sep-22	7284.15	7509.30	225.15	225.15	3.09%
Oct-22	6893.81	7653.25	759.44	759.44	11.02%
Nov-22	7094.53	7666.60	572.07	572.07	8.06%
Dec-22	7573.05	7481.91	-91.14	91.14	1.20%
Jan-23	7451.74	7595.97	144.23	144.23	1.94%
Feb-23	7771.70	7734.48	-37.22	37.22	0.48%
Mar-23	7876.28	7971.65	95.37	95.37	1.21%
Apr-23	7631.74	7671.18	39.44	39.44	0.52%
May-23	7870.57	7536.37	-334.20	334.20	4.25%
Jun-23	7446.14	7332.42	-113.72	113.72	1.53%
Jul-23	7531.53	7503.64	-27.89	27.89	0.37%
Aug-23	7699.41	7624.71	-74.70	74.70	0.97%
Sep-23	7439.13	7475.03	35.90	35.90	0.48%
Oct-23	7608.08	7672.75	64.67	64.67	0.85%
Nov-23	7321.72	7755.04	433.32	433.32	5.92%
Dec-23	7453.75	7723.04	269.29	269.29	3.61%
Jan-24	7733.24	7755.53	22.28	22.28	0.29%
Feb-24	7630.57	7871.14	240.57	240.57	3.15%
Mar-24	7630.02	8026.04	396.02	396.02	5.19%
Apr-24	7952.62	8127.46	174.84	174.84	2.20%
May-24	8144.13	7776.66	-367.47	367.47	4.51%
Jun-24	8275.38	7598.75	-676.63	676.63	8.18%
Jul-24	8164.12	7616.15	-547.97	547.97	6.71%

10.38 L4M: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	5590.55	-579.19	579.19	9.39%
Aug-20	5897.76	5175.60	-722.16	722.16	12.24%
Sep-20	5963.57	5647.12	-316.45	316.45	5.31%
Oct-20	5866.10	5505.45	-360.65	360.65	6.15%
Nov-20	5577.27	6355.01	777.74	777.74	13.94%
Dec-20	6266.19	6646.99	380.80	380.80	6.08%
Jan-21	6460.52	6478.39	17.87	17.87	0.28%
Feb-21	6407.46	6239.35	-168.11	168.11	2.62%
Mar-21	6483.43	6431.08	-52.35	52.35	0.81%
Apr-21	6713.63	6727.31	13.68	13.68	0.20%
May-21	6969.81	6939.78	-30.03	30.03	0.43%
Jun-21	7022.61	7203.35	180.74	180.74	2.57%
Jul-21	7037.47	7256.37	218.90	218.90	3.11%
Aug-21	7032.30	7297.49	265.19	265.19	3.77%
Sep-21	7119.70	7322.30	202.60	202.60	2.85%
Oct-21	7086.42	7285.09	198.67	198.67	2.80%
Nov-21	7237.57	7205.56	-32.01	32.01	0.44%
Dec-21	7059.45	7161.85	102.40	102.40	1.45%
Jan-22	7384.54	7590.48	205.94	205.94	2.79%
Feb-22	7464.37	7678.93	214.56	214.56	2.87%
Mar-22	7458.25	6960.73	-497.52	497.52	6.67%
Apr-22	7515.68	7028.18	-487.50	487.50	6.49%
May-22	7544.55	7070.22	-474.33	474.33	6.29%
Jun-22	7607.66	7001.42	-606.24	606.24	7.97%
Jul-22	7169.28	6993.83	-175.45	175.45	2.45%
Aug-22	7423.43	6884.13	-539.30	539.30	7.26%
Sep-22	7284.15	7043.73	-240.42	240.42	3.30%
Oct-22	6893.81	7159.93	266.12	266.12	3.86%
Nov-22	7094.53	7217.56	123.03	123.03	1.73%
Dec-22	7573.05	7433.64	-139.41	139.41	1.84%
Jan-23	7451.74	7430.53	-21.21	21.21	0.28%
Feb-23	7771.70	7610.55	-161.16	161.16	2.07%
Mar-23	7876.28	7460.03	-416.25	416.25	5.28%
Apr-23	7631.74	7501.42	-130.32	130.32	1.71%
May-23	7870.57	7238.56	-632.01	632.01	8.03%
Jun-23	7446.14	7092.81	-353.33	353.33	4.75%
Jul-23	7531.53	7013.36	-518.17	518.17	6.88%
Aug-23	7699.41	7136.18	-563.23	563.23	7.32%
Sep-23	7439.13	7181.34	-257.79	257.79	3.47%
Oct-23	7608.08	6936.70	-671.38	671.38	8.82%
Nov-23	7321.72	7341.69	19.97	19.97	0.27%
Dec-23	7453.75	7350.65	-103.10	103.10	1.38%
Jan-24	7733.24	7342.37	-390.87	390.87	5.05%
Feb-24	7630.57	7474.69	-155.88	155.88	2.04%
Mar-24	7630.02	7473.21	-156.81	156.81	2.06%
Apr-24	7952.62	7529.96	-422.66	422.66	5.31%
May-24	8144.13	7626.43	-517.70	517.70	6.36%
Jun-24	8275.38	7479.40	-795.98	795.98	9.62%
Jul-24	8164.12	7473.48	-690.64	690.64	8.46%

10.39 L5M: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	6580.04	410.30	410.30	6.65%
Aug-20	5897.76	6438.06	540.30	540.30	9.16%
Sep-20	5963.57	5138.87	-824.70	824.70	13.83%
Oct-20	5866.10	6027.58	161.48	161.48	2.75%
Nov-20	5577.27	6467.51	890.24	890.24	15.96%
Dec-20	6266.19	6518.52	252.33	252.33	4.03%
Jan-21	6460.52	6778.71	318.18	318.18	4.93%
Feb-21	6407.46	6942.65	535.19	535.19	8.35%
Mar-21	6483.43	6788.47	305.04	305.04	4.70%
Apr-21	6713.63	6705.94	-7.69	7.69	0.11%
May-21	6969.81	6693.91	-275.90	275.90	3.96%
Jun-21	7022.61	6492.39	-530.22	530.22	7.55%
Jul-21	7037.47	6697.51	-339.96	339.96	4.83%
Aug-21	7032.30	6896.91	-135.39	135.39	1.93%
Sep-21	7119.70	6980.24	-139.46	139.46	1.96%
Oct-21	7086.42	7169.10	82.68	82.68	1.17%
Nov-21	7237.57	7254.25	16.68	16.68	0.23%
Dec-21	7059.45	7272.55	213.10	213.10	3.02%
Jan-22	7384.54	7191.14	-193.40	193.40	2.62%
Feb-22	7464.37	7332.77	-131.60	131.60	1.76%
Mar-22	7458.25	7333.77	-124.48	124.48	1.67%
Apr-22	7515.68	7186.55	-329.13	329.13	4.38%
May-22	7544.55	7383.64	-160.91	160.91	2.13%
Jun-22	7607.66	7353.51	-254.15	254.15	3.34%
Jul-22	7169.28	7236.85	67.57	67.57	0.94%
Aug-22	7423.43	7349.16	-74.27	74.27	1.00%
Sep-22	7284.15	7392.15	108.00	108.00	1.48%
Oct-22	6893.81	7471.22	577.41	577.41	8.38%
Nov-22	7094.53	7465.84	371.31	371.31	5.23%
Dec-22	7573.05	7430.15	-142.90	142.90	1.89%
Jan-23	7451.74	7298.78	-152.96	152.96	2.05%
Feb-23	7771.70	7136.49	-635.21	635.21	8.17%
Mar-23	7876.28	7243.46	-632.82	632.82	8.03%
Apr-23	7631.74	7507.73	-124.01	124.01	1.62%
May-23	7870.57	7727.37	-143.20	143.20	1.82%
Jun-23	7446.14	7735.02	288.88	288.88	3.88%
Jul-23	7531.53	7655.48	123.95	123.95	1.65%
Aug-23	7699.41	7491.61	-207.80	207.80	2.70%
Sep-23	7439.13	7530.08	90.95	90.95	1.22%
Oct-23	7608.08	7427.31	-180.77	180.77	2.38%
Nov-23	7321.72	7441.21	119.49	119.49	1.63%
Dec-23	7453.75	7567.02	113.27	113.27	1.52%
Jan-24	7733.24	7622.48	-110.76	110.76	1.43%
Feb-24	7630.57	7648.91	18.34	18.34	0.24%
Mar-24	7630.02	7733.20	103.18	103.18	1.35%
Apr-24	7952.62	7849.31	-103.31	103.31	1.30%
May-24	8144.13	7990.26	-153.87	153.87	1.89%
Jun-24	8275.38	8069.42	-205.96	205.96	2.49%
Jul-24	8164.12	7891.57	-272.55	272.55	3.34%

10.40 L6M: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	6737.25	567.51	567.51	9.20%
Aug-20	5897.76	6695.47	797.71	797.71	13.53%
Sep-20	5963.57	6622.55	658.98	658.98	11.05%
Oct-20	5866.10	6046.09	179.99	179.99	3.07%
Nov-20	5577.27	5900.91	323.64	323.64	5.80%
Dec-20	6266.19	5887.44	-378.75	378.75	6.04%
Jan-21	6460.52	6410.18	-50.34	50.34	0.78%
Feb-21	6407.46	6797.23	389.77	389.77	6.08%
Mar-21	6483.43	7087.91	604.48	604.48	9.32%
Apr-21	6713.63	7127.43	413.80	413.80	6.16%
May-21	6969.81	6877.05	-92.76	92.76	1.33%
Jun-21	7022.61	7182.43	159.82	159.82	2.28%
Jul-21	7037.47	6864.29	-173.18	173.18	2.46%
Aug-21	7032.30	6653.63	-378.67	378.67	5.38%
Sep-21	7119.70	6720.89	-398.81	398.81	5.60%
Oct-21	7086.42	6911.36	-175.06	175.06	2.47%
Nov-21	7237.57	7279.44	41.87	41.87	0.58%
Dec-21	7059.45	7544.32	484.87	484.87	6.87%
Jan-22	7384.54	7668.51	283.97	283.97	3.85%
Feb-22	7464.37	7303.48	-160.89	160.89	2.16%
Mar-22	7458.25	7174.87	-283.38	283.38	3.80%
Apr-22	7515.68	7083.51	-432.17	432.17	5.75%
May-22	7544.55	7085.38	-459.17	459.17	6.09%
Jun-22	7607.66	7715.06	107.40	107.40	1.41%
Jul-22	7169.28	7722.41	553.13	553.13	7.72%
Aug-22	7423.43	7579.49	156.06	156.06	2.10%
Sep-22	7284.15	7480.93	196.78	196.78	2.70%
Oct-22	6893.81	7290.71	396.90	396.90	5.76%
Nov-22	7094.53	7244.90	150.37	150.37	2.12%
Dec-22	7573.05	7126.03	-447.02	447.02	5.90%
Jan-23	7451.74	7108.43	-343.31	343.31	4.61%
Feb-23	7771.70	7023.60	-748.10	748.10	9.63%
Mar-23	7876.28	6856.80	-1019.48	1019.48	12.94%
Apr-23	7631.74	7024.66	-607.08	607.08	7.95%
May-23	7870.57	7170.96	-699.61	699.61	8.89%
Jun-23	7446.14	7507.68	61.54	61.54	0.83%
Jul-23	7531.53	7560.44	28.91	28.91	0.38%
Aug-23	7699.41	7627.10	-72.31	72.31	0.94%
Sep-23	7439.13	7516.79	77.66	77.66	1.04%
Oct-23	7608.08	7605.07	-3.01	3.01	0.04%
Nov-23	7321.72	7586.57	264.85	264.85	3.62%
Dec-23	7453.75	7320.96	-132.79	132.79	1.78%
Jan-24	7733.24	7474.48	-258.76	258.76	3.35%
Feb-24	7630.57	7420.47	-210.10	210.10	2.75%
Mar-24	7630.02	7407.39	-222.63	222.63	2.92%
Apr-24	7952.62	7632.06	-320.56	320.56	4.03%
May-24	8144.13	7752.10	-392.03	392.03	4.81%
Jun-24	8275.38	7904.94	-370.44	370.44	4.48%
Jul-24	8164.12	8190.83	26.71	26.71	0.33%

10.41 L9M: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	6670.52	500.78	500.78	8.12%
Aug-20	5897.76	6690.00	792.24	792.24	13.43%
Sep-20	5963.57	6796.17	832.60	832.60	13.96%
Oct-20	5866.10	6985.47	1119.37	1119.37	19.08%
Nov-20	5577.27	7001.35	1424.08	1424.08	25.53%
Dec-20	6266.19	6521.31	255.12	255.12	4.07%
Jan-21	6460.52	6030.52	-430.00	430.00	6.66%
Feb-21	6407.46	5921.37	-486.09	486.09	7.59%
Mar-21	6483.43	5576.61	-906.83	906.83	13.99%
Apr-21	6713.63	6643.16	-70.47	70.47	1.05%
May-21	6969.81	7097.95	128.14	128.14	1.84%
Jun-21	7022.61	7551.66	529.05	529.05	7.53%
Jul-21	7037.47	7505.78	468.31	468.31	6.65%
Aug-21	7032.30	7400.16	367.86	367.86	5.23%
Sep-21	7119.70	7115.93	-3.77	3.77	0.05%
Oct-21	7086.42	6653.85	-432.57	432.57	6.10%
Nov-21	7237.57	6561.28	-676.29	676.29	9.34%
Dec-21	7059.45	6567.55	-491.90	491.90	6.97%
Jan-22	7384.54	6668.03	-716.51	716.51	9.70%
Feb-22	7464.37	6943.25	-521.12	521.12	6.98%
Mar-22	7458.25	7266.37	-191.88	191.88	2.57%
Apr-22	7515.68	7465.13	-50.55	50.55	0.67%
May-22	7544.55	7170.60	-373.95	373.95	4.96%
Jun-22	7607.66	7071.30	-536.36	536.36	7.05%
Jul-22	7169.28	7064.27	-105.01	105.01	1.46%
Aug-22	7423.43	7377.57	-45.86	45.86	0.62%
Sep-22	7284.15	7921.67	637.52	637.52	8.75%
Oct-22	6893.81	7908.02	1014.21	1014.21	14.71%
Nov-22	7094.53	7774.72	680.19	680.19	9.59%
Dec-22	7573.05	7781.17	208.12	208.12	2.75%
Jan-23	7451.74	7585.60	133.86	133.86	1.80%
Feb-23	7771.70	7515.63	-256.07	256.07	3.29%
Mar-23	7876.28	7394.18	-482.10	482.10	6.12%
Apr-23	7631.74	7429.65	-202.09	202.09	2.65%
May-23	7870.57	7249.94	-620.63	620.63	7.89%
Jun-23	7446.14	6909.61	-536.53	536.53	7.21%
Jul-23	7531.53	6873.71	-657.82	657.82	8.73%
Aug-23	7699.41	7051.46	-647.95	647.95	8.42%
Sep-23	7439.13	7549.21	110.08	110.08	1.48%
Oct-23	7608.08	7570.54	-37.54	37.54	0.49%
Nov-23	7321.72	7747.98	426.26	426.26	5.82%
Dec-23	7453.75	7748.00	294.25	294.25	3.95%
Jan-24	7733.24	7839.24	106.00	106.00	1.37%
Feb-24	7630.57	7875.53	244.96	244.96	3.21%
Mar-24	7630.02	7642.05	12.03	12.03	0.16%
Apr-24	7952.62	7581.46	-371.16	371.16	4.67%
May-24	8144.13	7560.38	-583.75	583.75	7.17%
Jun-24	8275.38	7526.19	-749.19	749.19	9.05%
Jul-24	8164.12	7544.88	-619.24	619.24	7.58%

10.42 L12M: Predictions (lowest MSE of 40)

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	7162.41	992.67	992.67	16.09%
Aug-20	5897.76	7343.23	1445.47	1445.47	24.51%
Sep-20	5963.57	7086.86	1123.29	1123.29	18.84%
Oct-20	5866.10	7038.28	1172.18	1172.18	19.98%
Nov-20	5577.27	6941.40	1364.13	1364.13	24.46%
Dec-20	6266.19	6966.31	700.12	700.12	11.17%
Jan-21	6460.52	6984.47	523.95	523.95	8.11%
Feb-21	6407.46	7035.67	628.21	628.21	9.80%
Mar-21	6483.43	7128.66	645.23	645.23	9.95%
Apr-21	6713.63	6976.66	263.03	263.03	3.92%
May-21	6969.81	6538.56	-431.25	431.25	6.19%
Jun-21	7022.61	6697.44	-325.17	325.17	4.63%
Jul-21	7037.47	6785.29	-252.18	252.18	3.58%
Aug-21	7032.30	7035.48	3.18	3.18	0.05%
Sep-21	7119.70	7255.53	135.83	135.83	1.91%
Oct-21	7086.42	7387.39	300.97	300.97	4.25%
Nov-21	7237.57	7328.56	90.99	90.99	1.26%
Dec-21	7059.45	7277.89	218.44	218.44	3.09%
Jan-22	7384.54	7076.97	-307.57	307.57	4.17%
Feb-22	7464.37	6901.78	-562.59	562.59	7.54%
Mar-22	7458.25	7023.06	-435.19	435.19	5.84%
Apr-22	7515.68	7149.73	-365.95	365.95	4.87%
May-22	7544.55	7319.86	-224.69	224.69	2.98%
Jun-22	7607.66	7499.79	-107.87	107.87	1.42%
Jul-22	7169.28	7491.22	321.94	321.94	4.49%
Aug-22	7423.43	7344.54	-78.89	78.89	1.06%
Sep-22	7284.15	7287.50	3.35	3.35	0.05%
Oct-22	6893.81	7271.11	377.30	377.30	5.47%
Nov-22	7094.53	7385.40	290.87	290.87	4.10%
Dec-22	7573.05	7580.96	7.91	7.91	0.10%
Jan-23	7451.74	7455.86	4.12	4.12	0.06%
Feb-23	7771.70	7534.12	-237.58	237.58	3.06%
Mar-23	7876.28	7636.12	-240.16	240.16	3.05%
Apr-23	7631.74	7795.85	164.11	164.11	2.15%
May-23	7870.57	7939.05	68.48	68.48	0.87%
Jun-23	7446.14	7934.29	488.15	488.15	6.56%
Jul-23	7531.53	7981.26	449.73	449.73	5.97%
Aug-23	7699.41	7939.09	239.68	239.68	3.11%
Sep-23	7439.13	7692.08	252.95	252.95	3.40%
Oct-23	7608.08	7735.10	127.02	127.02	1.67%
Nov-23	7321.72	7692.32	370.60	370.60	5.06%
Dec-23	7453.75	7853.33	399.58	399.58	5.36%
Jan-24	7733.24	7819.49	86.25	86.25	1.12%
Feb-24	7630.57	7852.16	221.59	221.59	2.90%
Mar-24	7630.02	7793.81	163.79	163.79	2.15%
Apr-24	7952.62	7730.26	-222.36	222.36	2.80%
May-24	8144.13	7904.95	-239.18	239.18	2.94%
Jun-24	8275.38	7481.40	-793.98	793.98	9.59%
Jul-24	8164.12	7669.75	-494.37	494.37	6.06%

10.43 ARIMA: Predictions

Time	FTSE100	Prediction	Error	Absolute Error	% Error
Jul-20	6169.74	6013.58	-156.16	156.16	2.53%
Aug-20	5897.76	6048.88	151.12	151.12	2.56%
Sep-20	5963.57	6003.62	40.05	40.05	0.67%
Oct-20	5866.10	5948.40	82.30	82.30	1.40%
Nov-20	5577.27	5966.39	389.12	389.12	6.98%
Dec-20	6266.19	5962.75	-303.44	303.44	4.84%
Jan-21	6460.52	6042.34	-418.18	418.18	6.47%
Feb-21	6407.46	6003.22	-404.24	404.24	6.31%
Mar-21	6483.43	6005.00	-478.43	478.43	7.38%
Apr-21	6713.63	5979.29	-734.34	734.34	10.94%
May-21	6969.81	6051.10	-918.72	918.72	13.18%
Jun-21	7022.61	6029.33	-993.28	993.28	14.14%
Jul-21	7037.47	5965.25	-1072.22	1072.22	15.24%
Aug-21	7032.30	5999.93	-1032.37	1032.37	14.68%
Sep-21	7119.70	5953.64	-1166.06	1166.06	16.38%
Oct-21	7086.42	5897.77	-1188.65	1188.65	16.77%
Nov-21	7237.57	5914.77	-1322.80	1322.80	18.28%
Dec-21	7059.45	5910.45	-1149.00	1149.00	16.28%
Jan-22	7384.54	5989.07	-1395.47	1395.47	18.90%
Feb-22	7464.37	5949.24	-1515.13	1515.13	20.30%
Mar-22	7458.25	5950.07	-1508.18	1508.18	20.22%
Apr-22	7515.68	5923.64	-1592.04	1592.04	21.18%
May-22	7544.55	5994.51	-1550.04	1550.04	20.55%
Jun-22	7607.66	5972.01	-1635.65	1635.65	21.50%
Jul-22	7169.28	5907.01	-1262.27	1262.27	17.61%
Aug-22	7423.43	5940.94	-1482.49	1482.49	19.97%
Sep-22	7284.15	5893.74	-1390.41	1390.41	19.09%
Oct-22	6893.81	5837.12	-1056.69	1056.69	15.33%
Nov-22	7094.53	5853.22	-1241.31	1241.31	17.50%
Dec-22	7573.05	5848.12	-1724.93	1724.93	22.78%
Jan-23	7451.74	5925.86	-1525.88	1525.88	20.48%
Feb-23	7771.70	5885.25	-1886.45	1886.45	24.27%
Mar-23	7876.28	5885.21	-1991.07	1991.07	25.28%
Apr-23	7631.74	5857.99	-1773.76	1773.76	23.24%
May-23	7870.57	5927.99	-1942.58	1942.58	24.68%
Jun-23	7446.14	5904.69	-1541.45	1541.45	20.70%
Jul-23	7531.53	5838.83	-1692.70	1692.70	22.47%
Aug-23	7699.41	5871.96	-1827.45	1827.45	23.73%
Sep-23	7439.13	5823.90	-1615.23	1615.23	21.71%
Oct-23	7608.08	5766.47	-1841.61	1841.61	24.21%
Nov-23	7321.72	5781.71	-1540.01	1540.01	21.03%
Dec-23	7453.75	5775.81	-1677.94	1677.94	22.51%
Jan-24	7733.24	5852.70	-1880.55	1880.55	24.32%
Feb-24	7630.57	5811.28	-1819.30	1819.30	23.84%
Mar-24	7630.02	5810.38	-1819.64	1819.64	23.85%
Apr-24	7952.62	5782.34	-2170.28	2170.28	27.29%
May-24	8144.13	5851.50	-2292.63	2292.63	28.15%
Jun-24	8275.38	5827.38	-2448.00	2448.00	29.58%
Jul-24	8164.12	5760.68	-2403.44	2403.44	29.44%

ARIMA(p,d,q)			***SARIMA(p,d,q,s)***				
p	d	q	p	d	q	s	AIC
0	1	0					5854
1	1	0					5856
0	1	1					5856
1	1	1					5857
0	1	0	0	1	0	12	5970
0	1	0	1	0	1	12	5855
1	1	1	1	0	1	12	5855
1	1	1	0	1	1	12	5730
0	1	1	1	0	0	12	5858

10.44 Script (Python): BASE

```
import numpy as np
import pandas as pd
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from sklearn.preprocessing import StandardScaler
import keras_tuner as kt

def load_data(file_path):
 df = pd.read_csv(file_path)

 X = df.iloc[:, 2:].values
 Y = df.iloc[:, 1].values

 return df, X, Y

def build_model(input_dim):
 model = keras.Sequential([
  layers.Dense(512, kernel_regularizer=keras.regularizers.l2(0.001)),
  layers.LeakyReLU(alpha=0.01),
  layers.BatchNormalization(),

  layers.Dense(256, kernel_regularizer=keras.regularizers.l2(0.001)),
  layers.LeakyReLU(alpha=0.01),
  layers.BatchNormalization(),
  layers.Dropout(0.3),

  layers.Dense(128, kernel_regularizer=keras.regularizers.l2(0.001)),
  layers.LeakyReLU(alpha=0.01),
  layers.BatchNormalization(),
  layers.Dropout(0.3),

```

```
[34]       layers.Dense(64, kernel_regularizer=keras.regularizers.l2(0.001)),
[35]       layers.LeakyReLU(alpha=0.01),
[36]       layers.BatchNormalization(),
[37]       layers.Dropout(0.3),
[38]
[39]       layers.Dense(1)
[40]      ])
[41]
[42]      model.compile(
[43]       optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
[44]       loss='mse',
[45]       metrics=['mae']
[46]      )
[47]      return model
[48]
[49]    file_path = 'data.csv'
[50]    df, X, Y = load_data(file_path)
[51]
[52]    missing_indices = df[df['VAR_0'].isna()].index
[53]
[54]    X_train_complete = df.dropna(subset=['VAR_0']).iloc[:, 2:].values
[55]    Y_train_complete = df.dropna(subset=['VAR_0']).iloc[:, 1].values
[56]
[57]    X_scaler = StandardScaler()
[58]    X_train_complete = X_scaler.fit_transform(X_train_complete)
[59]
[60]    Y_scaler = StandardScaler()
[61]    Y_train_complete = Y_scaler.fit_transform(Y_train_complete.reshape(-1, 1)).flatten()
[62]
[63]    early_stop = EarlyStopping(monitor='val_loss', patience=20, restore_best_weights=True)
[64]    reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.5, patience=10, min_lr=1e-5)
[65]
[66]    model = build_model(input_dim=X_train_complete.shape[1])
[67]    history = model.fit(
[68]     X_train_complete, Y_train_complete,
```

```
[69]      epochs=200,
[70]      batch_size=32,
[71]      verbose=1,
[72]      validation_split=0.2,
[73]      callbacks=[early_stop, reduce_lr]
[74]     )
[75]
[76]     if not missing_indices.empty:
[77]      X_missing = X_scaler.transform(df.iloc[missing_indices, 2:].values)
[78]      Y_predicted_scaled = model.predict(X_missing)
[79]
[80]      Y_predicted = Y_scaler.inverse_transform(Y_predicted_scaled)
[81]
[82]      df.loc[missing_indices, 'VAR_0'] = Y_predicted.flatten()
[83]
[84]      print("Predicted missing 'VAR_0' values:")
[85]      print(df.loc[missing_indices, ['TIM', 'VAR_0']])
[86]
[87]     df.to_csv("Generated_Data_filled.csv", index=False)
```

10.45 Commentary (Python): BASE

```
[1]      Import NumPy libraries for calculations
[2]      Import Pandas libraries for data frames
[3]      Import TensorFlow for the architecture
[4]      Import Keras from the TensorFlow library for model construction
[5]      Import Keras Layers for model construction
[6]      Import EarlyStopping for training
[7]      Import StandardScaler for normalising variables
[8]      Import Keras Tuner for hyperparameter tuning
[9]
[10]     Defines a function to load and process data from the CSV data file
[11]     Reads and converts CSV data into a Pandas dataframe
```

```
[12]
[13]    Selects all the feature columns, ignoring the first 2 (time and FTSE100 Index price)
[14]    Sets VAR_0, the FTSE100 Index price, as the dependent "target" variable
[15]
[16]    Returns the data frame built in [10] to [14]
[17]
[18]    Opens and defines the function that contains the neural network
[19]    The model is sequential
[20]    Layer 1: 512 neurons, adds a Ridge regularisation penalty, factor 0.001
[21]    LeakyReLU activation layer
[22]    Normalises the activations, stabilising training and improving output
[23]
[24]    Layer 2: 256 neurons, adds a Ridge regularisation penalty, factor 0.001
[25]    LeakyReLU activation layer
[26]    Normalises the activations, stabilising training and improving output
[27]    Dropout rate, set to reduce overfitting
[28]
[29]    Layer 3: 128 neurons, adds a Ridge regularisation penalty, factor 0.001
[30]    LeakyReLU activation layer
[31]    Normalises the activations, stabilising training and improving output
[32]    Dropout rate, set to reduce overfitting
[33]
[34]    Layer 4: 64 neurons, adds a Ridge regularisation penalty, factor 0.001
[35]    LeakyReLU activation layer
[36]    Normalises the activations, stabilising training and improving output
[37]    Dropout rate, set to reduce overfitting
[38]
[39]    Layer 5: 1 neuron (output)
[40]    Closes the definition of the function that contains the neural network, opened on [18]
[41]
[42]    Opens the compilation of the model
[43]    Uses a prebuilt Adam optimiser with a learning rate
[44]    Mean square error as the loss function
[45]    Mean absolute error as a metric
```

```
[46]    Closes the compilation of the model, opened in [42]
[47]    Returns the compiled model, opened in [42]
[48]
[49]    File path for my data
[50]    Loads the data in a dataframe built in [10] to [14]
[51]
[52]    Although unnecessary, as the data is clean, this checks for missing data
[53]
[54]    Selects all the rows with missing Var_0 (FTSE100 Index price) values
[55]    Extracts Var_0 values for training, from [54]
[56]
[57]    Scales input variables (macroeconomic variables)
[58]    Standardises input variables (macroeconomic variables)
[59]
[60]    Scales Var_0 (FTSE100 Index price)
[61]    Standardises Var_0 (FTSE100 Index price)
[62]
[63]    Calculates validation loss (based on accuracy), and if it does not improve, the model terminates
[64]    Reduces the learning rate if the validation loss (based on accuracy) plateaux's
[65]
[66]    Builds the neural network based on the definitions in [18] to [40]
[67]    Opens the model fitting function
[68]    Feeds the model the training data
[69]    200 epochs (iterations)
[70]    Takes data points in groups of 32 at a time
[71]    For ease of use, return the training progress to the terminal for the user
[72]    Validation splits the data so that 20% is used for testing.
[73]    Terminates the model early based on [63] and [64]
[74]    Closes the model fitting function, opened on [67]
[75]
[76]    Although unnecessary, as the data is clean, checks for missing data
[77]    Although unnecessary as the data is clean, standardises missing data
[78]    Although unnecessary, as the data is clean, it predicts missing data
[79]
```

[80] Converts predictions to the original scale
[81]
[82] Enter the missing values into the dataset, predicted on [76] to [80]
[83]
[84] Gives the user an output in the terminal
[85] Gives the user an output in the terminal
[86]
[87] Enter the CSV file and input the predictions into the spreadsheet for further analysis

10.46 Script (Stata): ARIMA

```
[1]     gen date = tm(1984m1) + _n - 1
[2]     format date %tm
[3]     tsset date, monthly
[4]
[5]     gen D_VAR_0 = D.VAR_0
[6]     corrgram D_VAR_0, lags(12)
[7]
[8]     arima VAR_0, arima(1,1,1) sarima(0,1,1,12)
[9]     estat ic
[10]
[11]    predict VAR_0_fit, y
[12]
[13]    tsappend, add(49)
[14]
[15]    predict VAR_0_forecast, y dynamic(`=tm(2020m7)')
[16]
[17]    tsline VAR_0 VAR_0_forecast
```

10.47 Commentary (Stata): ARIMA

[1] Creates a variable called data, so that the first row is Jan 84, 2nd is Feb 84, etc
[2] Format [1] as monthly data
[3] Formats [1] as a time series

```
[4]
[5]     First difference of VAR_0
[6]     Correlogram (ACF and PACF) for 12 lags of D_VAR_0 created in [5]
[7]
[8]     Fits an ARIMA AR 1, FD 1, and MA 1 (based on AIC values computed in [9], done before running the code via trial and error)
[9]     Model information of [8]
[10]
[11]    Generates in-sample, fitted values (model predictions)
[12]
[13]    Does [11], 49 times
[14]
[15]    Produces dynamic forecasts, starting in July 2020
[16]
[17]    Plots the FTSE100 Index and then adds forecasted values
```

10.48 Script (Stata): Autocorrelation Tests

```
[1]     gen time = m(1984m1) + _n - 1
[2]     format time %tm
[3]     tsset time
[4]
[5]     dfuller VAR_0, lags(12)
[6]
[7]     gen d_VAR_0 = D.VAR_0 if !missing(D.VAR_0)
[8]
[9]     dfuller d_VAR_0, lags(12)
[10]
[11]    ac VAR_0
[12]    pac VAR_0
[13]    ac d_VAR_0
[14]    pac d_VAR_0
[15]
[16]    wntestq VAR_0
[17]
```

```
[18]    arima VAR_0, arima(1,1,1)
[19]
[20]    predict e, resid
[21]
[22]    tsline e
[23]
[24]    wntestq e
[25]
[26]    ac e
[27]    pac e
[28]
[29]    regress d_VAR_0 L.d_VAR_0
[30]    estat dwatson
[31]    bgodfrey, lags(12)
[32]
[33]    estat ic
[34]
[35]    predict yhat
[36]    gen residual_check = VAR_0 - yhat
[37]    tsline VAR_0 yhat
```

10.49 Commentary (Stata): Autocorrelation Tests

[1] Creates a monthly time variable starting from January 1984.
[2] Format [1] as monthly data
[3] Formats [1] as a time series
[4]
[5] Augmented Dickey-Fuller for stationarity in VAR_0, with 12 lags
[6]
[7] First difference of VAR_0, removing missing values caused by differencing
[8]
[9] Augmented Dickey-Fuller on differenced data
[10]
[11] Plot autocorrelation of VAR_0

```
[12]    Plot the partial autocorrelation of VAR_0
[13]    Plot autocorrelation of Differences
[14]    Plot partial autocorrelation of Differences
[15]
[16]    Portmanteau (Ljung-Box) test for autocorrelation in the original (non-stationary) series (reject H0, not white noise)
[17]
[18]    ARIMA(1,1,1)
[19]
[20]    Predicts the residuals
[21]
[22]    Plots [20]
[23]
[24]    Portmanteau (Ljung-Box) test for residuals
[25]
[26]    Plot autocorrelation of residuals
[27]    Plots the partial autocorrelation of residuals
[28]
[29]    Regresses differenced series with its lag for the Durbin-Watson test
[30]    Durbin Watson
[31]    Breusch-Godfrey test for 12 lags
[32]
[33]    AIC and BIC output (so I can check for ARIMA model fit)
[34]
[35]    Predicts the values from the ARIMA model
[36]    residual check populated with = actual - predicted residuals
[37]    Plots a line graph of [36]
```

10.50 Script (Stata): Correlation of MEVs to MEVs

```
[1]     drop TIM
[2]     drop VAR_0
```

```
[3]     correlate
```

10.51 Commentary (Stata): Correlation of MEVs to MEVs

[1] Drop the variable TIM
[2] Drop the variable VAR_0
[3] Correlation matrix of the remaining variables

10.52 Script (Stata): Correlation of FTSE100 Index to MEVs

```
[1]     forvalues i = 1/26 {
[2]      local varname = "VAR_`i'"
[3]      quietly correlate VAR_0 `varname'
[4]      display "Correlation between VAR_0 and `varname' is " string(r(rho), "%9.4f")
[5]     }
```

10.53 Commentary (Stata): Correlation of FTSE100 Index to MEVs

[1] Opens the loop from i = 1 to i = 26
[2] "varname" stores the "VAR_i" value for each iteration
[3] Calculates the output of each correlation, but does not output it
[4] Outputs in a neat table: "Correlation between VAR_0 and VAR_i is ..."
[5] Formats the output to 3 decimal places
[6] Closes the loop opened in [1]

10.54 Script (Stata): Regression of FTSE100 Index to MEVs

```
[1]     forvalues i = 1/26 {
[2]      regress VAR_0 VAR_`i'
[3]     }
```

10.55 Commentary (Stata): Regression of FTSE100 Index to MEVs

[1] Opens the loop from i = 1 to i = 26

```
[2]     Regresses VAR_0 against the "VAR_i" value for each iteration
[3]     Closes the loop opened in [1]
```

10.56 Script (Stata): Summary Statistics

```
[1]     drop TIM
[2]     summarize
```

10.57 Commentary (Stata): Summary Statistics

```
[1]     Drop the variable TIM
[2]     Generate summary statistics on the remaining variables
```

10.58 Script (Python): Spline

```
[1]     import pandas as pd
[2]     import numpy as np
[3]     from scipy.interpolate import CubicSpline
[4]
[5]     df = pd.read_excel("input.xlsx", header=None)
[6]
[7]     values = df.iloc[:, 0].tolist()
[8]
[9]     x_original = np.arange(0, len(values) * 3, 3)
[10]    x_interp = np.arange(0, len(values) * 3 - 2, 1)
[11]
[12]    spline = CubicSpline(x_original, values)
[13]    y_interp = spline(x_interp)
[14]
[15]    new_df = pd.DataFrame(y_interp, columns=["Expanded Values"])
[16]    new_df.to_excel("q2moutput.xlsx", index=False)
```

10.59 Commentary (Python): Spline

```
[1]     Import Pandas libraries for data frames
[2]     Import NumPy libraries for calculations
[3]     Import CubicSpline from the Scipy Interpolate library for quadratic splining
[4]
[5]     Reads and converts CSV data into a Pandas dataframe
[6]
[7]     Extracts the first column from the dataframe into a list in Python
[8]
[9]     Creates an array of X values spaced three units apart (3 months in a quarter)
[10]    Interpolates the values from [9]
[11]
[12]    Creates a cubic spline function of the values from [9|
[13]    Applies the function created in [12] to the values from [10]
[14]
[15]    Converts the new data into a pandas dataframe
[16]    Exports the new data into an Excel document labelled "q2moutput" (Quarterly2Monthly)
```

10.60 Script (Stata): Durbin Watson

```
 [1]    import excel "C:[FILE DIRECTORY HERE|", sheet("csv_training") firstrow
 [2]
 [3]    gen date = tm(1984m1) + _n - 1
 [4]    format date %tm
 [5]    tsset date, monthly
 [6]
 [7]    arima VAR_0, arima(1,1,1)
 [8]
 [9]    predict VAR_0_fit, y
[10]    gen residuals = VAR_0 - VAR_0_fit
[11]
[12]    regress residuals L.residuals
[13]    estat dwatson
```

10.61 Commentary (Stata): Durbin Watson

```
 [1]    Imports data
 [2]
 [3]    Generates a data variable from January 1984
 [4]    Formats [5]
 [5]    Declares [5] as a time series
 [6]
 [7]    Fits an ARIMA(1,1,1) into VAR_0
 [8]
 [9]    Generates in-sample fitted values from [9], saving them in a new variable "Var_0_fit"
[10]    Calculates the residuals
[11]
[12]    Regression of residuals and their lagged variables
[13]    Performs a Durbin-Watson
```